

Private Lessons in Drawing the World

Tim Dubrovsky

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My parents left this world in the same year, together, as if agreeing without words that neither would take another step without the other. They went the way old trees go in a forest storm: not with drama, not with noise, but in a way that shifts the shape of the whole landscape. With them ended the last living thread of a family that had spent generations defending its wide-windowed apartments along the Yauza River and its hidden summer dachas deep in the woods from the watchful state — the power that scratches at every lock.

Our family line reached back to the owners of famous mechanical workshops in Imperial Moscow and to their stern mothers, who ran their income houses with the calm, commanding confidence of ship captains. Their portraits still sit in the old albums, staring out as if they've seen every revolution and already decided not to be impressed.

My grandfathers were part of Stalin's massive empire — essential, disciplined, and exhausted. One ran a giant aircraft plant where fighter planes rose from steel frames into sleek machines under the glow of factory lamps. The other moved through the shadowed hallways of military counterintelligence, putting down rebellions in western Ukraine after the war, smothering the last sparks of resistance in a place history had already burned raw.

On my mother's side, the women were the quiet backbone of everything. My grandmother and great-grandmother, both born into prosperous families from Moscow's old merchant guilds, spent their lives building small pockets of warmth inside a city that grew colder and more suspicious by the year. They were always with me, doing their best not to meet the eyes of the new Moscow — the hungry, sharp-edged Moscow — as if even a glance could invite trouble.

So that October, with the windows plastered in soggy yellow leaves, I ended up alone in the emptied apartment, sorting through papers that fell apart in my hands. My father's Olympic medals. My mother's paintings. The forbidden manuscripts of Soviet writers who weren't allowed to write. All of it lay around me like pieces of a constellation whose stars had gone dark. I kept searching for some kind of sign, a trace left behind by my parents as they walked, somewhere far beyond me now, hand in hand across their own Champs-Élysées. Not the one in Paris, but the quiet one people imagine when they think of paradise: an open, endless meadow where the living can follow only in memory.

I hoped they had left a note. Hidden in the blue-flowered Chinese porcelain vase that my mother loved. Tucked between the books. Anywhere.

In the old hiding place beneath the floorboards, I found only what they had prepared years earlier: tight bundles of money and children's toys wrapped in crisp paper. Nothing else. No final line. No last word caught on a page.

I'm not even sure what I expected. A confession about the carnage of World War II? A map through the darkness my grandfathers survived: arrests, purges, the usual fallout of building a "just society"? Or some urgent reminder, something unfinished, something meant only for me? But there was nothing. Just silence settling over the rooms, and the sliding of golden leaves down the glass.

I'm sixty now, and I've more or less fulfilled my part of the deal: you will never have to live within that meat-grinder socialist fantasy that tells you, *"Here you go, kids: take the land, the tools, the money, the sweat — everything built by people who show up, think, and make things work — and 'redistribute' it to those who contribute nothing but noise. Call it fairness. Call it equality."* It's the same tired illusion every soft-handed regime sells: punish the builders, reward the bystanders, drain the engine to fuel the passengers, and then act surprised when the train stops moving. Orwell already wrote that book for us¹, better than I ever could, so at least I don't have to start explaining from page one. I don't know if we'll have enough time to sit down and really talk it all through. Our gap in age, plus the language barrier, gets in the way: by the time we inch our way to the point, something important will slip between our fingers or simply be forgotten.

So instead of long, awkward tête-à-tête lectures, I'm asking you to draw with me. Close your eyes and imagine an art studio with canvases on easels, fat and skinny brushes, the smell of linseed oil, a paint-spattered palette, and bright morning light pouring in through tall, glass windows.

If you're willing, let's mix the colors now and stretch the canvas. Carefully, in detail, we'll paint everything we already know about the world around us. Then, with the finest brushes, we'll sketch in the outlines of the continents and lay down a grid of coordinates. That's how we'll figure out where we are, and where to move next.

People look up at the night sky and often assume the universe is too big, too strange, too far beyond human comprehension. Better to curl up, accept its mysterious rules, and stop asking questions. But the truth is far kinder and far more interesting. Everything around us is governed by only a handful of fundamental laws. Learn them, even roughly, and the world stops feeling like a riddle and starts behaving like an understandable, predictable machine.

We don't have infinite space here, so I'll leave you pointers to a few excellent books — places where the right words and the deeper explanations live. If some paragraphs feel too dense at first, just skip them. You'll come back later with clearer eyes. And your own kids, someday, will probably understand even more than either of us can imagine right now. This is only the beginning.

Modern science is barely four centuries old. It began the day Galileo, in Padua, ordered a set of beautifully polished lenses from the Low Countries and assembled them into a telescope. Almost immediately, he saw what no one before him had seen clearly: craters etched into the Moon, sunspots drifting across the Sun's glowing face, and the Sun itself turning like a vast, slow wheel. At night, the Milky Way was no longer a hazy myth but a luminous river of countless stars.

Here's the fun parallel history rarely mentions: around the same time, Antonie van Leeuwenhoek — a Dutch cloth merchant and self-taught lens-grinder — looked through similar lenses *the other way* and discovered an entirely different frontier. In a drop of water, in a piece of onion skin, in a speck of dust scraped off his desk, he found algae, protozoa, and cells — an entire micro-universe swarming under his fingertips.

The technology of glass manufacturing and polishing led to the creation of optical lenses and split the world open in two directions at once. Outward: the cosmos. Inward: the cell. Suddenly, the old picture of reality shattered. Everything living was made of cells, and everything above was made of stars.

Breakthroughs like that kept coming: steam and gasoline engines, electricity, rifles, steel, antibiotics, and the atomic bomb. Each invention reshaped civilization, but something quieter and just as important happened alongside it: the inventors themselves changed. Thought and persistence began to turn into success, success into wealth, and wealth into longer, more comfortable lives. With surplus resources, they educated their children, built private laboratories, and founded universities. As knowledge accumulated, they gained better tools, ships, and weapons, and for the first time looked across the oceans with the means to act on their ideas. They built empires according to their own logic and laws, not inherited tradition. Shaped by science, strengthened by industry, and protected by wealth, they gradually became a new kind of people...

Still, we're getting a bit ahead of ourselves. Before we go any further, I need to warn you about something. Always, and I mean always, be careful.

Remember: these drawing lessons are for you, for your future kids, and for no one else. You've read *count Odoevsky's* old fairy tale about *the Black Hen*, right? Exactly. Some secrets are gifts; some are responsibilities; and some are simply wasted on people who aren't ready for them. So why bother explaining human evolution, the rise of civilizations, or the weird logical knots inside modern "fairness" theories to every frantic passerby drifting in and out of your life? Just smile, wave politely, and nod along when your left-leaning professors or self-appointed *progressives* unleash another stream of word salad. It's not worth bruising your forehead trying to reason with them. Even Galileo knew better: sometimes you keep your head down, stay silent, and let the Earth keep turning.

I've tried to smooth the rough edges just enough to preserve the meaning without inviting unnecessary drama. I've also removed nearly all personal details; otherwise, I'd have to explain a hundred irrelevant things or lie by omission, and neither would help you.

Besides, you don't need much of that. You're built from the same blueprint, shaped by the same circuitry and stubborn genes that run through me and through others who kept remaking themselves despite the world, changing so deeply and deliberately that they emerged on the other side as something quietly, unmistakably new.

Before we get our hands messy with the colors again, we need to talk about two big ideas — easily the most important theories science has produced in the last twenty or thirty years.

Only recently have we finally gathered enough data, precision, and intellectual muscle to start understanding how things actually work.

The first theory connects the strange quantum rules that govern elementary particles with the architecture of matter itself — galaxies, stars, planets, everything stitched into the fabric of the universe. The second is about energy, symbiosis, and evolution. It traces the origin of life and explains the long silence that came before *the Cambrian explosion* — that sudden burst, 540 million years ago, when almost every major form of life appeared out of nowhere.

Now let's put the biggest canvas onto the easel and begin with broad, thick brushstrokes, everything we know about stars and planets. I keep returning to the painting metaphor because the images in our minds are never exact copies of reality. No one has actually seen the inside of a star or watched DNA fold itself into chromosomes.

But if you understand the idea behind each thing, and sketch the connections between them yourself, you'll always know where you stand, where to go next, and why. You'll see the right path long before most people even look up.

*With thoughts beyond the reaches of our souls...
Say, why is this? Wherefore? What should we do?²*

Chapter 1. The Egg, or the Point of Singularity

One morning, approximately 13.8 billion years ago, something strange happened. Some other dimension suddenly laid an Egg, or the Egg emerged from another world where entirely different laws apply. We have no idea why this happened, what kind of world it was, or what forces shaped the Egg. It just suddenly appeared, and that was it.

And then, in a microscopic blink far shorter than a single heartbeat of our current second, the unimaginable happened: the Egg began to swell exponentially, at a monstrous speed. This was the explosion that gave birth to our Universe.

Physicists believe that in those first instants, the Egg was the single point where all the anti-gravity in existence was concentrated. “*Anti-gravity*” is the term Alan Guth, the father of modern cosmology, used for the kind of repulsive energy thought to have triggered the expansion of everything. Albert Einstein anticipated something similar with his cosmological constant, which he introduced to explain why the Universe remains open instead of collapsing into a single cosmic trash compactor.

Put simply, anti-gravity is the force that keeps the *Big Bang* from becoming the *Big Crunch*. We still don’t know what this force actually is, or what kind of field carries it. Gravity, at least, plays fair: matter bends space, bent space pulls matter, and everything feels the squeeze.

But in the Egg’s first instant, things were different, almost mythic. It seemed the Egg had not yet severed its connection with the world from which it emerged. The Egg, like a vacuum cleaner, began sucking energy and matter from its maternal world, pumping in force and substance the way a living heart pulls blood from veins.

In a tiny shred of time, it ballooned by dozens of orders of magnitude. Its expansion outran even the speed of light. Yes, relativity says nothing within our Universe can travel faster than light, but that limit applies only to motion *through* space. The Egg wasn’t moving *through* space. It was *creating space*³.

The speed limit of relativity does not apply to that earliest instant, when matter and energy were becoming the fabric of space itself, when the universe was too young to follow rules, yet old enough to begin inventing them.

Then, in a single flicker of time, the cord snapped

Whatever connection the Egg still had to its maternal world — its original reservoir — broke cleanly, like a rope pulled past its limit. By then, the deed was done. The energy had

already poured through, flooding our newborn Universe. The channel closed, the pump shut off, and everything went quiet.

The Egg — now swollen beyond any useful metaphor — continued to expand, but more calmly and steadily. The first explosive phase, inflation, was over. The second, slower phase had begun.

How Everything Here Is Put Together

What comes next might feel like a steep climb, but you're allowed to skip several pages and come back later. However, without this new vocabulary, we won't be able to squeeze ourselves into modern physics, and I won't be able to explain, cleanly and honestly, where everything came from.

The world we live in is stitched together by just a handful of fundamental forces. You can line them up from strongest to weakest like this:

1. The Strong Nuclear Force

The heavyweight champion of the Universe. This is the force that releases terrifying amounts of energy in nuclear reactions, and the one that keeps stars burning for billions of years. It also glues quarks into protons and neutrons, and then packs those into the nuclei of every element you've ever heard of (We'll talk about quarks soon).

2. The Weak Nuclear Force

Much gentler, operating only at subatomic distances. It governs radioactive decay and plays a quiet but essential role in how elements form inside stars.

3. Electromagnetism

The one that makes almost all of chemistry and biology possible. It binds atoms together into molecules. It shapes how light behaves — how it travels, bends, scatters, warms your face, and fills your camera roll.

4. Gravity

The softest whisper of the group — but ironically, the one people talk about the most. Gravity acts anywhere where there is mass or energy. It threads through the entire cosmos, choreographing everything from the tumble of tiny asteroids to the vast swirl of galaxy clusters, immense cities made of stars.

By the time the Egg finished its first sprint of expansion, it already contained every bit of energy and matter the Universe would ever possess. All of it — everything that would later become stars, planets, oceans, clouds of hydrogen, your bones, your phone — was blended

into a uniform, perfectly smooth, unimaginably hot and dense soup. Inside that furnace, the four fundamental forces hadn't yet split apart.

They were fused into a single super-force

A single rule. A single law governing everything. A single chord that would later separate into the four-note harmony holding the cosmos together.

Drifting into the first hundred-billionth of a second after the Big Bang

The Egg has blown apart — its job is done — and what remains is a newborn Universe, already swollen to a billion kilometers across. That's an absurd number, impossible to hold in the mind, but imagine a space that balloons from subatomic to astronomical before you can blink. And as this infant cosmos races outward, its temperature plummets. It falls just far enough for something extraordinary to happen.

The Higgs bosons — tiny, indivisible particles responsible for mass — begin to freeze

Only a moment ago, they moved freely, like sparks dancing inside a white-hot furnace. But now, as the cosmic temperature sinks past a critical threshold, they lock into place. They “condense,” forming a new kind of field that fills all of space.

Physicists call this the *Higgs field*³. But you can think of it more simply: Imagine a gigantic cloud of scalding steam suddenly cooling into liquid water. The molecules are the same both before and after, but now they cling to one another. They form a heavier, denser medium. The substance itself has not changed, but its behavior has, radically.

This is exactly what happened in that early cosmic instant: the Universe changed its *state*. Particles that once zipped around effortlessly now felt resistance. They slowed. By slowing, they acquired mass, because interaction with the Higgs field is what gives most particles their weight in the first place. Before this moment, everything in the Universe was perfectly symmetrical and perfectly smooth, like a featureless sheet of white light.

The Higgs field broke that symmetry

With symmetry shattered, something else happened — something even more important for the story of matter, life, and everything that would follow.

The four fundamental forces finally split apart

They were once a single unified super force, indistinguishable from each other. However, as the Universe cooled, this unity fractured. The strong nuclear force separated first, powerful and compact. Then, the weak nuclear force peeled off. After that came electromagnetism. Gravity, forever the quiet sibling, had already emerged earlier, woven into the very structure of spacetime itself.

For the first time, the Universe had structure. Rules. Distinctions. Boundaries⁴.

The cosmic orchestra that had been playing one pure note suddenly broke into four different instruments — each with its own tone, its own range, its own role in shaping everything that would come next. Matter could now form. Energy could flow. Space could stretch and curve under the influence of gravity. The Universe was no longer a glowing, uniform blur. It had become a place where things could *happen*.

So, what is all this stuff around us actually made of?

Or, more precisely: *what was matter made of in the very first moments after it hatched from the Egg?* Modern physics gives us a surprisingly compact answer. Our most successful framework, the *Standard Model*, tells us that everything you see, touch, breathe, or ever will encounter is built from just *twelve tiny, indivisible particles*:

- six quarks
- six leptons

That's it. Twelve building blocks. They're like the alphabet of the Universe: a small, neat set of symbols from which every word, sentence, and story of matter can be written.

But an alphabet alone isn't enough. Letters must interact — they need grammar, verbs, force, and motion. So, these twelve matter particles *communicate* by tossing back and forth other particles, called *force carriers*. These messenger particles *are* the forces. When two particles “feel” each other, it simply means they have exchanged one of these messengers.

The messengers:

- *Photons*: the carriers of electromagnetism. They hold atoms together, shape magnetic fields, and illuminate your room at night. A fridge magnet sticks because it's constantly trading photons with the refrigerator door. The same particles that make your desk lamp glow also choreograph magnetic forces. It's weird and beautiful.
- *Gluons*: the carriers of the strong force. These are much harder to picture. Think of them as a kind of cosmic glue so intense that it traps quarks permanently inside protons and neutrons. No quark has ever escaped this grip. If photons are calm negotiators, gluons are bouncers who never, ever let anyone leave the club.
- *The W and Z bosons*: carriers of the weak force. They govern radioactive decay and nuclear reactions that power the Sun. You'll never visualize them; they flicker in and out of existence too fast. But without them, the Universe would be dark and sterile.

Spin: the strange internal rotation

Matter particles — quarks and leptons — belong to a family called *fermions*, defined by their *half-integer spin* ($\frac{1}{2}$).

Force carriers — photons, gluons, W and Z bosons — are *bosons*, which have *integer spin (1)*.

Spin isn't really spinning in the everyday sense; it's more like a built-in quantum twist. But this little twist determines everything about how particles behave, how they stack, and how they build complex structures like atoms, molecules, cells, neurons, and eventually... you.

From twelve particles to everything

With just this compact menu of particles — twelve fermions and a handful of bosons — you can assemble:

- all stable matter in our daily lives,
- all the exotic matter found inside collapsing stars,
- all fleeting particle showers seen in accelerators,
- every atom in every planet,
- and every chemical reaction your body performs every second.

It's a strange thought, but true: Your bones, your phone, thunderclouds, blue giants, black coffee, nebulae, and the rings of Saturn are built from the same twelve building blocks. Everything else is just arrangement and interaction.

By the time the Higgs field “switched on,” the Universe had already stretched far beyond anything we can picture. Once that field froze into place, like a mist turning to frost, it split the four fundamental forces apart. That was the moment when the particles of the newborn Universe stopped behaving like identical twins and began acting as distinct individuals.

Leptons, quarks, and the W and Z bosons suddenly *acquired mass*. Not because someone handed them a heavy backpack, but because mass is simply a measure of how strongly a particle interacts with the Higgs field, which now fills every cubic inch of space like an invisible ocean. Move through that ocean, and you feel resistance. That resistance *is* mass.

Some particles plow through like trucks; others slip through like minnows. Only photons glide untouched, ignoring the Higgs field entirely, which is why they remain massless and forever travel at the speed of light.

This entire cast of particles, and all the rules of their dance, is what physicists now call the *Standard Model*. If you ever want to dive deeper into it on your own, there are wonderful books written by people who spend their lives thinking about quarks, leptons, and symmetry breaking. They can tell you more, in many more words, than I possibly could in a single chapter³⁻⁶.

How We Know Any of This

Because obviously no one was standing next to the Egg with a clipboard. To figure out what happened in the first fractions of a second of the Universe's existence, scientists had to learn

how to *recreate those impossible conditions* — the crushing densities and blistering temperatures present right after the Big Bang. And since we can't haul the cosmos into a laboratory, we built the laboratory big enough to *resemble the cosmos*.

Deep beneath the fields outside Geneva, Switzerland, lies a 27-kilometer ring of superconducting magnets, the *Large Hadron Collider*. This is CERN's flagship machine and the most ambitious scientific instrument humans have ever assembled. Inside this buried tunnel, protons are pushed to a hair's breadth below the speed of light and slammed into each other head-on.

Why? Because Einstein left us a cheat code: $E = mc^2$. Energy can become mass.

The more energy you pack into a collision, the hotter and denser that tiny point of space becomes, and the heavier the particles that can emerge from it. For the briefest instant — far shorter than a trillionth of a second — the collider produces tiny, blazing-hot droplets of reality where the conditions astonishingly resemble those inside the original Egg.

In those droplets, long-extinct particles flicker back into existence. Particles that lived only in the earliest heartbeat of the Universe. Particles that the cosmos abandoned billions of years ago. One of these was the Higgs boson, finally spotted in 2012 — an artifact of the primordial world, revived just long enough for detectors to whisper, '*I was here.*'

This is how we reconstruct the birth of everything: by recreating, in miniature, the forces and temperatures that once shaped the Universe itself.

In the end, we have learned *two ways* to peer back into the first flickers of space and time. One is by recreating relic particles — ghosts of the primordial world that once filled the Egg. We bring them back, briefly, through the violent artistry of the collider.

The *second path* couldn't be more different: it doesn't smash anything. It simply watches.

Think of giant space telescopes — astronomical leviathans parked at the Lagrange points (specific positions in space where objects sent there tend to stay put), far from Earth's noise and heat, floating in gravitational stillness and staring outward for years at a time. The newest among them, the *James Webb Space Telescope*, is basically a time machine with a golden mirror.

Here's the key idea you absolutely must grasp: the speed of light — about 300,000 kilometers per second, a number so huge it barely fits in the mind — is still *finite*. It's fast, but not infinitely fast. That means we can *never* see distant galaxies as they are "right now." We see them as they *once were*, because the light they emitted billions of years ago is only reaching us today. So, when Webb takes a picture, it's not capturing the present. It is opening a window into the deep past. Through that window, we can watch galaxies assembling themselves bead by bead, like dew forming on a spiderweb. We can see the moments right after the Universe crawled out of the Egg.

This is the astonishing part: the two approaches — the collider’s violent recreations of lost particles, and the telescope’s patient collection of ancient light — *agree*. The data from reawakened relic particles and the whispers of galaxies born billions of years ago fit together like two halves of a broken tablet. Piece by piece, they’re giving us a coherent picture of how matter, energy, and time first came to be.

And yes, I know — it all sounds suspiciously like the opening pages of *Genesis*, or like one of those origin stories from older mythologies. Humans have been telling themselves tales of cosmic beginnings long before equations existed. You’re welcome to imagine a Creator hovering somewhere beside the Egg if it gives you comfort. But to me, adding a divine artist to this picture feels a bit like drawing cartoon eyes on machines and letting them chatter with each other — charming, sure, but no more necessary than in the movie *Cars*.

Reality is already miraculous enough.

Physics today is standing in the middle of a quiet earthquake, a moment of deep transformation. We finally found the last missing piece of the puzzle, the particle responsible for giving matter its mass: the *Higgs boson*. On paper, that discovery completed the *Standard Model*. Tie a ribbon on it, call it done.

Except... the very moment the theory reached its greatest triumph, the universe handed us a to-do list so overwhelming it bordered on comedy. We still don’t understand the true dynamics of inflation — what powered that impossible first expansion or how it slowed and settled. Despite a century of trying, we still can’t fuse all the fundamental forces, gravity included, into a single, seamless framework.

The biggest mystery of all is one that shouldn’t even exist: Why did all the antimatter disappear? According to the equations that govern the motion of electrons (e^-) around atomic nuclei — the equations written down by Paul Dirac, one of the architects of quantum physics — there should also exist electrons with opposite charge: *positrons*. His math wasn’t poetic; it was blunt, mechanical, and absolutely clear. Sure enough, a couple of decades later, researchers at Caltech produced actual positrons in the lab.

From that moment on, particle physics became a two-column ledger: every charged particle in one column, its antimatter twin — same mass, opposite charge — in the other.

Now the question burns brighter than ever: If the early universe created matter and antimatter in equal amounts, where did all the antimatter go? Did it annihilate instantly, colliding with matter and disappearing in a flood of pure energy? Possibly. Except that the amount of energy released in such a universal annihilation should have turned the cosmos into a blinding wall of radiation. Clearly, we’re still here, made of atoms, not ashes.

This property of antimatter, that an electron meeting a positron vanishes in a flash, leaving behind two high-energy photons, set fire to the human imagination. The entire genre of science fiction practically grew out of it. You remember the long-running saga of *Starship Enterprise*? Warp cores powered by matter-antimatter reactions, infinite energy on tap...

Well, science fiction aside, modern accelerators really do create antimatter — *positrons*, at least. Not much of it, just tiny showers of particles produced at absurd financial and energy costs. But still: they exist. They are real. And the universe must have had a reason for inventing them. Finding that reason is one of the great unfinished stories of modern physics.

By the time we reach the *second* stage of inflation, the newborn universe is already a titan, about one hundred billion kilometers across, and still swelling. Picture a soap bubble expanding through nothing, except this bubble isn't just pushing into empty space; it is *carrying space with it*. Its shimmering edge is the frontier of our reality. Beyond that boundary lies... nothing. No light, no energy, no space, no time. Not darkness — because even darkness needs a stage to stand on. Just absolute absence.

Inside the bubble, meanwhile, everything is boiling.

The temperature, though dropping fast, is still measured in *trillions* of degrees. And the entire cosmic volume is filled with a bizarre, seething substance: a quark-gluon plasma in which the ingredients of matter sprint around faster than light itself (That's allowed here — nothing is moving *through* space faster than light. Instead, space *itself* is still reshaping, stretching, and rewriting its own rules in real time).

At these ridiculous temperatures, the *strong nuclear force* hasn't yet “switched on.” It exists, but it cannot manifest. The plasma behaves like a perfect gas — smooth, homogeneous, and frictionless. Quarks and gluons slide through each other like ghosts at rush hour, barely interacting, ignoring the constraints that will later bind them.

But give the universe just a few more microseconds, let the temperature fall just a bit, and everything changes.

Free quarks suddenly “freeze,” locking themselves together with gluons into compact, stable triplets. This is the moment when the first *protons* appear. Two up-quarks, each carrying a charge of $+\frac{2}{3}$, join with a single down-quark with a charge of $-\frac{1}{3}$. Add them up, and you get exactly +1: the electric charge of a proton (H^+).

The structure of the proton is so elegant, so energetically perfect, that once formed, it simply never falls apart. Unlike nearly every other configuration of matter in the universe — which decays, transforms, or disintegrates over milliseconds or millions of years — a proton just... stays. In fact, over 13.8 billion years, almost none of the original protons have decayed at all. They are as close to immortal as physics allows.

Soon after, the proton's neutral sibling appears: *the neutron*. It is built from a different combination — two down-quarks ($-\frac{1}{3}$ each) and one up-quark ($+\frac{2}{3}$) — which together give it zero net charge. Heavier, uncharged, but equally fundamental.

During this whole era, protons and neutrons — and their mirror opposites, antiprotons and antineutrons — are being born nonstop in the blazing density of the early universe. But creation comes with a catch: whenever matter meets antimatter, the two annihilate

instantly into a flash of photons. Yet the furnace is so hot, so impossibly dense, that new particle–antiparticle pairs pop right back into existence, keeping the cycle going. Birth, collision, annihilation, rebirth — like sparks inside a blast furnace that never cools.

Within this frantic churn, a tiny imbalance begins to grow.

At first, it's microscopic — just a whisper of asymmetry between matter and antimatter. But as the universe expands and cools, that whisper becomes destiny. Little by little, antiprotons and antineutrons fail to keep up. They annihilate faster than they can reappear. The imbalance compounds. When the universe finally moves past this furnace stage, only *matter* survives.

This quiet statistical tilt — barely measurable, yet absolutely decisive — is why you, your atoms, your thoughts, your entire world exist at all.

As soon as the temperature drops below the threshold needed to create proton–antiproton and neutron–antineutron pairs, that furious cycle simply... stops. The furnace is still blisteringly hot, but not hot *enough* to conjure new heavy particles out of raw energy. Matter and antimatter have had their duel, the imbalance is set, and the stage shifts.

But the universe isn't done improvising.

There's still plenty of energy, enough to produce vast numbers of *electron–positron pairs*. And now *they* take the spotlight, running the same birth–collision–annihilation loop that protons and neutrons just performed. Only this time, the dance is faster, lighter, almost jittery. Electrons are tiny — the lightest stable elementary particles in existence, without any inner structure, no hidden quarks inside them. They're nearly two thousand times lighter than the triplets of quarks they will soon orbit in neat atomic shells.

So now, only a few microseconds after the Big Bang, the young universe is suddenly teeming with everything it will ever need to build stable matter: protons, neutrons, electrons, and photons by the oceanful. The whole toolkit is finally in place.

And yet... it's still too early for atoms. Too hot, too wild, too dense. Everything keeps colliding, dissolving, and reforming, caught inside a cosmic pressure cooker that hasn't finished cooling. A little more patience is required. The universe is almost ready to take its first real shape — but not quite.

By the end of the *first minute*, the universe has already made a decisive choice: for every neutron still hanging around, there are roughly seven protons. The imbalance is set, and the chemistry of everything that will ever exist is already tilted toward hydrogen.

Now the energy density has dropped just enough for something remarkable to begin. Protons and neutrons start to *fuse*, literally sticking together to form the first atomic nuclei. This marks a turning point in cosmic history. The temperature and density of the expanding universe have finally fallen into the same range we see today in the cores of stars. And just as in stars, the *strong nuclear force* awakens and goes to work.

When a proton grabs onto a neutron, the two lock together to form *deuterium*, a heavier isotope sibling of hydrogen. When two deuterium nuclei collide, they can merge into a brand-new *helium nucleus*. The entire cosmic factory runs hot and fast: in just *three minutes*, every stable light nucleus that could form... does.

By that time, the universe is already thousands of light-years across, a blistering sphere of expanding radiation. But those temperatures and densities are fleeting. Very soon, they'll drop too low to sustain any more nuclear reactions. The window closes, permanently.

This is why helium is such powerful evidence for the Big Bang. Stars can fuse hydrogen into helium — but not in the enormous quantities we observe across the universe today. Even if every star that ever existed burned hydrogen nonstop for nearly fourteen billion years, they still couldn't produce the amount of helium we actually observe. The only explanation is that most helium was made in the universe's first few minutes, not in the long, slow furnaces of stars.

The light nuclei created in those early moments have barely changed over the eons. They were the universe's first building blocks, laid down before galaxies, before stars, before planets—before anything had a name. The heavier elements, the stuff of planets and people, would come much later, cooked in the deep nuclear cores of massive stars and released in explosions bright enough to shake the cosmos.

But the universe's first ingredients — the primordial ones — were born right there, in the first three minutes of everything.

For the next few *hundred thousand years*, nothing “dramatic” happens — at least not compared to the fireworks of the first minutes. Space simply keeps stretching, thinning, cooling. Time slows down into something that resembles duration instead of an explosion.

During this long dusk, the universe is filled with a murky fog: a swirling soup of elementary particles and freshly formed nuclei, all immersed in a blinding sea of photons and electrons. Imagine a cosmic storm cloud where everything collides so constantly that nothing can fully emerge or see past its own nose.

Somewhere inside this haze, ***dark matter*** is already present — silent, invisible, refusing to play by the same rules as everything else. Photons, the future carriers of light, are not free yet either. Each one is constantly colliding with electrons, trapped by the extreme density of the early universe. There is no such thing as “light traveling through space.” There is only *light being trapped*, absorbed, and re-emitted in an endless shuffle.

And about that dark matter. Today, we're pretty sure it makes up *about a quarter of the universe*, yet we barely understand what it is. It must have formed during the second phase of inflation, when the universe was still absurdly hot and dense, yet its behavior doesn't quite match anything else in physics. It cooled, yes, like everything else, but it cooled into something different. Something that neither glows nor absorbs, neither reflects nor blocks. Something our instruments can't see directly.

The best way to picture it is as an *invisible scaffold*, a delicate and flexible spiderweb stretched across the young universe. It doesn't interact with ordinary matter through light; photons ignore it completely. But it *does* have mass, and mass means gravity. And gravity shapes everything.

Where the invisible web thickens — where its threads knot and cross — that's where ordinary matter will eventually gather, like wind-blown seeds caught in a net. Those knots of dark matter will become the blueprint, the underlying *matrix* on which the visible universe will grow.

Soon — though “soon” here means hundreds of millions of years — the first stars will ignite inside the deepest pockets of that web. And the first galaxies will spin themselves into existence like tiny whirlpools forming in a cosmic tide.

The universe is still dark at this point. But the stage is built, the scaffolding is raised, and the script is quietly being written in the gravity of things we cannot even see.

Everything in the newborn universe changes the moment the temperature drops to about 3000°C , roughly half the temperature of the Sun's surface. At that point, the electrons, suddenly robbed of much of their kinetic energy, can no longer tear themselves free from protons with every collision. Electromagnetic attraction finally takes the lead, and negatively charged electrons settle into stable orbits around the first protons.

This is the moment when the *first atoms* appear — hydrogen and helium, forming everywhere at once. The blazing plasma thins into a vast ocean of cooling gas, swallowing up protons, neutrons, and the remaining free electrons. From this softening glow rise enormous gas clouds, the raw material from which the first stars and galaxies will eventually be born.

Matter begins to lose its electric charge because the positively charged protons are now locking into place with negatively charged electrons. Neutral atoms are more stable and less frantic than the charged chaos that came before.

And for the photons — those particles of light that had been trapped for hundreds of thousands of years — this is the moment of release. The universe turns *transparent* almost instantly, and everything floods with light. The last whisper of that first blaze, the glow of the universe's childhood, is still visible today as the *Cosmic Microwave Background (CMB)*.

That's how matter and light finally took shape. In that same moment — *380,000 years after the Big Bang* — radiation and matter parted ways forever, drifting into two parallel destinies: one shaping the visible structures of the cosmos, the other carrying the oldest story we can still read in the sky⁶.

Stars

As soon as the era of matter began, the great cosmic tempo slowed even further. Now it was gravity's turn — the fourth and weakest of the fundamental forces, yet the only one stubborn enough to act everywhere mass or energy dares to exist. Bit by bit, gravity took command of the enormous clouds of hydrogen and helium that had only just condensed from the blazing primordial plasma.

Meanwhile, hidden in the dark like a giant ghostly scaffold, the vast web of dark matter — now far more abundant than ordinary matter — stretched outward for hundreds of light-years around the point where the Egg once appeared.

Tiny ripples of uneven density — small imperfections that must have existed inside the Egg itself — were pulled and stretched in a kind of cosmic taffy pull. In the pockets created by this stretching, clouds of hydrogen and helium began to collect. The process was painfully slow, requiring hundreds of millions of years of patient celestial housekeeping.

At last, in the densest knots of the invisible web, something new began to take shape: spherical giants of unimaginable mass, hundreds of times heavier than our Sun. Gravity, tireless and never in a hurry, kept drawing hydrogen and helium toward these dark-matter anchor points. The gravitational pull of these forming bodies was astounding. It squeezed the gas, tightening its grip decade after decade, century after century, until the hydrogen inside grew hotter and ionized — breaking apart once again into free protons and electrons.

And then, finally, the universe lit its first candles.

The very first blue stars burst alight — their outer layers still gaseous, but their cores blazing with the same kind of scorching plasma that filled the young Universe itself. Gravity continued pressing inward, heating the stellar core to tens of millions of degrees. At such furious temperatures, the nuclei of hydrogen atoms finally surrendered to fusion, joining together to forge new elements.

Each fusion event released an avalanche of energy — torrents of heat racing outward as streams of photons and other particles. These newborn stars did not arrive as solitary hermits: they clustered together, gathering into the early shapes of what would one day become galaxies⁷.

The First Blue Stars

The very first blue stars were made of nothing but hydrogen and helium — the only ingredients the young Universe had on its menu. And these stars weren't just large; they were leviathans, true cosmic heavyweights, swelling to nearly *150 times the mass of our Sun*.

Inside them, temperatures soared to unimaginable extremes. Nuclear reactions raged so fiercely that the stars burned through their fuel at a reckless pace, like bonfires built entirely from dry pine needles — brilliant, blazing, and disastrously short-lived.

Our modest little Sun can shine peacefully for billions of years. The first blue giants barely lasted a single million. They lived fast, died loud, and each time one reached the end of its turbulent life, it exploded, throwing into space the new elements it had forged deep within its core.

And so — star after star, explosion after explosion — the Universe spent another quiet two hundred million years enriching itself, seasoning the cosmic broth with heavier and heavier elements.

Layered Stars and the Birth of Heavy Elements

In the next generations of stars, nuclear fusion continued climbing the elemental ladder. Their interiors began to resemble an onion — not because stars make you cry (though some astronomers might disagree), but because they truly formed in layers.

On the outside, the “skin” of the star remained the same blend of hydrogen and helium. But move inward — or rather, imagine zooming inward — and each layer grows hotter than the one above it. These inner regions act like a factory floor where lighter elements from the outer layers are pushed deeper, cooked, compressed, and transformed into heavier ones:

- hydrogen becomes helium,
- helium becomes carbon,
- and carbon becomes oxygen, all of it piling up in the core.

However, stars of *moderate mass*, anything less than eight Suns, simply cannot achieve the extreme temperatures and pressures needed to fuse carbon and oxygen into even heavier elements. So, when their hydrogen and helium shells finally burn out, the outward radiation pressure drops. Gravity, which has been waiting patiently for its chance, takes over.

The star begins to *pulse*. Its outer material collapses inward toward the core, heats up just enough to ignite briefly, releases energy, and then gets thrown outward again — only to cool, fall back, and repeat the cycle. Little by little, the star swells into a *red giant*, its surface expanding and cooling, its brightness flickering with every breathlike oscillation.

At the same time, a strong stellar wind blows much of the star’s mass into space, scattering it like shimmering cosmic dust. What remains — after the outer layers drift away — is the dense, smoldering heart of the star: a *white dwarf*, still glowing but no longer burning, the quiet ember left behind after a giant’s long and complicated life.

The Death of a Star

The fate of a star is written, almost cruelly, in its mass. If a star has more than ten Suns’ worth of material, the pressure and temperature inside soar past all imaginable limits, reaching a billion degrees and climbing. At such infernal heat, nuclear reactions are no longer limited to the simple ingredients of the early Universe. The star begins burning through a whole palette of elements. Hydrogen and helium, the lightweights, are consumed

first. Then heavier nuclei step onto the stage: carbon, nitrogen, oxygen, neon — the star fuses them all in its relentless drive to stay in balance.

However, the chain eventually reaches a dead end.

When silicon is fused into iron, the cosmic assembly line grinds to a halt. Iron is the brick wall of stellar alchemy: it *cannot* release energy through further fusion. And so, the heart of a massive star — now a furnace that no longer produces fuel — collapses catastrophically. Gravity, which had been balanced for millions of years by the outward pressure of fusion, suddenly finds no resistance. It crushes the core with terrifying speed, shrinking it by a factor of hundreds of thousands. The star folds inward on itself like a building losing all its internal supports.

Everything above the core — all those bright, swirling layers — hangs for an instant in space. Then gravity yanks them down toward the center, toward the tiny, impossibly dense object the core has become. This violent collapse briefly reignites every nuclear reaction the star ever knew, and the unleashed energy detonates the star from within.

We call that blinding flash a *supernova*, a burst so bright it can outshine an entire galaxy for days. The explosion hurls a colossal cloud of gas and heavy elements, tens of solar masses' worth, racing into space at more than 10,000 kilometers per second. The shockwave can ripple through the cosmos for centuries; we still see such glowing remnants from Earth today.

From Stellar Ashes to Planets and People

Countless explosions like these enriched the cosmic dust with carbon, silicon, iron, and everything else the Universe needed to start building more intricate structures. From this dust emerged smaller stars and their planets — including our Sun and the Earth beneath your feet. The intricate nuclear gymnastics happening inside stars of different masses are beautifully explained in *What Makes the Earth Move?* by David Bercovici⁶, while Zack Scott's illustrations and diagrams bring the story vividly to life⁷.

But the most astonishing part of the tale is this:

Every atom in your body — the calcium in your bones, the oxygen in your glass of water, the iron flowing in every heartbeat — was forged inside ancient, long-dead supernovae.

And one day, the old red supergiant star Betelgeuse, just 600 light-years away, will add its ashes to the mix. When it finally explodes, likely within the next million years, it will shine in our night sky as brightly as the full Moon for several months. A final, spectacular farewell.

Neutron Stars

Some stars outgrow the Sun so dramatically that, once they exhaust their nuclear fuel, they collapse into something even stranger. If a star's mass is *20–30 times greater than the Sun's*,

its final act is to transform into a *neutron star* — a sphere only 10–20 kilometers across, yet somehow packing in *two Suns' worth* of material.

Picture an atomic nucleus scaled up to the size of a city. That's a neutron star. It is made almost entirely of neutrons squeezed so tightly together by the strong nuclear force that matter, as we know it, stops behaving like matter. As if that weren't dramatic enough, these stars spin around their axis at incredible speeds, sometimes hundreds of times per second.

When a massive star collapses, it shrinks so suddenly that its magnetic field gets hugely amplified. Field lines that once spread out over millions of kilometers are crushed into a space only the size of a small town. Packed so tightly together, the magnetic field becomes *billions of times stronger* than that of a normal star.

Now imagine that the neutron star's magnetic poles don't line up with the way it spins. When that happens, electrons on the surface are forced toward the poles, creating a powerful beam of energy. As the star spins, this beam sweeps through space like the world's most overpowered lighthouse. But instead of visible light, it sends out radio waves.

When this rotating beam crosses Earth's line of sight, we detect pulses — perfectly timed, eerily regular. That cosmic lighthouse is called a *pulsar*.

You might wonder why I suddenly jumped from supernovae to neutron stars; your head is probably full enough already with all these new terms. But without neutron stars, I can't explain where the *heavy and rare-earth elements* came from — all those that line up in the periodic table *after* iron. These elements are everywhere in your life: essential in enzymes, electronics, catalysts, and radioactive processes.

The mystery of their origin only became clear in 2017, when two neutron stars wandered too close to each other, and an unavoidable cosmic catastrophe began.

As they spiraled towards each other at nearly the speed of light, they warped the fabric of space itself and generated a gravitational wave, a ripple that lasted for dozens of seconds and was detected here on Earth, even though the collision happened *130 million light-years away*. Moments later, a gamma-ray burst recorded by the *Fermi Gamma-ray Space Telescope* revealed that the collision had created a black hole.

But the real surprise came weeks later.

Astronomers detected a small glowing cloud in the aftermath — a nebula made of heavy atoms. A literal cloud of *gold, platinum, uranium* and many other elements, blasted into space at unimaginable speeds. It was the first direct, undeniable proof that elements heavier than iron are forged only in events as cataclysmic as the collision of neutron stars.

Ask your mother for the ring with the emerald, designed by the brilliant jeweler *Jean Schlumberger*, who worked with the house of *Tiffany & Co.* in the last century: it is made from those golden clouds.

Black Holes

They are, without question, the most astonishing objects in the Universe. In rare cases, when a star's mass exceeds thirty Suns, its final collapse produces what John Wheeler famously named a *black hole* in 1967. At that stage, even neutrons can no longer resist gravity. They crumble into quarks and gluons, which are forced into an unimaginably tiny volume.

Remember the Egg? Just as light could not escape the early Universe packed into a single blazing point, no light or radiation can escape the region where the mass of many hundreds of Suns is now squeezed together. In the end, there is only one thing we can confidently say about a black hole: *it possesses enormous mass*. Everything else, everything inside, obeys physical laws we do not yet know.

We detect black holes only indirectly, by watching how they interact with ordinary matter. If a black hole orbits close to a massive star, tidal forces tear away vast amounts of stellar material. The stripped, ionized gas is whipped into motion by the black hole's gravity and, as it crosses the boundary into a different state of space-time, it forms a brilliantly glowing *accretion disk* around the hole — visible across a wide range of wavelengths, even from Earth.

The large black holes that sleep in the centers of nearly all galaxies behave like deep-sea anglerfish: every so often, they swallow smaller, careless black holes that drift too close.

The Expansion of the Universe

Since our world has been expanding ever since the moment of the Big Bang, it's natural to wonder what awaits it ahead. Will the expansion continue forever — or will the great scattering of galaxies eventually slow down and stop?

All physical observations insist on the same surprising fact: *the expansion of the Universe isn't slowing down at all — it's speeding up*. And that completely contradicts everything we thought we understood about how stars and galaxies should behave. For a long time, scientists believed that gravity — the great cosmic anchor — was the only force that mattered at truly large distances. Under that assumption, gravity should have been tugging on everything, gradually breaking the expansion, and eventually pulling the Universe back inward under its own weight.

So, the discovery of accelerated expansion was a shock — a clear sign that something was missing from our picture of the world.

In other words, there had to be *another force* at work, something that pushed against gravity and kept driving the Universe farther and farther away from its birthplace. Physicists gave this mysterious effect a rather unfortunate name: **dark energy**.

Dark matter and dark energy are called “dark” not because they’re related, but simply because neither interacts with light. They cannot be seen, touched, or photographed; they ignore photons completely. We know almost nothing about them — except that mathematics stubbornly insists they must exist.

Unlike gravity, which we feel every time we drop a pencil or fall off a skateboard, dark matter and dark energy reveal themselves only on the scale of galaxies. And dark energy — a force that acts over immense distances — seems to have had little influence until the Universe grew large enough for it to matter.

Now, with our best estimates of the Universe’s size, we can say something astonishing:

- about 70% of its total content (meaning mass *plus* energy) is dark energy,
- about 25% is dark matter,
- and the remaining 5% is ordinary matter (the stuff that stars, planets, and you and I are made of).

It appears that dark energy began to overpower gravity around *4 billion years ago*, long after the Solar System had already formed. Since we’ve found ourselves speaking of our cosmic neighborhood... let’s take a closer look at it.

The Sun and the Planets

Our Sun was born from an ordinary-looking cloud of gas and dust. Its story began when a shockwave from a nearby supernova, carrying with it a treasure chest of newly forged chemical elements, slammed into that cloud and set it spinning like a cosmic potter’s wheel. The rotation flattened the cloud into a disk, and powerful gravity at its center squeezed the gas until nuclear fusion finally ignited.

Thus, the death of a massive ancient star gave the push needed for a small new one to be born.

The Sun is an almost perfect sphere of incandescent plasma, made mostly of hydrogen and helium, wrapped in a dynamic magnetic field and rotating around its axis roughly every 25 days. The surface simmers at around 6000°C, but deeper inside, the temperature climbs dramatically. At the star’s core, the furnace where fusion takes place, the heat exceeds *15 million degrees*.

At such outrageous temperatures, the nuclei of ionized hydrogen (essentially bare protons with a positive charge) move fast enough to overcome their natural electrical repulsion. When they collide, they stick together and begin the process of fusion, eventually forming helium, whose nucleus contains two protons and two neutrons. Fusion releases enormous amounts of energy because some of the mass of the original particles is converted directly into energy. You remember the famous equation:

$E = mc^2$, where E is energy, m is mass, and c is the speed of light.

When the Sun formed from its gas-and-dust cradle, it didn't swallow all the surrounding material. What was left behind settled into a *protoplanetary disk*, and the planets later grew out of that disk. As the disk cooled, gas condensed into solid particles. Near the Sun, temperatures were so high that only metals and silicon-based compounds (silicates) could stay solid. These materials had been forged earlier in a supernova and delivered by its shockwave.

From them, the inner rocky planets took shape: *Mercury, Venus, Earth, and Mars*.

Farther from the Sun, where space was cooler and rich in gas, something different happened. Within about ten million years, the gas giants *Jupiter and Saturn* assembled themselves. And at the distant icy edges of the Solar System, where it was cold enough for water, nitrogen, and carbon dioxide to freeze, the final pair of planets formed: *Uranus and Neptune*, built from ice and born last.

The Newborn Earth

The newborn Earth looked nothing like the familiar blue world we know today. Its atmosphere was an unbreathable haze of methane, carbon dioxide, water vapor, and hydrogen. And the surface? A churning ocean of molten lava, kept from cooling by the constant bombardment of asteroids, comets, and even stray planets⁸.

One of those rogue planets was *Theia*, a Mars-sized protoplanet. Its collision with Earth was catastrophic: a huge portion of Earth's mantle was ripped away and hurled into space, while Theia itself was completely destroyed. The debris spiraled in chaotic orbit for a while, then gradually settled and gathered into a single body.

That is how the *Moon* was born⁹.

All this unfolded around 4.6 billion years ago, roughly nine billion years after the Big Bang.

Comets continued to deliver water to Earth, but the planet was still too hot — the water instantly evaporated, feeding the atmosphere with more steam. Then, after several million more years, the surface finally cooled enough for atmospheric vapor to condense.

And that's when the *great flood* began.

Rain fell for millions of years, filling the lowlands, smoothing the jagged crust, and covering the entire planet with a global ocean. Once a day, colossal tsunamis swept across this young water world. At the time, Earth spun much faster than it does now, and the newly formed Moon hung low over the horizon, tugging fiercely at the seas.

As the planet cooled, it began to separate into layers. The lighter materials — silicon, aluminum, and oxygen — rose to the top, forming a kind of frothy crust that would later become the *lithosphere*. The denser elements, like iron and nickel, sank downward and gathered into the planet's core.

Earth's Restless Interior

Today, the planet's core is a spinning sphere made of two layers: a *liquid outer core* and a *solid inner core*, both composed mainly of iron and nickel. The core remains blisteringly hot because of the immense pressure deep inside the planet and the steady decay of radioactive elements.

As the Earth rotates, its metallic core acts like a gigantic dynamo, generating a magnetic field whose invisible lines thread through the crust and extend far into space, where they shield the planet from charged particles in the solar wind.

The heat rising from the core warms the entire planet from within, driving slow, powerful currents of magma. These currents soften the lower parts of the lithosphere, while the rigid but lightweight crust above it cracks into enormous slabs. These slabs, the *lithospheric plates*, float atop the molten basalt below. When they drift apart, new oceans open between them. When they collide, scrape sideways, or slide one beneath another, the results shape the face of the planet: mountains rise, trenches deepen, and on the ocean floor, enormous rifts split open, releasing glowing rivers of lava that form brand-new oceanic crust.

This gradual formation of new sea floor is called *seafloor spreading*, and it is the driving force behind the movement of Earth's tectonic plates.

A few million years later, vast regions of ocean floor — together with the sediments that accumulated on them — are dragged back down into the planet. They melt, rise again, and feed chains of volcanoes near the coasts. An excellent example lies not far from your home: the stratovolcanoes Mount St. Helens and Mount Rainier in Washington, and the stratovolcanoes Lassen Peak and the Butte Mountains in California.

In geological terms, these giants erupted just moments ago, at a time when the *Farallon Plate*, lying beneath the Pacific Ocean, was slowly sliding under the North American continent — a process known as *subduction*. Riding this slowly sinking plate like an escalator, entire island chains were carried from the ocean toward the continent, eventually stitching themselves into the patchwork we now call California¹⁰. Then, having delivered its cargo, the Farallon Plate finally vanished beneath the continent.

The Breathing Lithosphere

The lithosphere — meaning all the tectonic plates together — seems to breathe over geologic time. Every few hundred million years, the continents roam across the globe and coalesce into a single massive supercontinent. Eventually, pressure from rising magma deep within the Earth causes this supercontinent to fracture, sending its pieces drifting apart once more.

The most recent supercontinent, *Pangaea*, formed about 250 million years ago. It was preceded by *Rodinia* and, before that, by *Columbia*. Everything you might want to know

about this grand planetary cycle is explained in Ted Nield's wonderfully written book *Supercontinent*¹¹.

Matter and Space

Now let's take a look at where we find ourselves today, *13.8 billion years* after the Big Bang. The matter around us is ancient and cold — nothing like the blistering, overwhelmingly dense substance of the Universe's infancy. Today, matter is mostly *empty space*. Almost all the mass of any atom sits in its nucleus.

If you could enlarge an atom to the size of a football field, and place its nucleus — a dense knot of fused protons and neutrons — at the center, that nucleus would be no bigger than a tiny bead from a bracelet. The electrons wouldn't orbit like planets. Instead, they would form shimmering probability clouds stretching all the way to the goalposts.

And yet, as Albert Einstein showed us, space is not emptiness at all. It's more like an invisible fabric woven together with matter and energy. What we experience as gravity is simply the folding and bending of this fabric. Massive objects tug the fabric inward, and smaller objects have no choice but to move along the curved paths created by those folds.

For us — living on a small planet tucked safely into a calm corner of our galaxy — everything feels warm and welcoming: a sky full of air, vast oceans, soft clouds. Yet the moment you step beyond the reach of our atmosphere and the magnetic field — two protective blankets shielding our world from ultraviolet light and streams of energetic particles — you are met with the raw cold of space: about -270°C , stretching unbroken for light-years in every direction.

Our parent star remains young enough to grant us a couple of billion more years for reflection and the gradual refinement of body and mind. Don't worry — there's still plenty of hydrogen left inside; the Sun won't be running out anytime soon.

Our small Solar System, together with the spiral galaxy we call the Milky Way, slowly circles the supermassive black hole known as *Sagittarius A** — a monster weighing *four million times* more than our Sun. It constantly devours the stars that wander too close, but we're in no danger. We live on the quiet outskirts of the galaxy — *26,000 light-years* away from all that chaos.

And that, for now, is everything we've managed to fit onto our first canvas.

Chapter 2. The Origins of Life

Let's take a look at our second canvas and examine the history of our own planet. The question at the center of this chapter — the one that quietly follows humanity through every age — is simple and enormous all at once:

How did life appear on Earth, and what is life, really?

The foundation of life on Earth is *carbon*¹². No other element in the entire periodic table can build molecules of such staggering variety — in shape, size, and function. A carbon atom has a remarkable talent: it can form bonds not only with many other elements but also with *other carbon atoms*, and it can do this up to four times at once.

This lets carbon assemble itself into almost anything:

- long winding chains,
- closed rings that snap together like bracelets,
- branching structures as intricate as coral,
- and three-dimensional frameworks that resemble tiny molecular sculptures.

Because of this flexibility, carbon can build proteins, carbohydrates, phospholipids, and the nitrogenous bases that form the alphabet of life — DNA (*deoxyribonucleic acid*) and RNA (*ribonucleic acid*).

Only molecules made from carbon can meet the two essential requirements of life:

- the ability to reproduce themselves,
- and the ability to change and adapt to their environment.

You already know where the atoms needed for this came from. Oxygen, carbon, nitrogen, and phosphorus were all forged in the heart of a massive star that exploded long before our Solar System formed. Its ashes, drifting through space for millions of years, eventually helped seed the cloud of gas and dust that became Earth.

One more ingredient was indispensable: water. Comets and asteroids, like cosmic couriers, carried it from the cold outskirts of the Solar System and delivered it to the young planet.

The Evolution of Life

The evolution of life depends, to a very large extent, on natural selection and on the inheritance of acquired traits. Every one of these traits is recorded in our genes: long, resilient chains of DNA, folded inside the nucleus of each cell in a very specific way and written in a four-letter alphabet: A–T, G–C.

Everything else in us is built from proteins.

DNA is the library, the blueprint, the grand architectural plan that dictates the development of every organism, from a single bacterium to a human being. But to translate the information in DNA into living matter, the cell needs an active set of messengers called RNA.

Here's how the system works:

- *Messenger RNA (mRNA)* makes short, temporary copies of the necessary sections of DNA.
- *Transfer RNA (tRNA)* ferries the correct amino acids — the building blocks of proteins — to the ribosomes.
- *Ribosomes* themselves are incredibly sophisticated molecular machines that assemble proteins according to the instructions carried by mRNA.

Only proteins have the nearly infinite structural and functional variety needed to sustain life. They form your nails and teeth, your muscles and skin, your bones and delicate enzymes. And, of course, proteins are required for *DNA replication* itself, the doubling of DNA, and for the synthesis of all types of RNA.

Here we meet a profound puzzle: To make DNA, you need proteins. But to make proteins, you need DNA. So how did this entire intricate system come together in the first place?

The First Spark of Life

Today, most biologists agree that in the very beginning, it was RNA that played the central role. RNA molecules are simpler than DNA and, despite being less stable, they are stable *enough* to serve as reliable carriers of information.

It isn't too bold to imagine that on the early Earth, or perhaps on some other planet around some other star, the first forms of life appeared spontaneously as RNA molecules. This idea is strongly supported by the discovery of organic compounds on comets, including the nitrogenous bases that form RNA itself.

Like DNA, RNA can replicate, meaning it can make copies of itself. Once something can make copies of itself, it becomes subject to natural selection, because some copies will work better than others and will be more likely to persist. Following Richard Dawkins, we can call any such self-reproducing unit a "replicator." Replicators can change over time, become more effective, and compete with one another—and that process, in essence, is evolution¹⁴.

An RNA polymer (a chain of ribonucleic acid units) is extraordinarily flexible. It can fold into all sorts of shapes, many of which resemble the complex forms of proteins. Some folded RNAs can even accelerate biochemical reactions, just like protein enzymes do. These catalytic RNAs are called *ribozymes*.

Even more impressive: RNA can store and copy information, transport amino acids, and help assemble proteins. In other words, RNA forms the bridge between:

- the blueprint (information), and
- the living body (function)

And with that, we can offer our first, slightly rough, but remarkably complete, definition of life:

What Life Is (Version 1.0)

- The main actors are replicators — the carriers and guardians of information.
- The information is written in a template (a matrix) made of DNA or RNA and stored within a protein body.
- The template is reproduced with extraordinary precision using RNA and specialized enzymes.
- Based on this template, a protein organism (a function) is synthesized.
- The transcription and translation of DNA into proteins happen through RNA and enzymes.
- And in the outside world, natural selection goes to work, favoring organisms best adapted to their environment — or, if you want a more dramatic image, the protein “functions” of replicators cross swords with each other, like Hamlet dueling Laertes.

Where Did the Building Blocks Come From?

But where, on the young Earth, could the raw materials for such astonishingly complex molecules have come from? Simple: *from asteroids!*

Let’s look at one of the most exciting recent discoveries, the newest report on the asteroid Bennu, studied by NASA’s spacecraft OSIRIS-REx and published not long ago in *Nature Geoscience*.

The samples brought back from Bennu contained *nucleic acids and amino acids* — the very building blocks of DNA, RNA, and proteins. Alongside them, scientists found sugars, including ribose, which play the starring role in the “RNA world” hypothesis — the idea that RNA was the first molecule capable of storing and copying information at the dawn of life.

Did these building blocks form directly on the asteroid itself, or did Bennu inherit them from somewhere even older and stranger? We don’t know yet.

But one thing is absolutely clear: asteroids like Bennu carry within them a complete toolkit of molecules necessary for life to appear. Further, fragments of such celestial bodies rained down regularly onto the early Earth and onto the other rocky planets of the inner Solar System.

How Life May Have First Appeared

And now let’s try to imagine how life could have begun on a young, restless planet.

Back then, volcanoes were far more active than they are today. The Earth's crust was thinner, the oceans were shallower, and the very idea of tectonic plates was only beginning to take shape. In many places, volcanic materials in streams of *hot, mineral-rich water* rose through the surface from deep within the Earth.

When these scalding fluids met the cooler ocean water, several things happened at once:

- rising streams cooled,
- pressure dropped,
- and minerals dissolved within these fluids began to precipitate, settling into the pores, cracks, and crevices of the rocky seafloor.

It was in environments like these, scientists believe, that the *first cells* and the *first membranes* may have formed.

To produce millions of microscopic cells—tiny chambers containing early replicators and enclosed by the first inorganic iron–sulfur membranes—all that was required was a difference in redox potential and acidity. In simpler terms, it took a proton concentration gradient between the cooler seawater above and the hot volcanic fluids slowly rising from within the Earth.

And such gradients *certainly* existed. Don't worry, I'll explain everything step by step.

Why Proton Gradients Mattered, and What They Are

Why were proton gradients so crucial for the very first cells — and what *are* they, exactly? You'll meet these terms again and again as we move through the story of life, so let's take a moment to understand them calmly and clearly.

Remember how we spent a long time talking about *protons* (H^+) earlier in the book, back when we were painting the birth of the Universe? You can think of them simply as *hydrogen nuclei* — tiny, positively charged particles present in every liquid environment. They determine *acidity*: the more protons surrounding a cell, the more acidic the environment becomes.

The measure of proton activity/concentration in a solution is its acidity, measured by pH (the “potential of hydrogen”). pH is defined as the negative base -10 logarithm of the proton concentration, expressed in moles per liter:

$$pH = -\log_{10} [H^+]$$

For example:

- Sour lemon juice has a pH of about 2,
- while neutral human blood has a pH of 7.4

Because the scale is logarithmic, lemon juice is *five orders of magnitude* more acidic than blood, meaning it contains *100,000 times more protons* per unit volume.

Now let's talk about gradients.

Imagine a massive concrete dam on a river. Water behind the dam sits high above the valley. Guided through a pipe, that water rushes downward under the pull of gravity, turning the blades of a steel turbine in the power station below. The spinning turbine generates electricity, and the spent water flows back out into the river.

That difference — between high water and low water, between potential energy and motion — *is the gradient*. It's what creates flow.

You can picture a gradient in other ways too: a streak of color fading from deep purple to pale pink, or a drop of dye spreading through clear water.

A gradient simply means *a difference between "here" and "there."* And differences like this — in height, concentration, charge, and pressure — are what make the universe *move*.

Back to the First Cells at the Bottom of the Sea

Now let's return to our early cells on the ocean floor, each one already surrounded by the first primitive membranes. If the concentration of protons inside that semipermeable membrane was lower than the concentration outside, then voilà — *a gradient*, and with it, a potential *source of energy* for every task of cellular metabolism.

It didn't take long for the first cells to arrive at a genius solution: they invented *protein turbines*: molecular machines embedded directly in their membranes. By letting the stream of protons flow through them, these turbines could generate energy. We'll talk later about what kind of energy that is and how exactly these turbines work.

Their membranes grew more sophisticated over time. The earliest RNA-based replicators lived inside the naturally sized pores of volcanic rock, using the proton gradients created by hot hydrothermal fluids washing over the seafloor. Eventually, they evolved a way to protect themselves from the outside world: they built *sturdy walls* made of proteins, polysaccharides, or phospholipids.

Now safely sheltered, they were free to do what mattered most: *make copies of themselves and extract energy* to support the transcription and translation of their RNA.

Separated from the chemical chaos of the bubbling ocean around them, these quiet little volcanic rock pockets became *islands of order*. The replicators refined their energy factories and learned to pinch off tiny vesicles — small bubbles wrapped in fragments of the parent membrane — each carrying a bit of their precious information.

Over time, the replicators no longer depended solely on the pores of the rock. They began to build *their own portable homes* — protein survival machines in which they could work

uninterrupted. Later, they replaced their RNA templates with a more stable molecule, DNA, and upgraded their metabolic machinery.

And that, most likely, is how the first *archaea* were born.

All of this, remember, happened in *complete darkness*, at the bottom of the ocean. There was plenty of water, heat, and dissolved minerals — but not a trace of oxygen anywhere on Earth.

Why Were They Called Archaea?

Why were these organisms given the name *archaea*? In the late 1970s, geologist Jack Corliss and his team made a series of deep-sea dives in the submersible Alvin, descending into the darkness near the Galápagos Islands. There, where two lithospheric plates pull away from each other, towering jets of superheated water rise through cracks in the seafloor — the dramatic hydrothermal vents known as black smokers.

These hydrothermal fluids can exceed the boiling point of water, yet they do not boil — the immense pressure at the ocean floor keeps them in liquid form. They are saturated with minerals and loaded with volcanic gases such as hydrogen, hydrogen sulfide, and carbon dioxide.

In this alien environment — pitch-black, scalding, and toxic by our standards — scientists discovered single-celled organisms that looked a bit like bacteria, but were unlike anything known at the time. They were later named *archaea*, from the Greek word for “ancient,” after Carl Woese and George Fox showed in 1977 that these microbes represented a completely distinct branch of life.

Archaea and bacteria together are known as *prokaryotes* — organisms without a nucleus. Their structure is extremely simple. They rarely form colonies, and they never develop into multicellular beings. They are the minimal, rugged survivors of Earth’s earliest era.

Cyanobacteria

One of the most astonishing things about life on Earth is how quickly it appeared. Life emerged in the hot volcanic flows at the bottom of the ocean a mere *100 million years* after the planet itself formed — at a time when comets and meteorites the size of small mountains were still raining down from space, the same kind that carved the largest craters on the Moon.

Over time, life slowly made its way upward from the seafloor into the sunlit surface waters. By about *3.4 billion years ago*, delicate green organisms made of thin, mucous threads and films were already building the first great reefs — small, rounded hills called *stromatolites*. They were composed mostly of *cyanobacteria*, creatures that used sunlight to synthesize their own food.

Cyanobacteria are the most successful and enduring bacteria to ever inhabit our planet: they ruled Earth for three billion years. On a hot summer day, you can still find dense greenish layers of cyanobacteria in shallow ponds near your home — they have changed astonishingly little.

Few people can truly imagine a lineage with a history stretching back billions of years. To persist for so long, the informational matrix inside a replicator must be written with celestial fire and must resonate like *the Passacaglia in C Minor, BWV 582*, as performed by the great Karl Richter on the organ of Ottobeuren. And if you listen again to Karl Richter — especially that magical section where the high registers of the old organ shimmer like tiny bells — let's try to craft another definition of life, shorter and more elegant: *Life is an informational matrix that unfolds and perfects itself through selection over time.*

Now, let's take a closer look at what lies inside a cyanobacterium, and where all the oxygen around us came from.

Oxygen — the First Energy Revolution

So, cyanobacteria became the most important group of Earth's early "sun-eaters." They still had no cell nucleus, but they *invented true photosynthesis*, a way to capture light from the nearest star and turn it into an enormous flow of usable energy, releasing oxygen as a by-product.

To achieve this, they built one of the most intricate molecules in the history of life: *chlorophyll*. Chlorophyll looks almost like a piece of delicate metal lace wrapped around a single magnesium atom placed precisely at its center. Its task is to catch incoming photons of sunlight and convert their energy into a form the cell can use.

Here's what happens next:

The energy of absorbed light is used to *pull apart two molecules of water (H_2O)* and remove their electrons (e^-). This reaction releases:

- protons (H^+) and
- oxygen gas (O_2)

The freed electrons are immensely valuable: they carry the energy needed to build the cell's main energy currency — *ATP* (we'll discuss it in detail in the chapter "*Respiration, or the Unified Principle of the Proton Pump*").

At the very end of this pathway, oxygen is produced, and cyanobacteria simply vent it into the atmosphere.

But that's only half the magic.

The remaining energy captured by chlorophyll is used to take the carbon atom in atmospheric carbon dioxide (CO_2), replace one of its oxygen atoms with two hydrogens, and

— through a long chain of elegant biochemical steps — turn it into sugars (and release more free oxygen in the process). In other words:

The electrons freed from water during photosynthesis are *stored inside the chemical bonds of sugars* — tiny, portable packets of energy.

Why do cells need sugars? Because sugars are another form of energy storage.

Later, both plants and animals will rely on these stored sugars. During *cellular respiration*, these sugars “burn” in the presence of oxygen to release usable energy.

As this happens, the electrons originally taken from carbon during photosynthesis drop down to a lower energy level when they are captured by oxygen atoms. As they fall, they release energy in the form of *heat*. This is the same heat that keeps warm-blooded mammals warm.

As oxygen captures these electrons, it becomes water, while the carbon from the sugars is released as carbon dioxide. These CO₂ and water molecules are the main “exhaust” products of respiration.

For the First Archaea, Oxygen Was a Catastrophe

For the first archaea and anaerobic bacteria, who lived safely under the heavy canopy of an oxygen-free atmosphere, the appearance of oxygen was nothing short of a catastrophe.

At first, cyanobacteria produced very little free oxygen, barely a whisper in the air. But over the course of a billion years, its concentration slowly rose, eventually reaching 21%. Around 2.1 billion years ago, this triggered *the first mass extinction* in Earth’s history.

Then something unexpected happened: the oxygen level suddenly dropped again, plummeting to about 2%. We don’t know exactly why, but most likely a burst of tectonic activity dragged enormous quantities of organic matter — generations upon generations of dead microorganisms — beneath the seafloor. Over millions of years, those buried remains transformed into oil and natural gas.

But the oxygen didn’t stay low for long. Soon it flooded the atmosphere again, rising to such levels that even the rocks couldn’t hide from it: iron in them turned into rust, and carbon became limestone.

At the same time, *methane and carbon dioxide* vanished from the atmosphere, locked away inside newly formed mountain ranges. And with the greenhouse gases gone, oxygen kept accumulating until it reached roughly the level we see today — about 20% of the *atmosphere*.

But you probably remember that methane and carbon dioxide are the two key ingredients of Earth’s atmospheric blanket — the greenhouse gases that trap heat. Without them, the planet suddenly plunged into its *first and most severe ice age*. Glaciers kilometers thick

advanced from the poles toward the equator and eventually met in the middle. For nearly *300 million years*, Earth was a snowball. And almost all living things, as you might guess, perished.

But the survivors — the hardy few — were pushed toward further innovation. Somewhere below the ice, near underwater volcanoes, new *aerobic bacteria* emerged. They reconfigured their metabolism to use oxygen rather than fear it.

And so, without exaggeration, we can say: *cyanobacteria reshaped all life on Earth*. The extinction of the dinosaurs, compared to the First Oxygen Revolution, was a minor, almost trivial episode.

Energy

All the lush diversity of life on Earth — every fish, fern, mushroom, beetle, dinosaur, and human — appeared only within the last *600 million years*. That's roughly one-sixth of the planet's entire history. For the billions of years before that, Earth was home to nothing but bacteria and archaea.

So, we must ask: Was there some kind of barrier that held evolution back — something that had to be overcome before life could become multicellular and truly take off?

Yes. And that barrier was **energy**.

Neither bacteria nor archaea were able to generate enough of it to move quickly, hunt, grow large, or build complex bodies. Both groups were, in a sense, sluggish, peaceful, and a bit monotonous — tiny chemical workshops drifting quietly through the ancient oceans.

This continued until the cyanobacteria arrived. They mastered one of the most difficult tricks in all of biology: *capturing sunlight and turning it into usable energy*.

As far as we know, oxygen-producing photosynthesis exists only in:

- cyanobacteria,
- green algae,
- and plants.

But algae and plants are *impostors* — their photosynthesis is performed not by their own machinery but by *chloroplasts*, which are actually photosynthetic cyanobacteria living inside large eukaryotic cells. These ancient cyanobacteria generate sugars for their hosts in exchange for protection and stability.

This partnership — a deep symbiosis — provided enormous evolutionary advantages for both sides in a world increasingly saturated with oxygen. And as plants and algae became the base of the global food chain, they began supplying all other organisms with the nutrients produced through photosynthesis¹³.

In the end, the fundamental difference between *animals* and *plants* comes down to this:

Plants make food out of light, air, and water. We animals make food out of them.

And, as a bonus, plants produce oxygen. But what happens to all the sugars stored inside a cell? Let's take a look.

Respiration, or the Universal Principle of the Proton Pump

Let's break down the essential process of *respiration* — the one that turns sunlight and oxygen into the energy that powers muscles, jaws, wings, and fins.

In its simplest form, *respiration is the oxidation of food by oxygen to release energy*. But at a deeper level, respiration is built on an ancient mechanism: *the proton pump*.

Remember how the earliest cells invented special *protein turbines* embedded in their membranes — turbines that generated energy by letting a stream of protons flow through them? Well, once Earth's atmosphere filled with oxygen, those turbines got something like *turbo boost*.

All respiration in animals takes place in small cell organelles called *mitochondria*. Inside them, in a chain of oxidation–reduction reactions carried out by elaborate protein complexes known as *cytochromes*, the energy stored in sugars is released and used to *pump protons* (H^+) across the inner mitochondrial membrane.

This creates a *difference in proton concentration*, an electrochemical imbalance — about 150 millivolts of electrical potential. That potential is the force that drives protons back through the membrane. And that flow of protons powers one of the most extraordinary enzymes in biology: *ATP synthase* (also called *ATPase*).

This enzyme really does resemble a tiny turbine. Hidden inside is a *rotating rotor*, which spins when protons rush through it. As the rotor turns, ATP synthase — anchored directly in the membrane — manufactures the universal energy currency of life: *ATP, adenosine triphosphate*. Cells can store ATP for hours and spend it whenever needed.

The whole system works like a watermill or a dam:

- in the mill, the wheel turns as water flows from a reservoir down the chute;
- in the mitochondrion, ATP synthase turns as protons flow from one side of the membrane to the other.

It is the same ancient principle — the *proton pump* — scaled up into the engine of every animal on Earth.

Photosynthesis: A Mirror Image of Respiration

Photosynthesis in plants is built on almost the *same* blueprint as respiration. As you remember from the chapter on *Energy*, this process is handled by special organelles called *chloroplasts*.

To pump protons across the chloroplast membrane, plants use *sunlight itself*. And the actual pumping works almost exactly like what happens during respiration in mitochondria.

In both cases, life relies on a universal, beautifully simple idea — the battery principle:

- Electrons travel down an electron transport chain, passing from one molecule to another like runners in a microscopic relay race.
- Their final destination — the “electron acceptor” — is *oxygen*.
- And all along the way, the energy released by electron jumps is used to *push protons* (H^+) *across an inner membrane*.

Once enough protons accumulate on one side, the *proton pump* kicks in and unleashes a torrent of protons through the molecular turbine known as *ATP synthase*. This turbine turns, clicks, hums — and manufactures ATP, the universal energy currency of every living cell. This mechanism is so fundamental that it might as well be a signature of life on Earth, as iconic as the double helix of DNA or the amino-acid alphabet of proteins.

Cyanobacteria — the ancient masters of sunlight — eventually became today’s chloroplasts, the power plants of plant cells. And mitochondria — those tiny organelles that churn out ATP by burning oxygen — exist in *every* eukaryotic cell, whether plant or animal.

Together, chloroplasts (a.k.a. former cyanobacteria) and mitochondria (a.k.a. former aerobic bacteria) form the *primary energy cycle of the planet*:

- *Chloroplasts* turn sunlight into chemical energy and release oxygen.
- *Mitochondria* use that oxygen to extract energy from food.

If you simplify the entire scheme to its bare bones, life on Earth starts to look like a two-stroke engine — sunlight in, energy out¹⁵.

But hold on — how did cyanobacteria become chloroplasts?

You might be wondering: *How exactly did cyanobacteria end up inside plant cells?*

The honest answer is: we don’t know the full story yet. What we *do* know is that this must have happened very early — at the dawn of plant evolution — when a larger cell swallowed a cyanobacterium but never digested it. Instead, they formed a partnership that became permanent.

And while the origins of chloroplasts are still partly shrouded in scientific mystery, the other great alliance of early life — the union of *archaea* and *aerobic bacteria* — is much

better understood. That partnership completely rewired the energy economy of early eukaryotic cells, set the stage for multicellular organisms, and ultimately ignited the event known as the *Cambrian Explosion*. We mentioned that explosion back at the start. Now it's time to trace its roots. In the next section, we'll talk about this remarkable union — the symbiosis that changed everything.

The Symbiosis of Archaea and Bacteria, or the Second Energy Revolution

At the base of the great Tree of Life — a trunk as massive and intricately braided as the roots of a banyan — biologists recognize three primary domains: *archaea*, *bacteria* (both lacking a nucleus), and *eukaryotes*, whose cells *do* contain nuclei.

All eukaryotes — which means every plant, every mushroom, every animal, and every one of us — are, in a sense, *chimeras*, composite beings born from one of the most extraordinary partnerships in Earth's history: the alliance between an archaeon and a bacterium. From this unlikely union, practically all modern complex life began.

Let's imagine how such a thing could have happened.

About *two billion years ago*, just after Earth thawed from its global "Snowball" phase, a certain nimble archaeon — probably an ordinary scavenger nibbling on whatever organic bits it could find — swallowed a tiny aerobic bacterium. This bacterium already knew how to breathe oxygen, but for reasons we still don't fully understand, the archaeon failed to digest it.

The little bacterium survived. Then — with sublime audacity — it began dividing *inside* its astonished host.

The symbiosis was so wildly successful that this hybrid cell — half archaeon, half bacterium — became the ancestor of *all modern eukaryotes* with mitochondria. That includes every plant leaf you've seen, every mushroom you've stepped over, every lion, octopus, oak, and person walking the Earth today¹⁶.

It's no coincidence that this merger happened precisely when oxygen levels in the oceans and atmosphere had started to rise again. The engulfed bacterium — now proudly known as the *mitochondrion* — used dissolved oxygen to break down sugars and fats supplied by the host, releasing bursts of chemical energy. And, crucially, it handed over most of that energy to its host in the form of ATP.

This new system of cellular respiration was far more efficient than the old anaerobic (oxygen-free) pathways used by earlier life. Sugars and fats are extraordinarily powerful and portable energy sources — far superior to the sunlight that cyanobacteria had relied on while floating motionless at the ocean's surface¹⁷.

To compare the two strategies, imagine a flimsy wooden biplane from World War I trying to compete with a modern fighter jet equipped with twin turbojet engines and a roaring

afterburner. The arrival of mitochondria was precisely that kind of technological leap — a sudden and overwhelming advantage. It was this leap that unlocked the energy required for speed, movement, muscle, nerves, and eventually multicellular complexity.

In other words, this chimera was a bombshell

From that unlikely merger emerged a *new kind of giant cell*, one equipped with a nucleus — a protected vault where its DNA was stored. The nucleus became a pristine library, a sort of central administration office: the place where memory, blueprints, and heritage were kept safe behind a nuclear membrane, neatly separated from the noisy factory floors of the cytoplasm.

Outside this nucleus, the large and intricate cell swallowed food from the environment and shuttled fuel inward to hundreds of miniature power stations — the mitochondria.

In essence, a eukaryotic cell is a chimera, a composite creature built from several ancestral cells welded together. And once these hybrid cells learned to harness the full power of their internal mitochondria, they overcame every remaining energetic barrier. Suddenly, evolution had the fuel to attempt something bold: *multicellular organisms*.

Division of labor arose. Cells began specializing. And from that great cellular guild emerged muscles, stomachs, eyes, claws, warm blood, and eventually — brains.

If you want a wonderfully clear and entertaining account of how body parts evolved step by step, look no further than Neil Shubin's book *Your Inner Fish*¹⁹.

There is, however, another crucial detail we should clarify — a logistic miracle every multicellular organism must solve: How do trillions of cells learn to talk to one another?

They communicate using “words” written in molecules that pass from one cell to the next. For example, a cell can broadcast a message by releasing signaling molecules — mediators — into its surroundings. These mediators latch onto receptors embedded in the outer membrane of a neighboring cell, the one the message is intended for. Binding to these receptors triggers a cascade of molecular events inside the target cell, often reaching all the way into its nucleus.

Once the signal arrives in the nucleus, it can switch certain genes on or off. And just like that, the receiving cell changes its behavior. A message from another cell may tell it to self-destruct, to divide, or to begin producing an entirely new substance.

Now that we understand cellular turbines and fuel, let's peek behind the curtain and see who, exactly, is running the show inside the cell.

Immortal Genes

Genes leap from one body to the next, manipulating those bodies as they please and for their own benefit, slipping away long before the mortal shells that carried them grow old and die. The chromosomes inside any given body vanish with it, the way a hand of cards disappears once a round is played. But the cards themselves don't perish during the shuffle.

The cards are the genes

Genes are not destroyed during crossing-over; they merely switch partners and continue their long voyage forward. They are the **replicators**, and we are the elaborate, temporary machines they build in order to survive. Once we have completed the only task they truly care about — staying alive long enough to produce children — we are dismissed. But genes... genes come from geological time. They are here for millions of years.

The lifespan of a *physical* DNA molecule is surprisingly short — perhaps a few months, a few years, certainly not the full duration of a human life. But DNA as an *information pattern*, as an endlessly renewed copy of itself, can persist for hundreds of millions of years.

Here is how Richard Dawkins explains it: a gene — or active replicator — is a stretch of the genome, a segment of a chromosome that encodes a trait essential to the organism. It might instruct the building of a protein that carries oxygen, or an enzyme that breaks down sugar — anything that advances its own favored version of a trait, improving how that trait works.

A particular version of a gene is called an *allele*. Naturally, for every gene there exist alternative alleles — versions that weaken its effect or even shut it down entirely. And these alleles, the alternative forms of a single gene, are perpetually competing for their place in the next generation.

This matters a great deal, so let me tell the story once more — more slowly this time. And first — in memory of my father, who spent his life in this world of slender racing shells and impossible discipline — let me offer, in full, a beautiful analogy from Richard Dawkins' *The Selfish Gene*:

“A single oarsman cannot win the Oxford and Cambridge boat race on his own. He needs eight companions. Each is a ‘specialist’ of a sort, always occupying the same position in the boat, acting as stroke, bows, or whatever. Rowing is a cooperative enterprise, although some individuals may be better at it than others. Suppose a coach wants to choose a crew from a pool of candidates, among whom there are strokes, bows, and other specialists. Imagine that he proceeds as follows. Each day he makes up three new trial crews, by randomly shuffling candidates among the positions in the boat, and then races the crews against each other. After some weeks he notices that certain individuals seem often to end up in winning boats. These he earmarks as good oarsmen. Other candidates tend to find themselves in losing crews, and are eventually rejected. But even a top-class oarsman may sometimes find himself in a losing crew, perhaps because the other members are poor, or

simply through bad luck — for example, because of a head wind. Good oarsmen are only, on average, more likely to end up in winning boats.”

“The oarsmen are genes. Rivals for each seat in the boat are alleles capable of occupying the same locus. Fast rowing corresponds to the ability to construct a body that succeeds — that is, that survives. The wind is the external environment. The pool of alternative candidates is the gene pool. Insofar as the survival of each individual body is concerned, all its genes are in the same boat. Many ‘good’ genes may find themselves in bad company — that is, in a body that also contains a lethal gene which kills the whole organism in early life. In such a case the good gene perishes along with the rest. But that is only one body. Copies of the good gene live on in other bodies which do not contain the lethal gene. Many genes go to the bottom because they are unlucky enough to share a body with bad genes; many perish because of other misfortunes, such as being struck by lightning. But by definition, luck and misfortune are distributed at random, and a gene that consistently loses is not merely unlucky — it is a bad gene.”

“One of the qualities of a good oarsman is the ability to row in harmony with the rest of the crew. This may be as important as strong muscles. The general quality that successful genes must possess is a tendency to postpone the death of their survival machines at least until after reproduction. Perhaps some of your cousins, or the brothers and sisters of your grandparents, died young, but none of your direct ancestors did. Ancestors simply do not die young!”

In an old photograph, my father’s two-man shell skims across the water, the oars bent in an arc of almost unbearable effort. I don’t even know where that moment was captured anymore. But I am certain of this: wherever their river runs now, whatever unfamiliar banks may rise around them, those two will still pull ahead of everyone else — and cross the finish first.

Sexual Reproduction and the Rise of the Cell Nucleus

Bacterial cells usually reproduce by splitting in half, making two identical copies of the parent cell. This simple act of division happens astonishingly fast — that speed *is* their survival strategy. Living at such a frantic tempo, bacteria cannot afford the luxury of collecting new or unnecessary genes. The moment they find a bit of food, they divide with reckless enthusiasm; the rest of the time, they sit quietly, waiting and starving. Any introduction of new genetic material is accidental, fragmentary, and utterly chaotic. Prokaryotes — bacteria without a nucleus — often just swap snippets of DNA the way schoolchildren pass crib notes under their desks during an exam.

Multicellular organisms live by a completely different script. Our defining feature is that we are made of diploid cells. That means every cell in your body carries two copies of each chromosome — and therefore two copies of every gene. But those copies are not simple duplicates. Paired, or homologous, chromosomes are two independent archives, each filled

with its own versions — its own *alleles* — of the same set of genes. One archive comes from your mother. The other from your father.

Every person carries a whole collection of errors or random mutations in their genome — *single nucleotide polymorphisms*, or SNPs. But because genes are so numerous, any specific mistake is incredibly unlikely to appear in both copies at once. If you have two versions of every gene, odds are that whenever one contains a mutation, the other remains intact — and nothing disastrous happens.

What matters here is understanding that we don't simply inherit genes from our ancestors; we receive half our genome from one independent source and half from another. Compared to bacteria, we are living chimeras — not mere vessels of ancestral intent, but bearers of a unique, wildly shuffled combination of parental instructions and impulses.

This great genetic shuffle has a name: *sexual reproduction*.

Sexual reproduction offers a way to shuffle your genes with someone else's and to generate new, potentially successful combinations — and the more variations, the better.

To make this possible, each parent produces special reproductive cells called *gametes*: eggs in females and sperm in males. When these gametes form, a key role is played by **crossing-over**. At one stage of cell division, the homologous chromosomes — say, in your mother, one chromosome inherited from her mother and the other from her maternal father — line up face to face and intertwine.

Then something remarkable happens: they are physically cut by specialized enzymes, and the matching segments — the alleles — are exchanged. Other enzymes stitch these swapped fragments back together, creating a new, never-before-seen version of the chromosome. The result is a gamete — your mother's egg cell — carrying a fresh, reshuffled genetic manuscript.

When two gametes meet (your mother's egg and one lucky sperm from your father), a beautifully choreographed exchange of hereditary material takes place. And on top of that, everything doubles. The child now operates with two full sets of chromosomes — one from the mother and one from the father. Each of those sets, in turn, after crossing-over, carries half a library from each grandparent.

Of course, all this genetic choreography is energetically expensive and requires the perfectly coordinated work of countless enzymes. And such delicate processing of information cannot happen out in the general workshop — the cytoplasm — where thousands of chemical reactions rage like an industrial plant. It needs its own quiet office, sealed away from the noise.

That is how eukaryotes acquired the *cell nucleus*. We still don't know exactly how it appeared. Perhaps it began as a partnership of two different archaeal species, one cell

ending up inside another and gradually taking over the role of genetic archivist. Or perhaps a mishap during cell division produced an internal compartment that turned out to be extraordinarily useful. One way or another, the nucleus emerged — and once it did, evolution treated it as a treasure worth keeping.

So why go through all the trouble of sexual reproduction?

First, it is a powerful tool for cleaning harmful mutations out of the genome. When gametes form, the genetic material from mother and father is cut into small fragments and then reassembled from only half of those pieces. There is always a chance that the damaged stretches of DNA simply won't make it into the child's genome.

Second, the constant increase in genetic variation in every generation sent the diversity of life skyrocketing. Sexual reproduction means that organisms can choose whose genes to combine with their own. In other words, **sex gives us choice**.

Once animals mastered sexual reproduction, they began, in a sense, to “play God.” One of the primary functions of sex is partner selection based on genetic quality. A female of any species — insects, birds, humans — evaluating a male by his indicators of fitness (leadership, strength, intelligence) is conducting, in her own way, a sophisticated genetic screening.

Sometimes the female's sensory preferences do the choosing: a peacock's shimmering tail, the hunting ability of a human male, or even the color of his eyes. And once such a preference appears, evolution amplifies it. This launches an intense selection pressure favoring the ornaments, abilities, and courtship rituals that happened to impress the female. The brilliant plumage of parrots, the enormous human brain, speech, and abstract thought — all of these are products of **sexual selection**.

Over millions of years, the divergence of mating preferences drove species apart, branching and re-branching life until most of the biodiversity on Earth emerged from this slow, persistent, romantic tug-of-war.

The Birth of Complexity

How can blind chance — a mutation, a copying error in a strand of DNA — give rise to organisms of such breathtaking elegance and precision? Charles Darwin offered the answer more than 160 years ago: **natural selection**.

Mutations themselves do nothing more than stir up randomness within a population. But selection is the opposite of random. The individuals least suited to their environment are steadily filtered out; the ones best suited survive and pass on the genetic blueprints that made their success possible. If a mutation, even a tiny one, helps the heart pump blood more efficiently, that advantage is inherited. If it interferes with circulation, it vanishes along with the organism that carried it.

In every generation, only a small fraction of individuals manage to survive long enough to reproduce — the lucky, the persistent, the clever, the well-adapted. Over hundreds of generations, sheer luck cancels itself out. Natural selection works like a ratchet: a simple mechanical device that allows motion only in one direction. Random variation wanders all over the place, but the ratchet of selection locks in the improvements, step by step. And what emerges over deep time really does look like a staircase rising toward greater and greater complexity.

Yet small mutations are not the only way a genome can change. To build more intricate organisms, you eventually need something fundamental: more genes, more instructions, more “shelf space” in the cellular library.

A tiny bacterial genome could never code for a human being — or for the endless genetic differences we see among humans. So where do extra genes come from?

They appear through a clever trick: **duplication**. A gene — or sometimes an entire genome — gets copied. Suddenly the organism has two versions of the same instruction manual. One copy keeps doing its original job; the other is free to evolve, to take risks, to stumble into a new function: the beginnings of a more complex eye, a larger brain, a sonic radar for dolphins and bats.

These jumps are not “Darwinian” in the narrow sense. They aren’t slow refinements of an existing design. They’re big, bold leaps across genetic space, sudden restructurings that give evolution fresh material to sculpt. Once a duplicated gene exists, ordinary natural selection can begin polishing it — generation by generation — until it becomes indispensable.

And if a duplication turns out to be useless? Then natural selection sweeps it away, tidying up the genome like a meticulous librarian. But when duplication works, it becomes evolution’s invitation to invent — to play — to push life into realms it had never explored before.

Remember that ancient archaeon — the one that swallowed an aerobic bacterium and, in doing so, unlocked an endless source of energy in an atmosphere newly flooded with oxygen? Give evolution just a couple hundred million years, and this unlikely partnership blossoms into something astonishing: internal membranes, specialized organelles, intricate metabolic cycles, sex, a sprawling genome, multicellularity, predation — and eventually such magnificent inventions as flight, vision, hearing, echolocation, brains, and even consciousness itself.

From the very beginning, eukaryotes went all-in on complexity. The moment our distant archaeal ancestor joined forces with a mitochondrion, it signed an evolutionary pact written almost literally in blood — crimson not only because of oxygen but because of hemoglobin, the molecule that ferries that oxygen from lungs to tissues, to the cells, and into their mitochondria. The more energy these proto-cells burned, the larger, more intricate, and more capable they became. By breathing oxygen and not dying, they jumped aboard an evolutionary express train — and there was no getting off. Our vast, warm-blooded,

multicellular bodies burn fuel at the edge of biochemical possibility, and our oversized brain is the energy-hungry crescendo of all life on Earth.

By the way, the theory of how multicellularity emerged is surprisingly simple. Bacteria invented new tricks for devouring one another — and for not being devoured. A multicellular body is, among other things, a way to become big. And becoming big is one of evolution's oldest and most reliable strategies for not ending up as someone else's lunch. So, multicellularity may have begun as a defense mechanism.

Of course, having a multicellular body is expensive — outrageously expensive. But the advantages are immense: it not only makes you harder to eat, it allows you to become the eater; to move swiftly across the world, to run, crawl, swim, hunt. And every one of those abilities demands oceans of energy.

In short, the entire story of evolution can be sketched like this: cyanobacteria spent roughly a billion years flooding the atmosphere with oxygen; anaerobes died off; Earth plunged into a catastrophic global glaciation — the “Snowball Earth.” Then came aerobic bacteria, which, in symbiosis with ancient archaea, became mitochondria. From them arose eukaryotes; from eukaryotes — the first simple multicellular beings (probably algae and sponges). And after that, elegant multicellular bodies exploded across the planet. This began about 540 million years ago and is known as the *Cambrian Explosion*.

When We First Learned to Flee the Dark

In the Cambrian, the world grew teeth. Predators appeared, improbable and terrible, like *Anomalocaris canadensis* — creatures that seemed to have slipped out of some ancient nightmare, all grasping limbs and alien eyes. Against such things there were only two answers: to hide, or to flee. And for either, one needed more than luck. One needed a body that could move, and a mind that could choose.

We were no longer quite worms, but not yet fish. We stood at the threshold of something new. We became chordates.

To escape, we grew within ourselves a strange, half-liquid rod — soft, yet resilient — the first quiet promise of a spine. It gave the body an axis, a direction, a sense of forward and back. Around it, muscles gathered and learned to move with purpose, to bend the body into swift, deliberate motion. The sea, which had once carried us, now became something we could push against.

Our mind was still dim, but the first glimmers of sight had appeared. Eyes, simple and uncertain, opening like small windows into a world that had suddenly become dangerous.

And somewhere in that same distant age, something even more improbable occurred. The genome of our remote ancestor, *Pikaia*, doubled. And then, in a quiet act of excess, it doubled again.

Copies upon copies of the same instructions, scattered through the darkness of the cell. Some remained as they were, steady and unchanged. Some broke, falling silent. But others... others began to wander.

They changed. They took on new roles. They became the first hints of gills, of a heart, of a more intricate brain — the slow assembling of a body that could do more than merely exist¹⁸⁻¹⁹.

It is difficult not to see a kind of fragile hope in that moment. From redundancy came possibility. From excess, invention. From something copied twice over, a future began to take shape. And in those quiet waters, long before memory, we were already becoming ourselves.

Everything that followed — trilobites, dinosaurs, the dodo, saber-toothed cats, mammoths — has already been beautifully described elsewhere²⁰, and we don't need to retell it here.

I think we've finished our second great canvas — and a difficult one at that. The hardest part is behind us. Everything that comes next will be light, airy, almost like pink-colored marshmallow. I promise.

Chapter 3. Human Evolution

Now we're looking at our third canvas — smaller, humbler, and far less cosmic than the previous two. This one holds everything we currently know about the origins of humans and their long, dusty exit from Africa.

Earth itself, meanwhile, refuses to sit still. Its lithosphere and climate behave like a restless stage crew, constantly rearranging the scenery. If you rewound geological time in fast-motion, you'd see continents sliding back and forth in every imaginable configuration, smashing together and peeling apart, oceans opening and closing as casually as curtains in a school play.

And the engine behind all this planetary choreography — **plate tectonics** — also happens to be the quiet but decisive force steering our evolution. But let's first figure out exactly what geological moment we've landed in.

Global Cooling

The last 50 million years have been defined by a steady planetary chill, culminating in the ice ages that have been cycling in and out for the past 2.5 million years. This long decline in temperature began when India slammed into Eurasia, giving rise to the Himalayas right where the ancient Tethys Ocean once sparkled.

The rapid erosion of this colossal newborn mountain range sucked huge amounts of carbon dioxide out of the atmosphere, weakening the greenhouse effect and helping tip the planet toward colder climates.

As if that wasn't enough, Earth's orbit began to drift under the influence of three slow but relentless astronomical rhythms known as the *Milanković cycles*:

- *Eccentricity*: Earth travels around the Sun not in a perfect circle but in a slightly stretched ellipse, and this “stretchiness” grows and shrinks over a 100,000-year cycle due to gravitational nudges from other planets. As eccentricity increases, Earth's climate tends to harden.
- *Tilt*: Seasons exist because Earth's axis leans. When the Northern Hemisphere tilts toward the Sun, it's summer; when it tilts away—winter. But that tilt isn't fixed: it wanders by a few degrees over a 41,000-year period.
- *Precession*: Our tilted axis slowly swivels in space like a giant, patient spinning top, completing one full wobble every 26,000 years.

Together, these cycles periodically alter the amount of sunlight reaching specific regions. Every 100,000 years or so, the configuration becomes just right—and the planet nose-dives into another deep freeze. Around 2.5 million years ago, the celestial mechanics lined up perfectly with the new arrangement of tectonic plates, amplifying the cooling still further.

And What About the Modern Climate Crusaders?

At this point you might wonder: “*And what about that pale, troubled girl scolding world leaders from a UN podium, wagging her finger at the adults as if they were misbehaving schoolchildren. Remember Greta?*”

She is merely the latest face of a familiar impulse — the anxious chorus convinced that the planet is always on the verge of bursting into flames, freezing solid, or dissolving into acid rain. The modern climate crusader is simply the newest heir to a long dynasty of professional worriers. Their predecessors once trembled before the ozone hole; their successors will no doubt find something equally theatrical — perhaps the moral urgency of banning methane-fueled kitchen stoves, or a noble campaign to rescue whales from insufficiently sensitive humans.

“The apocalypse, as a genre, has never suffered from a shortage of enthusiasts.”

But let’s get to the point. The appearance of polar ice caps turned out to be the last straw. Continental drift, teamed up with the increasingly insistent Milankovitch cycles, tipped Earth into a recurring sequence of ice ages. The early ones were fairly gentle, but with every swing of that cosmic pendulum, they grew harsher and more dramatic. And yes — this is still happening. Each modern glaciation lasts about 100,000 years, punctuated by brief interglacial intermissions of ten to twenty thousand years, when the climate relaxes into something surprisingly warm — practically tropical even at high latitudes.

The peak of the last glacial period came 26,000 years ago. The entire northeastern stretch of North America lay beneath the Laurentide Ice Sheet, while the west groaned under the Cordilleran. Northwestern Europe disappeared beneath the Scandinavian Sheet. Whatever parts of the Northern Hemisphere weren’t entombed in ice turned into boundless steppes and wind-scoured tundra. Glaciers locked away so much atmospheric water that sea level during the last glacial maximum fell 120 meters below where it sits today.

For the past ten thousand years, we’ve simply been enjoying an interglacial vacation — and sea level right now stands *slightly above the long-term average* of the last two million years.

Now please — really, please — focus for a moment and read that last sentence again...

The Rise of *Homo erectus* in Africa

Modern humans trace their lineage back to the hominines — the upright-curious, tool-fiddling, fruit-munching apes who roamed Africa between five and two million years ago. Then, about two million years ago, something dramatic happened: all those earlier hominine species vanished, and out of that evolutionary bottleneck emerged a new player — our own genus, *Homo*.

And with *Homo erectus* the whole enterprise of being human suddenly went into overdrive.

This early model was taller than his competitors, smarter than his predecessors, and — crucially — the first to strike a match, metaphorically speaking. He tamed fire. And once you tame fire, you can pretty much declare yourself CEO of the savanna.

Naturally, none of this happened in a vacuum. Africa wasn't covered by ice sheets the size of continents the way Europe and North America were, but it *was* drying out. Forests shrank. Grasslands sprawled. The buffet-style life of a forest forager was over, and the savanna demanded a different résumé: endurance running, coordinated hunting, and the ability to avoid being eaten by anything with sharp teeth and faster legs.

So *erectus* did what any ambitious primate would do faced with a career shift — he adapted. He spread across Africa, following herds, inventing new tools, and slowly taking his place as a full-time predator. His anatomy already looked uncannily modern. He had sweat glands to prevent overheating during long chases under a brutal sun, and shoulders built for throwing — a design upgrade that allowed him to launch projectiles with alarming accuracy.

Even more intriguingly, *erectus* shared with us a long childhood, an increasingly sophisticated toolkit, and a social life that would not have looked entirely alien at a modern campsite. Anatomical changes and lifestyle choices fed each other in a feedback loop: more meat in the diet shrank the gut — a metabolic money-pit — freeing up calories for the other notoriously expensive organ: the brain.

And the brain *is* expensive. It makes up barely one-fiftieth of the body's mass but burns through a sixth of its total energy. So why did *erectus* suddenly start investing in mental real estate like a Silicon Valley start-up on the brink of going public?

Let me offer you Jeffrey Miller's wonderfully elegant idea — the “runaway brain” hypothesis²¹. In short: females were in charge of evolution, as usual, and they had opinions. And those opinions leaned toward clever males.

Courtship rituals, childcare competence, toolmaking finesse — the whole romantic circus — all favored individuals with slightly sharper minds. With every generation, brainier males had a better shot at passing on their genes. Sexual selection turned the *erectus* brain into a reckless spender, inflating it far beyond what was needed for basic survival. A compact, efficient ape-brain morphed into a sprawling, energy-gulping showpiece capable of such evolutionary extravagances as conversation, violin solos, and drawing animals on cave walls for no practical reason whatsoever.

It might *look* like all this intellectual fireworks display is about sharing knowledge or building culture. From the perspective of raw biology, though? It's basically a peacock tail made of neurons. A glimmering announcement of fitness, visible only to one extremely important audience: **the females deciding whose genes get to join forces with theirs.**

Like most predators who make their living on wide-open terrain, *Homo erectus* turned out to be a deeply social creature. And that sociality expressed itself in three unforgettable ways: unusual mating behavior, bursts of shocking brutality... and, surprisingly, cooking.

Once our ancestors discovered that food tastes better *and* kills fewer of its consumers when exposed to fire, the evolutionary consequences were immediate. Cooked meals packed more calories than raw ones and weren't riddled with parasites. Tribes that grilled, roasted, or simply warmed their dinners survived longer, reproduced more, and raised healthier children. Those who insisted on chewing through cold, sinewy, bacteria-rich slabs of flesh eventually exited the story — natural selection does not accommodate culinary stubbornness.

Where there are tribes, there are territories. And *erectus*, now bonded by firelight and shared meals, guarded their land with rigor that would put modern border patrols to shame. Unlike most mammals, they no longer had a “mating season”: their females were fertile year-round, which changed the rhythm of social life completely. We also know they favored long-term pair bonds — remarkably stable partnerships oriented toward raising children who took years to grow up.

And here comes a delightful evolutionary twist. Once males no longer needed to advertise themselves from a distance — no dramatic mating calls, no flamboyant displays — the physical differences between males and females began to fade. Over many generations, what later biologists would call sexual dimorphism softened. In other words, once pair-bonding became the norm, evolution quietly retired the peacock feathers, and everyone's appearance evened out.

The Evolution of *Homo erectus* in Europe and Asia

Eventually, *Homo erectus* became the first of our ancestors to wander beyond Africa — not because of some grand migratory plan, but simply because that's where the lunch was going. The endless savannas teemed with hoofed animals, and *erectus* just followed the herds wherever they drifted. One day you're jogging after antelopes in the Rift Valley, the next — surprise! — you're in Eurasia.

By about 1.5 million years ago, they had reached what is now China and the Caucasus. In the caves of Zhoukoudian near modern Beijing, and in Dmanisi in Georgia, archaeologists eventually uncovered the bones of their many regional offshoots. On the island of Java, *erectus* lingered astonishingly late — until roughly 100,000 years ago.

But the strangest fate befell the *erectus* population that settled on Flores, in Indonesia. There, *Homo floresiensis* emerged around a million years ago and, over time, shrank to hobbit-sized proportions — literally. These pocket humans hunted giant rats with spears. Their neighbors, the local Komodo-dragon relatives known as Flores monitor lizards, grew to a size intimidating even for a full-grown modern human. Imagine, then, what they looked like to people barely one meter tall, no matter how courageous. And yes, the dragons and

the giant rats are still alive today on Komodo and Flores. The tiny humans, alas, are not — volcanic eruptions on nearby islands finished them off...

By the way, nothing is stopping you from flying to Flores from Bali yourself. In the little port town of *Labuan Bajo*, you can always find a reliable captain with a small boat, ready to whisk you off to see the enormous Komodo dragons; to snorkel above untouched reefs near the miniature island of Kanawa; or to venture inland toward the majestic Kelimutu volcano, with its two crater lakes shimmering turquoise and green. You can easily reach the volcano by car from the town of Ende on Flores' southern coast.

I never managed to visit the great limestone cavern of Liang Bua — its roof long since collapsed — where those tiny ancient humans once lived. But you? I'm sure you'll make it.

Around 800,000 years ago *erectus* gave rise to a new and more refined model: *Homo heidelbergensis*. From them, about 250,000 years ago, emerged the Neanderthals in Europe²² and the Denisovans who spread across Asia. And yes — we'll be meeting them very soon.

The Rise of *Homo sapiens* in Africa

The first anatomically modern humans — *Homo sapiens* — appeared in East Africa somewhere between 300,000 and 200,000 years ago. They likely descended from *Homo heidelbergensis*, though not in one clean leap but through several evolutionary drafts and half-finished prototypes. And for the first 200,000 years, life wasn't even remotely glamorous: it was one long, grinding battle for survival²³.

While Eurasia was slowly icing over as Neanderthals spread across its frigid expanses, Africa was drying out. The forests retreated, the rains faltered, and even the tough old *Homo erectus* eventually vanished from the continent. What remained was a single fragile thread of humanity — a scattered handful of *sapiens*, perhaps as few as 300 individuals, perhaps as many as 3,000 — clinging to life on the swampy shores of a vast lake that once filled what is now the Kalahari Desert²⁴.

And then, mercifully, climate shifted. The rains crept back. Vegetation returned. Our ancestors trekked southward and fanned out along the Indian Ocean coastline. And there, roughly 100,000 years ago, something extraordinary flickered to life in their brains — a sudden cognitive spark that pushed the population forward.

Remember the section on "*The Birth of Complexity*," where we talked about whole genes doubling and giving evolution extra "workspace"? Something similar may have happened inside the skull of one particularly lucky *sapiens*. A set of favorable mutations appeared — and in such a tiny population, they could spread like wildfire.

Take the FOXP2 gene, the celebrity of early human genetics. Two ancient mutations in FOXP2 appeared after our lineage split from chimpanzees, and a third, much more recent one, is found in every living human but in none of the Neanderthals. Svante Pääbo's lab

showed that FOXP2 affects vocalization — mice carrying the human version literally squeaked differently. Of course, no single gene rewired the human mind, but it was part of a cluster of changes.

Thanks to techniques pioneered by Pääbo and David Reich, scientists have now compared thousands of ancient and modern human genomes — and uncovered a whole constellation of recent mutations: genes enabling adults to digest milk; genes that lightened skin, improving vitamin D production in colder climates; genes that offered some protection against malaria. There are probably dozens more we haven't even begun to understand, quietly shaping our behavior, metabolism, maybe even our thoughts²⁶.

Armed with their new cognitive toolkit, early *sapiens* wasted no time. They invented long-distance weapons: fire-hardened spear points, bows, arrows — technology that made them far more dangerous than their bodies alone would suggest. They learned cooperative hunting, turning themselves into clever, coordinated predators²⁵.

Still, the *sapiens* of that era were not the pinnacle of creation, just another stage in the unfolding of our species, still far from the reflective, talkative beings we are now. There was nothing uniquely magical about their brains — not yet. They were simply another step in a long chain of attempts, one more iteration on the road from ancient African *sapiens* to something entirely new: humans with richer memory, more intricate language, a deeper inner life.

Don't forget: cognitive evolution doesn't move in a neat staircase. It comes in waves — messy, uneven, overlapping waves²⁴. And each new wave, armed with a slightly sharper mind, often replaced the one before it. Which is a good reminder: staying close to the smartest kids in class has always been a winning evolutionary strategy. After all, it was just a few hundred unusually gifted *sapiens* (*the forerunners of our kind*) who eventually walked out of Africa — and went on to conquer the entire world.

The Last Exodus of *Homo sapiens* from Africa

Strictly speaking, *Homo sapiens* marched out of Africa more times than anyone can count, but most of those mini-expeditions left about as much historical weight as raindrop streaks across a dune. Pretty while they last — erased five minutes later. But this time? This time the story stuck.

Around 74,000 years ago, the Toba volcano on Sumatra blew itself to pieces with such enthusiasm that its ash drifted all the way to India and China — conveniently timestamping the archaeological record like a cosmic “you are here” sticker. By then, several groups of *sapiens* had already crept along the northern rim of the Indian Ocean, pushing eastward until they reached the coast of what is now southern China. Who exactly these people were, how many parties set off, or in what order they ventured out... well, that part of the script was lost ages ago.

What we do know is that this moment marks the first real split between the African and non-African branches of *Homo sapiens*. Today, the African branch is vividly represented by the *San* hunter-gatherers of southern Africa and the forest-dwelling *Pygmy* peoples south of the Sahara. The non-African branch, on the other hand, bears the unmistakable genetic fingerprint of a severe bottleneck — a tiny sliver of ancestors from whom every modern Eurasian descends. In other words, **all of us outside Africa trace our origins to a single, surprisingly small band of people equipped with surprisingly upgraded brains** who stepped out of Africa and never looked back.

By the time several *sapiens* groups reached Eurasia around 50,000 years ago, the world was anything but empty. It was populated by a whole roster of archaic humans — physically robust, cognitively distinct, and all distant heirs of earlier *Homo erectus* migrations. For convenience, we lump most of them under two banners: the Neanderthals (*Homo neanderthalensis*) and the Denisovans (*Homo denisovensis*).

Did these archaic humans mix with the newly arrived *sapiens*? Once upon a time, archaeology shrugged. Now, whole-genome sequencing and population genetics have answered with the subtlety of a foghorn: **yes — absolutely, enthusiastically, and quite a lot**. Different *sapiens* groups hybridized with different branches of the old *erectus* family tree wherever they met them²⁶.

Neanderthals and the Mysterious Denisovans

Neanderthals were, genetically speaking, a very different branch of the human family tree. Think heavy skulls with dramatic brow ridges, thick wide bones, and the kind of muscular build that would make a modern gym influencer weep with envy. Their brains were even a bit larger than ours — though probably not wired for speech in the way *sapiens* brains are. They certainly weren't stupid, however. They cared for their children, and for their elders too. In Shanidar Cave in northern Iraq, archaeologists found the burial of an elderly, half-blind man with almost no teeth and a useless arm — evidence that he had been looked after for years by his group.

At their peak, Neanderthal groups spread as far as Siberia and the Altai Mountains, mingling there with both Denisovans and early *sapiens*. According to David Reich's lab, which sequenced dozens of ancient skeletons from Europe and Siberia, early modern humans living at that time carried **6–9% Neanderthal DNA**. That's enormous. Radiocarbon dating places this interbreeding between 54,000 and 45,000 years ago. Even today, after tens of thousands of years of dilution, non-Africans (that is, Europeans and Asians) still carry **1.5–2.1% Neanderthal ancestry** — with East Asians holding slightly more than Europeans. Africans, of course, have none, because their ancestors never left Africa to meet these northern cousins.

In eastern Eurasia, *sapiens* moved north along the great rivers — the Mekong, the Irrawaddy, the Ganges — all the way to the Tibetan Plateau and the vast steppe stretching beyond it. In the west, they reached Europe around 50,000 years ago. Neanderthals had

dominated Europe for a quarter of a million years, successfully repelling every previous wave of outsiders. But when *sapiens* arrived, the tide turned fast: Neanderthals were pushed to the margins, and by 40,000 years ago they had vanished²⁷⁻²⁹.

Enter the Denisovans

Now let's talk about the Denisovans — whoever they were.

In 2012, in *Denisova Cave* in the Altai Mountains, practically the dead center of the Eurasian landmass, archaeologists discovered a tiny bone fragment: a sliver of a young girl's finger. When Johannes Krause sequenced it, the result shocked everyone. The bone belonged to a completely unknown group of archaic humans, probably relatives of *Homo erectus*. Five years later, once the full Denisovan genome was decoded, things got even stranger. It turned out Denisovans interbred with both Neanderthals and early *sapiens* — and that their closest living relatives today include *Aboriginal Australians* and *Papuans* of New Guinea.

They may all descend from an early wave of *sapiens* who left Africa around 120,000 years ago. According to David Reich's laboratory, 3–5% of the Papuan genome comes from Denisovans, with hybridization dated to 49–44 thousand years ago. That figure does not include the separate 2% of Neanderthal ancestry — a completely different chapter of archaic romance. Altogether, modern Papuans carry **5–8% DNA from archaic humans**.

What ultimately happened to the Denisovans in Asia is still a mystery. But I have a very bad feeling they met the same fate as the Neanderthals in Europe...

Real Biological Differences Between the Branches of *Homo sapiens*

So, what picture do we end up with? Long before the last Ice Age rolled in, a small band of people — equipped with new unusually capable brains and a talent for cooperative hunting — walked out of Africa, most likely through the Middle East. Once outside, this founding group split into several daughter lineages that fanned out across Eurasia with surprising speed.

- **The Eurasian branch** — some groups mixed with Neanderthals in Europe, eventually forming the lineage of European hunter-gatherers.
- **The East or Asian branch** — those who followed the coastline of the Indian Ocean — probably moved north along the great rivers, interbreeding along the way with both Neanderthals and, to a lesser extent, Denisovans.
- **Completely separate southern branches — the Australian and New Guinean lineages** — **carry the largest dose of archaic DNA**. These people likely left Africa among the very first waves, traveled eastward along the Indian Ocean, reached Burma and southern China, and then turned south toward Australia. Back then the region was part of a vast, now-submerged continent called Sunda, so the journey didn't require celestial navigation — just persistence and decent weather.

- **And finally, the branch with the richest genetic diversity of all — the African lineage.** It never left the continent and therefore never mixed with Neanderthals or Denisovans; its gene pool remained untouched by archaic cousins.

Now that paleogenetic has armed us with actual numbers — not fairy tales, not “maybe-it-was-like-this” campfire stories, but **full genomic reconstructions** of hundreds of *Homo sapiens* and archaic *Homo erectus* — the picture that emerges is, frankly, far messier than the tidy diagrams in high-school textbooks. The branches of the modern human family tree aren’t just slightly different shades of the same color; they’re more like distinct hues mixed on different continents under wildly different evolutionary pressures.

At this point, pretending otherwise feels... let’s say, charmingly naïve. The truth is simple: *long-separated human populations accumulated real, measurable biological differences.*

Not metaphysical, not ideological — strictly evolutionary. And the coming wave of large-scale genomic studies will make this even harder to ignore. Genes influence behavior, cognition, physical traits, metabolic quirks — the whole package — and natural selection has been editing that package in different directions depending on where your ancestors happened to be dodging predators or harvesting yams.

Some of these distinctions are already textbook examples: skin pigmentation tuned to latitude; body morphology shaped by climate; lactase persistence that allows some adults to drink milk without regretting it; genetic adaptations for high-altitude breathing among Tibetans; and population-specific susceptibilities to disease.

Take athletics: certain genetic variants linked to sprinting or endurance show up far more frequently among West African populations. That’s not a stereotype — that’s genotype.

And yet, every time these discussions surface, someone inevitably objects: “*Well sure, skin color and body shape vary... but cognition? behavior? That’s completely different!*” Except — biologically — it isn’t. Genes don’t politely stop at the neck. They operate everywhere, especially in the brain, which is, after all, just an organ made of cells following instructions encoded in DNA.

Consider this: if you take a very large sample of people, measure something as mundane as *how many years of education* each person completed, and then correlate those data with their genomes, will you find patterns? Well, Daniel Benjamin’s lab went ahead and tested that. They analyzed 400,000 Europeans and identified 74 genetic variants statistically associated with completing higher education. No crystal balls, no phrenology — just big data and population genetics. Their study appeared in *Nature*, issue 533, 2016.

Of course, no single gene “causes intelligence” — that’s not how biology works — but the point is unavoidable: **complex psychological and cognitive traits have genetic architectures, too.** And as genomic datasets become richer, we are going to learn a great deal more about how different populations adapted not just physically, but mentally and behaviorally, across tens of thousands of years.

Where did all these differences between human populations actually come from?

Because — as you surely remember — we all squeezed through the same genetic bottleneck, meaning our list of common ancestors is comically small. And yet, paradox: today the branches of *Homo sapiens* look wildly, even flamboyantly different. How?

Well, first of all, not everyone descended from that “lucky” little group that left Africa 50,000 years ago, carrying the cognitive-boost mutations we talked about earlier. For instance, the genomes of modern Papuans preserve traces of much older sapiens lineages — people who probably left Africa around 120,000 years ago, long before those later mutations swept through the main population.

Second, and this is the big one — once the branches of *Homo sapiens* spread out and became isolated, they landed in radically different ecological and climatic realities, each with its own evolutionary rules.

Take height. Joel Hirschhorn analyzed environmental influences on stature and concluded that natural selection itself nudged Southern Europeans toward being shorter on average than their Northern neighbors. Similar selective pressures shaped genetic variants influencing larger neonatal head size and a corresponding widening of the maternal pelvis — anatomical negotiations absolutely necessary for squeezing ever-bigger baby brains into the world.

So yes, evolution kept tinkering, sculpting populations in different ways depending on where they lived, what they ate, and what threats they had to outsmart.

And, unsurprisingly, those divergent branches eventually gave rise to vastly different cultures, technologies, and historical trajectories. Some societies went on to develop writing systems, cities, navigation, astronomy, and later — telescopes, caravels, and rockets. Others remained small-scale hunter-gatherer groups well into modern time. We’ll unpack that whole saga properly in the next two chapters, where civilizations finally take center stage.

The Wild Life of Ice-Age Europe

In the past few years, researchers have sequenced and analyzed more than fifty complete genomes of the earliest Europeans — people who lived between 45,000 and 7,000 years ago. This timespan includes everything: the brutal deep-freeze of the Last Glacial Maximum (from about 25,000 to 19,000 years ago), which pushed humans into the southern corners of the continent, and the slow, stubborn return northward that followed once the ice finally loosened its grip.

And here’s where things get beautifully weird. As *Homo sapiens* evolved, our brains swelled to such luxurious proportions that our children became absolutely helpless for a good five or six years. (Imagine a baby horse standing up and running within hours of birth — and then imagine our species, whose kids struggle to coordinate their limbs for half a decade.)

This evolutionary extravagance had enormous social consequences: it created grandmothers. Once brains got big and childhood stretched into a long apprenticeship, someone had to watch the little ones while the mothers kept the species going. And evolution, being the ultimate pragmatist, doesn't care *who* babysits — it only cares that the babysitting works. A post-reproductive woman who devoted herself to helping raise her grandchildren actually ended up leaving more descendants than one who kept having babies of her own and then competed with her daughters for food and care.

Tribes that failed to capitalize on this brilliant loophole? They vanished. Natural selection has no patience for missed opportunities.

And grandmothers weren't the only ones sticking around. There were grandfathers too — carriers of memories and techniques that stretched beyond a single lifespan. These older men and women could *teach* across generations. Grandfathers, grandkids, and everyone in between were likely already talking to one another much as you and I are doing now. A long childhood meant ample time to listen, mimic, practice, fail, and try again.

Meanwhile, the adults — freed from constant infant care — hunted for real. Not “wave a stick at a rabbit” hunting, but coordinated, high-risk expeditions after massive Ice Age animals that no one could bring down alone. Thanks to new tools and new cooperative strategies, they hauled back serious quantities of meat, and that steady protein bonanza kept fueling the expansion and refinement of the human brain.

The Rise of Gods and the Invention of Speech

New hunting strategies and new child-rearing habits — the direct outcome of several key mutations in the human brain — didn't just change what was on the menu. They changed **how people talked, thought, and obeyed**. Somewhere in this cognitive makeover, true articulated language appeared, along with something equally transformative: the instinctive submission to a leader — the seasoned hunter who alone knew *what* to hunt, *when* to strike, and *how* to keep the tribe alive another season.

He settled disputes, handed down rules, divided the kill. And then — inevitably — he died.

Maybe on a hunt, maybe from infection in a dark cave, maybe simply of age. But his death created a vacuum, and the new leader needed a way to anchor himself to the old authority. The cleanest solution? Create a story in which the old chief wasn't really gone — he had simply crossed into another world, a brighter and kinder one, from which he continued to watch over his hunters.

The elders and the artists picked up the job from there. They painted the walls of caves. They reenacted rituals. And gradually the departed chief, now imagined as silently overseeing the tribe from beyond, became a god.

The deep winding chambers of painted caves turned into temples, dwellings of this invisible guardian. The artists who orchestrated the ceremonies became the first priests. If you ever

get the chance, go see Lascaux in the limestone cliffs of the Vézère Valley. One glance at those 18-thousand-year-old compositions is enough to see just how dramatically the human brain had changed since our ancestors left Africa. The walls practically hum with cognition.

And once religion appeared, it quickly became the backbone of education — a portable moral code, a software package for teaching children how to behave. Later, when writing emerged in Mesopotamia, leaders began carving the first law codes onto clay tablets and stone slabs. They outlined moral imperatives, sprinkled them with examples and amusing cautionary tales, just to make the reading a little less tedious.

Of course, in the millennia that followed, the Abrahamic religions evolved into massive financial and administrative institutions — helpful tools for secular rulers trying to manage large, mostly illiterate populations. And papal courts, unsurprisingly, often ended up caring far more about collecting tithes than about cultivating genuine virtue.

Let's Be Fair: Luther, Love, and the Invention of Personal Faith

And yet — in the spirit of fairness — not all religious history is a tale of bureaucrats in funny hats counting tithes. There was Martin Luther, after all. In the early 16th century, he stood up and declared the central Catholic dogma bankrupt — that the Church, the Pope, and a battalion of clergy were somehow indispensable intermediaries between God and ordinary people. Luther insisted that salvation came through *faith alone*, a gift granted directly by God, not by ecclesiastical paperwork. Then he went on to do the unthinkable: he translated the Bible from Latin into German, handing his congregation the keys to the kingdom without asking permission from Rome.

And as if theological revolution weren't enough, Luther also married a nun — Katharina von Bora — who went on to have six of his children. Katharina wasn't seduced by incense or liturgy; she fell for the man's mind, for his ability to lead, to persuade, to articulate an uncompromising vision of the world. Intelligent women tend to be drawn to men who can think clearly and carry others with them — and I strongly suspect that these traits evolved in men largely because women kept selecting for them. Just imagine Katharina falling in love: abandoning her vows, stepping out of the cloister, raising a family with a man who argued with the Pope for a living. It's hard not to envy the kind of mentor she had. I wish I'd had someone half as inspiring when I was a kid...

And Then There's Buddhism — A Religion Unlike Any Other

But there is one more religion I absolutely must tell you about. One unlike any other — elegant, austere, almost mathematical in its clarity. Buddhism, conceived by Prince Gautama in one of the small northern Indian kingdoms around the 6th–5th centuries BCE. There is no god here. Only the idea of awakening — *bodhi* — your own personal path out of the cycle of suffering, toward a different plane of existence: nirvana.

If you ever have the chance, carve out a few days and go see three of the world's true wonders — places that, in my opinion, embody the essence of Buddhist philosophy. Start with the small but mesmerizing Bayon temple in Cambodia, near the town of Siem Reap. It was built to commemorate King Jayavarman VII's victory over the Chams in the Battle of Tonlé Sap. The temple itself is tiny, but everywhere — faces. The walls are covered with serene stone visages, probably of Jayavarman himself. By day they gaze outward with gentle smiles; by night, in the moonlight, those same faces shift, deepen, and grow impossibly solemn. One glance is enough to root you to the ground. It feels as if the king is still there — watching, judging, waiting.

The Second Wonder: Shwedagon, the Floating Mountain of Gold

Next comes the Shwedagon Pagoda in Yangon (aka Rangoon), the one Rudyard Kipling wrote about. At night this enormous, gold-lit stupa doesn't just stand above the city — it seems to float, like some celestial ship drifting over the dark rooftops. For centuries the Burmese kept adding layer after layer of gold leaf, each generation covering the last, until Shwedagon quietly transformed into the sacred center of the world.

High above, rubies and sapphires wink from the thin, spiraling spire; small bells chime in the wind; and below, circling clockwise, moves a river of worshippers — men, women, and crimson-robed monks folding briefly into the same slow tide. You can join them too: slip into the flow and make a few rounds of *kora*, holding in your palm a ring with a corn flower colored sapphire from the *Mogok mines* — it will help you focus, settle, breathe. And little by little, your thoughts will start moving in circles as well, smoothing themselves out until you feel that quiet, weightless calm that comes from letting the mind drift without resistance.

Even the famously shaggy Boris Johnson, when he once found himself near the Shwedagon Pagoda, suddenly slipped into reciting Kipling's "Road to Mandalay" from memory — the place does that to people:

*For the wind is in the palm trees, and the temple bells they say:
"Come you back, you British soldier, come you back to Mandalay!"
Come you back to Mandalay,
Where the old Flotilla lay:
Can't you 'ear their paddles chunkin' from Rangoon to Mandalay?*

And Finally — the Summit: Borobudur

And of course, try — really try — to reach the summit: Borobudur. The temple was raised by the rulers of the Mataram kingdom, the Shailendra dynasty, on the island of Java sometime between 750 and 850 CE. It stands on a high hill in the middle of a valley, pressed in on both sides by tall, sleeping volcanoes whose shoulders occasionally breathe thin feathers of grey smoke. The ascending terraces of black, porous lava embody the sacred mountain Meru. There are eight of them in all: five squared below, three round above. Borobudur is built this way for a reason. It's a message to you — or perhaps a reminder.

You begin on the lower levels, where you're immediately swallowed by the noisy, bustling world of nearby villages: market stalls, oxen and cart wheels, cardboard-box lunches, lottery tickets, all-inclusive resorts — more or less our civilization in miniature. Then you climb higher, toward priests and kings, where the walls flicker with whimsical Indian gods wearing elephant heads, demons with bulging eyes and too many arms, maharajas astride elephants — the whole lively, slightly absurd parade of myth and power.

And on the upper terraces you finally find yourself alone with the bodhisattvas, seated quietly in their carved stone latticed bells. They have already found their path and could have stepped into nirvana long ago — but for reasons known only to them, they chose to wait. For you. A light wind from the volcanoes brings morning cold, the scents rising from the valley, and a strange, profound calm. The little people below — absorbed in their fast food, tangled in their rulers, accompanied by their petty demons — are now far beneath you. And when they stop tugging at you from every direction, you suddenly notice something simple and unsettling: you alone choose the road you take, and therefore you are responsible for everything that follows.

And at the very top... I still feel only cold and fear. There, at what seems like the center of the world, stands a sealed stupa. And I have never managed to understand what lies inside.

Chapter 4. Life in Eurasia After the Ice Let Go

This chapter will be short — almost pocket-sized — but don't let that fool you. It's important. After all, you probably want to know where the Greeks, the Romans, the Europeans — and, yes, the Americans — actually came from.

Around 15,000 years ago, the continent-wide ice sheets that had smothered Eurasia began to retreat, slowly, like a tired giant sliding off the bed. The people who survived the last glacial millennium — tucked away in the Pyrenean, Apennine, and South-Ukrainian refuges where a human could ride out the cold without freezing solid — followed the great herds north again. These were hunter-gatherers, reclaiming Western and Eastern Europe after tens of thousands of years under ice.

Genetically, these groups looked nothing alike. They had spent the entire Ice Age separated by thousands of kilometers, each little refuge sheltering its own fragment of pre-glacial Europe. Ancient genomes from the Pyrenees (modern Spain) and the Apennines (modern Italy), analyzed by Josef Lazaridis in David Reich's lab, fall neatly into the first group of hunters who fanned out across Western Europe as soon as the glaciers loosened their grip. Meanwhile, people living in the northern Black Sea region for millennia — sites like *Kostenki* — formed the second group, the hunters who surged into Southeastern Europe and spread all the way up to the Baltic Sea.

And once the glaciers withdrew, Eurasia warmed dramatically. Forests returned, grasslands expanded, and enormous herds of hooved animals poured across the continent as if someone had removed a planetary “closed for winter” sign. Little by little, life for the hunters began to improve.

Agriculture

The most consequential thing to happen after the ice finally let go wasn't mammoths trudging north or forests creeping back across the continent. It happened far away, in the warm cradle between the Tigris and the Euphrates.

Roughly 12–10 thousand years ago, humanity staged its very **first technological revolution** — the invention of agriculture. And it began with a deceptively simple insight: if you gather seeds and *don't* eat them immediately — if you stash them away and drop them into the ground later — the earth will return your restraint tenfold.

Suddenly, food was no longer a fleeting stroke of luck but a renewable promise. Abundance appeared, population surged, and the genes of those early innovators — the people who cracked this strange new code of plant-taming — spread outward like ripples across a lake, reaching all the way to the shores of the newly filled Mediterranean Sea³⁰.

Let's walk through this transformation step by step — slowly, deliberately — because it's one of the most important cascades in the entire human story:

- **New cognitive abilities emerged** in those early communities (likely sparked by fresh mutations that sharpened the machinery of the brain).
- **A brand-new concept was born: agriculture** — the realization that every seed is a tiny vault, capable of producing whole handfuls of new seeds.
- **Resources multiplied.** More grain meant more children, and more children meant more chances for those new genes and ideas to survive.
- **Those genes spread together with the technology itself.** The next generation walked outward, carrying sacks of seeds and the memory of how to coax life from the soil.

And here's the key thing: Early farmers didn't conquer anyone. Their genes flowed across continents not through war, but through competence — through better tools, better food, better ways of thinking.

Step by step, cereal crops and the know-how to grow them traveled out of Mesopotamia, across the Anatolian peninsula (modern Turkey), into the Mediterranean basin and, eventually, deeper into Europe. This happened gradually, between 9,000 and 5,000 years ago. Once the farmers settled in those new lands, they mingled peacefully with the descendants of the Western hunter-gatherer groups — that first post-glacial population. For many years, scholars believed that it was on this blended genetic foundation that all later Mediterranean civilizations were built: Greece, Rome, and by extension the great tapestry of Europe itself.

If you ever feel like diving deeper into how the genes of the first farmers moved across continents, I'd point you toward the serious works of Richard Dawkins, Tom Higham, and Ian Morris — they unpack this saga in far more detail than we can here²⁸⁻³⁰.

The upper boundary of this agricultural pastoral scene is marked, fittingly enough, by a monument that looks as though it were dropped onto the Earth by absent-minded giants: *Stonehenge*. Built about five thousand years ago by a culture we still can't quite place, it stands on the Salisbury Plain like a riddle carved out of megaliths — a communal tomb, perhaps, or a sanctuary aligned with the sky. Whatever it was, it hints at a world where farming communities had already grown powerful enough to rearrange landscapes with stones the size of delivery trucks.

Fifteen years ago, most scholars would have told you that the question of European origins was essentially settled. But as always, certainty ages quickly. New paleogenetic data has rudely overturned the old, tidy story and replaced it with a far more tangled — and frankly bloodier — one. It turns out we forgot an entire branch of early farmers: those who lived on the territory of modern Iran.

These two agricultural populations — the Mesopotamian farmers between the Tigris and Euphrates and the Iranian farmers tucked behind their own mountain wall — were split by the massive *Zagros range* and, as a result, diverged genetically in a major way. I'm telling you all this because these two forgotten communities played a dramatic, almost operatic

role in the peopling of the entire Eurasian continent. None of the original four ancestral branches — the two hunter-gatherer lineages and the two farmer lineages — survived in “pure” form. They blended. But each left its genetic fingerprint and its technological gifts on the peoples of modern Europe and India.

So, here’s what happened.

Around the same time — roughly 9 to 5 thousand years ago — while the Mesopotamian farmers were shuffling gently toward the Mediterranean with bags of barley and wheat tucked under their arms, the Iranian farmers were rehearsing an entirely different exit strategy. Several waves of them surged northward. They threaded their way between the Black and Caspian Seas, crossed the Caucasus Mountains (or skirted them along the northeastern Black Sea coast), and spilled into the endless grasslands that stretched toward the huge river called *Itil’* — known to us today as the Volga, flowing into the northern Caspian.

There, on the open steppe, they drifted east and west, gradually blending with the second group of hunter-gatherers we met earlier — the ones who had repopulated Southeastern Europe after the ice withdrew.

And then the stars aligned.

By some improbable stroke of a climate good luck, cultural serendipity, and plain old human ingenuity, these mixed communities managed — almost all at once — to **invent the wheel**, **domesticate cattle and sheep**, and, most revolutionary of all, **tame the horse**.

A trifecta that would soon rewrite the destiny of half the planet.

Pastoralism, Horses, and the Wheel

The steppe is a vast, windswept belt of grass stretching thousands of kilometers from the *Carpathians* in the west to the *Altai mountains* in the east — too dry for farming, too cold in winter for comfort, and for most of human history populated only by small, scattered groups clinging to riverbanks and seasonal streams. And then, about five thousand years ago, everything changed at once: someone invented the wheel.

With a wagon hitched to a pair of oxen, you could finally move your entire world — women, children, tents, tools, hopes — from pasture to pasture. A completely new economy and rhythm of life emerged: **nomadic herding**.

The first great culture to embody this transformation left behind no cities, no palaces, no writing — only their burials. Archaeologists call them **the Yamnaya**, a name as blunt and earthy as the stones atop their graves. It comes from the Russian *yama*, meaning *pit*, a nod to the deep burial shafts. What’s left of them today are enormous burial mounds — *kurgans* — with a warrior at the center, lying on his side, head to the west, face to the north, often accompanied by a chariot, horses, and terrifying battle axes. But geneticists have recently

given us a clearer inheritance than archaeology ever could: full-genome analyses published in *Nature* (issue 522, 2015) show, with impeccable precision, that **these people were the primary ancestors of nearly all modern Europeans**.

Genetically, the Yamnaya were a near-equal blend of Iranian farmers and Eastern European hunter-gatherers — a fusion of two worlds, hardened by the demands of the steppe.

They weren't just herders, either. Evidence suggests they had a surprisingly advanced metallurgy scene and traded metals with the Sumerian cities to the south. We even know they brought copper and tin to *Uruk* — the ingredients desperately needed there to forge bronze. That iconic alloy of antiquity — copper plus tin, in a ratio of roughly ten to one — was invented in the first Mesopotamian cities (We'll get to them in the next chapter).

Their *battle axe* was a statement as much as a weapon: a heavy copper blade with a drilled socket for a wooden handle. A man received one when he entered the warrior class, and it remained with him until death. No gifting it, no passing it down. In the kurgans, these axes rest beside their owners like stern punctuation marks. Arrowheads are found too, hinting that the Yamnaya were archers as well, but nothing in those graves mattered as much as the horses.

Every dated burial from western Kazakhstan converges on the same conclusion: **steppe horses were domesticated about five thousand years ago**. From that moment on, the world tilts. Imagine it: two horses harnessed to a light war chariot with a pair of drivers. They could move faster than any human had ever moved across land — quicker than a sprint, smoother than a gallop. It was speed as a weapon, speed as a birthright.

And with that speed, the Yamnaya surged across the territories of the European farmers — from Kazakhstan to as far as Gibraltar — with astonishing ease. The expansion was almost entirely bloodless: peasants could hardly mount a defense, and even millennia later an infantry line could barely withstand a steppe cavalry charge. The chariots, light as the wind and twice as swift, carried their riders ever farther into the continent.

By the beginning of the second millennium BCE, the vanguard of the Yamnaya had already touched the *Rhine* in the west and the *Altai mountains* in the east — an arc of expansion so vast, it reads like mythology, but is now confirmed bone by bone, genome by genome.

Material traces of that era are modest. Only in Anatolia, at *Alaca Höyük* in modern-day Turkey, do we find deep shaft graves fit for kings — the closest thing the Yamnaya world had to monumental architecture. Everywhere else, the steppe people left a different calling card: *Corded Ware*. Across soft clay, still wet from shaping, someone dragged a twisted string or simple rope, leaving an unmistakable pattern. Five thousand years ago these vessels appeared from the Swiss Alps to England and France, all the way to the forest edges of western Russia — the archaeological shadow of a migration so immense it reshaped a continent.

And paleogenetic leaves no room for ambiguity: people buried with *Corded Ware* in what is now Germany were, on average, **75% Yamnaya** and only **25% descendants of earlier European farmers**.

Further south, on the Indian subcontinent, the same story unfolded — but written in sharper strokes. The Yamnaya people push into South Asia permanently altered the genetic landscape of northern India. Today, individuals belonging to the two highest *varnas* — Brahmins (priests) and Kshatriyas (warriors) — carry a disproportionately large share of steppe ancestry. But that story is too large to unpack right now. Let's just say this: the newcomers erased the ancient *Indus Valley Civilization*, including the thriving cities of *Harappa* and *Mohenjo-daro*. When archaeologists finally brushed the dust off those streets, they found them choked with the bones of people who had not died peacefully.

And what of the gods these steppe conquerors worshipped? What songs did they sing, what clothes did they wear? We don't know. But linguists are confident about one thing: they — which is to say, “we,” just a hundred-odd generations removed — spoke Proto-Indo-European, the ancestral tongue of nearly every European language, as well as Persian and Sanskrit.

Curious what a Yamnaya warrior looked like? Take a few steps toward the nearest mirror.

And of course, you understand that when I write “Iranian farmers,” “the farmers of the Tigris–Euphrates basin,” or refer to the Yamnaya who emerged in what is today western Kazakhstan, it does **not** imply that modern populations of Iran, Iraq, or Kazakhstan are “the ancestors of Europeans,” or have anything to do with Romans, Greeks, or contemporary Europeans. Entirely different peoples live there now — history has turned the wheel many times since then.

The Dark Side of Domestic Animals

You might wonder how a relatively small population of Yamnaya steppe herders — even armed with their terrifying new technologies of horses and chariots — managed to genetically overwrite nearly three-quarters of the dense farming societies stretching from today's Russia all the way to England. British rule in India or the Mongol domination of the subcontinent lasted for centuries and left scarcely a fingerprint in the genomes of modern Indians. So why were the Yamnaya different?

Because they brought with them something far deadlier than cavalry, metallurgy, or wheels. They carried **their livestock — and the diseases that came with them**.

The Yamnaya were the first humans to undergo full-scale exposure to the microbial horror show that comes bundled with cattle, sheep, goats, and horses: smallpox, for starters. Only those herder populations whose genomes happened to contain protective variants survived these ancient, slow-rolling pandemics. Over thousands of years, natural selection sculpted their immune systems into something hard, wiry, and battle-tested.

And if you peer closely into the human genome today, you can still see the scars. Viral shrapnel is embedded everywhere: scrambled fragments of ancient pathogens, duplicated segments, viral sequences grafted into our chromosomes and repurposed in desperate acts of evolutionary survival. Our DNA is essentially a war diary — annotated by viruses.

Take *Yersinia pestis*, the bacterium behind bubonic plague. Its traces appear again and again in ancient Yamnaya DNA. Steppe peoples were clearly exposed — often — and those who lived long enough to pass on their genes carried resistance with them. When they poured into Europe, they also brought their entire microbial entourage. The local farming populations had no immunity³¹.

And the nightmare didn't end quickly. These zoonotic pandemics haunted humanity for thousands of years. As recently as the late Middle Ages, the plague was still sweeping entire cities off the map.

Consider the example everyone learns in school — though usually stripped of its evolutionary context. **The Black Death.** Bubonic plague entered China around 1345 with caravans arriving from the steppe. By 1347 it had hit Constantinople, and from there it rode the grain ships to Genoa and Venice. Within a year it was everywhere. In just five years, the Black Death killed at least **one-third** of Europe and China — roughly 25 million people in Europe alone.

Call it what you want — natural selection, blind chance, or the whims of microbes — but sometimes evolution plays by rules that are brutal and deeply unfair.

Milk

It's worth pausing here to ask a simple question: how did the whole business of domesticating animals even begin? Most likely, the Yamnaya started doing it for the most practical reason imaginable — so they wouldn't have to sprint after their dinner across the endless steppe. Far more civilized, surely, to sit on a horse, whistle for the dogs, and keep your herds moving safely away from wolves. What to do with milk, however, they figured out only later.

The trouble was simple: they couldn't drink it.

Early herders — just like almost all humans back then — lacked lactase, the enzyme that breaks down lactose (milk sugar) into glucose and galactose. Without lactase, fresh milk punishes the digestive tract with admirable enthusiasm. So, the Yamnaya people weren't sipping milk by the fireside. Not yet.

But somewhere — in one small group, one family line — a mutation appeared. A gene that kept lactase active throughout adulthood. And the people who carried it instantly gained a tremendous advantage. Milk is portable, storable, calorie-rich... practically liquid survival.

The pressure of natural selection was so strong that today almost every European carries this lactase-persistence allele.

But now let's say the same thing again — only this time in the harder, colder language of fate. Across the Mediterranean world and the great sweep of the Eurasian steppe, **only those who carried this new gene — or chose their lovers wisely among its carriers — endured.** They weathered the famines, the long winters, the seasons when herds thinned and grain stores failed. Milk kept their children alive when nothing else would. As for the others... they slipped quietly out of history. Their babies could not drink the one food that might have saved them, and in the ruthless arithmetic of lean years, they simply never made it through.

Hold on to this truth, because it governs everything alive: **natural selection is not a metaphor. It is an axe blade.** It cuts through humans, dogs, butterflies — through every lineage under the sun — sparing only those whose bodies and brains happen to carry the right variations at the right moment.

And whether that moment is a drought, a plague, a winter storm, or a cup of warm milk... the rule is always the same: those who can endure, endure. Those who cannot, disappear.

What Remains When Everything Else Falls Away

It has always been this way. In every generation that came before, a man was expected to stand for his own.

It began long before memory, back in those dim ages when something in the skulls of australopithecines, wandering the African savannas, quietly began to grow. Their children were becoming astonishingly clever, and at the same time utterly helpless. They needed years in their mothers' arms before they could stand on their own feet. And a mother, her hands filled with a child, could no longer gather enough food, could not run as fast, could not always fight.

So, the balance shifted. What she could not do, someone else had to. The man she chose had to bring food. He had to stand between her children and the teeth in the dark. If he failed, the story ended there. Hunger, or claws, or both. And with that, his line would vanish, quietly erased from what came next. No cowards, no idlers carried forward. Only those who could provide, who could build, who could endure. And alongside them, the women who could see clearly, who could measure a man in a glance and choose well.

Natural selection. The old, merciless arithmetic.

This is how the human mind was shaped. Its motives. Its stubborn will. Not in a moment, but over millions of years, slowly, patiently, without ever asking permission.

And yet, look around. It has not stopped. The predators are mostly gone, yes. But something else has taken their place. We have turned, as a species, toward one another. The pressure

remains, only the rules have shifted. The selection continues, quieter perhaps, but no less relentless.

There are simply too many of us now. So, remember this, if nothing else stays with you: your women and your children are the only things that truly matter.

Chapter 5. Civilizations

Don't worry — there will be no tedious droning about dynasties or who stabbed whom behind which ziggurat. None of that. Bring your brightest paints and a clean canvas — we're starting our fourth picture. This time, we're going to sketch the essential gifts left to us by the great civilizations of the past.

But first — as René Descartes would insist — let's agree on our terms. What exactly do we mean when we say *civilization*?

- **First, it's the sum of technologies — collective brains working in sync.** If we've learned how to harness horses to chariots and thunder from Asia to Gibraltar, flattening everything in our path... if we've mastered bronze and forged weapons sharp enough to claim every city between the Tigris and Euphrates... then we are no longer just a wandering mob of well-fed raiders. No: we're the emissaries of an advanced civilization.
- **Second, civilization is the collective will of its people.** Will we fight to the last breath alongside Emperor Constantine XI Palaiologos, dying on the walls as we defend our city against the Ottomans? Or will we abandon these lands and disappear into the winds of history? (Yes, I'm talking about the final fall of Constantinople on May 29, 1453 — a date you really should remember, if only because it's one of history's most cinematic endings.)
- **And finally — civilization is a moral imperative.** Is there one God for all, whom we may address directly, as Martin Luther insisted? Or does the sky above us swarm with a whole pantheon of gods — dazzling, remote, and accessible only to a priesthood with excellent headgear? Do we live in a society where all are equal before the law, as modern Europe and the U.S. claim? Or is it more like classical Athens, where citizens were equal... but were also allowed to own slaves, who were — awkward detail — about a third of the population? Well, America didn't start out as a land of pure equality either. You've probably heard that Thomas Jefferson — yes, the man who penned the *Declaration of Independence* — owned a few dozen slaves. That's not a rumor. It's a receipt.

Every true civilization leaves behind monuments that punch right through time itself — the kind that make you stand still and remember you're very small: the Pantheon in Rome, the shimmering dome of Hagia Sophia in Constantinople, the pyramids of Cairo rising out of the desert like some geological hallucination.

Civilizations also leave their texts — the memory banks of humanity. Greek manuscripts survived even the inferno that destroyed the Library of Alexandria; monks copied them by hand for centuries, translating them into Latin, then into Persian, passing them along like a sacred flame.

The Romans, utterly dazzled by Greek culture, simply took the entire Greek pantheon wholesale and reissued it under Latin brand names. (A very early example of successful rebranding.)

And of course, civilizations leave behind something subtler than stone or parchment: **their genes** — dispersed quietly across continents, faint at first, but still visible in living populations thousands of years later.

Ethics, law, technology — that triad is what shapes the radius of a civilization's influence and the reach of its descendants. Not the size of its armies or the number of gold statues it left behind, but the imprint it made on the minds and bodies of the humans who followed.

Here is where the real trouble begins. Talking openly about genes — and their very direct relationship to how the brain works — has somehow become impolite in this short, itchy era of the “new ethics.” Yet the implication is painfully clear: **people are not equal in their abilities, their motivation, or their potential.**

I've heard those charming schoolroom tales you're told — the ones insisting that all children are born the same, and that if you took a little Aboriginal boy from the Australian outback and placed him in a good adoptive family, he might grow up to be a professor of mathematics. By now you understand this isn't how things actually work. It won't happen.

And so, to cover up this uncomfortable truth — that people differ, and that professors do not sprout from every seedling — someone dusted off an old linguistic trick. Go look at Orwell's *newspeak* when you get a moment. Today, that same newspeak goes by a far friendlier name: *political correctness*.

You've probably noticed the quiet swap inside DEI, the moment when *equality* — a clear idea about equal rights and equal starting chances — was rubbed out and replaced by *equity*, a fuzzy notion of “fairness” that no one seems able to define.

Under equality, a math professor and a garbage-truck driver had the same childhood opportunity to try for a university chair. One tried and succeeded; the other chose a different life. That was the whole point.

But under equity? Now the story goes like this: regardless of what each person achieved or failed to achieve, they are somehow owed roughly the same annual income and similar houses — adult outcomes forcibly flattened, as if history, effort, talent, circumstance, and sheer human variability never existed at all.

I've seen all of this before, long ago. As a child, I didn't realize I was living in the most “just” society on earth — that socialism, allegedly, had already arrived. My great-grandmother Apollinaria and my grandmother Serafima never mentioned it. Perhaps because they were the wife and daughter of a Moscow merchant-industrialist, a man who spent his entire life crafting precision instruments in his workshops on the old Abelmanovka street.

What I *do* remember are those slow, honey-thick summer evenings at our dacha in the Moscow suburb Kholshcheviky. The huge table draped in a linen tablecloth that smelled faintly of dust and flax. The tea that arrived with éclairs or a slice of Napoleon cake. And

then, once darkness settled over the garden, someone would read me fairy tales or teach me old merchant games.

One of those games was about the shrewd traders of Sukharevsky Market in old merchant Moscow. It always began with three iron rules for beginners:

- “Don’t say yes or no!”
- “Don’t name black or white!”
- “Evade every question!”

Apparently, this was how crafty dealers survived — and managed to move their wares — in a market where half the goods had “fallen off” someone else’s wagon.

I must have dozed off at that table, because in my memory the Mad Hatter and the March Hare from Tim Burton’s *Alice in Wonderland* wander straight into the kitchen where my grandmothers sit pouring tea. And the mad tea party never ended — it simply spilled into real life and has been going on, shamelessly, for the last thirty years. The three Sukharevsky rules glow on the wall like commandments written in fire. And words themselves wear thin like old copper coins — their meanings drifting, mutating, vanishing.

And I know — I *know* — all this nonsense is just a dream; the market itself was razed long ago. Yet sometimes I swear I’m walking along those vanished stalls, piled high with objects stolen from good Moscow homes. The scent of damp fur, from coats torn off the backs of people from that other, vanished world, hangs in the air like a ghost.

Well... social revolution. What can you do.

Activists of the “*new ethics*” now insist that the endless waves of people crossing our southern border illegally must be called *undocumented*, never *illegal*. And the vagrants and panhandlers who have overrun the once-enchanted city of San Francisco — nesting in colonies that look and smell uncannily like sea-lion rookeries along the Pacific coast — must now be described not as beggars, but as *experiencing homelessness*.

Apparently, these people must have had childhood traumas, they say — a *history of family abuse*, or whatever the fashionable phrase of the week is — and so they are *temporarily* living on the street. Without the trauma, of course, they would all be teaching mathematics somewhere by now.

Let me share with you a terrible secret — one you cannot repeat to the uninitiated, one you must never utter aloud, like the name of Lord Voldemort: people who refuse to work and end up living and dying on the street do *not* arrive there by accident. This *is* natural selection. Just like the milk example from the previous chapter.

Sure, we can keep contorting ourselves and inventing ever softer, more compassionate euphemisms to describe selection at work in human society — but we cannot abolish it. No decree, no slogan, no committee on inclusive language will ever make it go away...

Of course, I know that talking about natural selection, about races and nations, about peoples and civilizations, has become downright dangerous. The moment you open your mouth, you're branded a racist, a fascist, and some sort of horned demon who stole the childhood of a fragile girl named Greta... Which is precisely why you and I have to be careful — we have to choose our words with the precision of a watchmaker. Without the right words, we'll never understand why our society has quietly stratified into layers of very different people who, for reasons anyone can see but few dare acknowledge, barely mix at all.

Pretend the layers don't exist, insist everyone is the same, refuse to name what is plainly in front of you — fine. But then we'll keep trudging in circles like a donkey turning a millstone, ending each day chewing the same old hay in the same old stall.

So, to find the right words — or at least the words that won't get us burned at the digital stake — let's shift our attention to the supporting cast. Not to people, but to the beings who have lived beside us for thousands of years: the warm, lumbering cows, the stubborn sheep, the dogs who guard and guide them.

Where did *they* come from, anyway?

To get a closer look, let's pull out a piece of cardboard and sketch... dogs. Yes, dogs. A quick little study of canine life. They're simple, right? Loyal, furry, and blessedly free of the uncomfortable questions we've been instructed to avoid — the ones where you're not supposed to say "yes" or "no," or mention anything "black" or "white."

A safe place to start. Or at least safer than most.

So then — all dogs began as wolves. Real wolves. The ruthless, elegant predators you can still see pacing behind the glass at the zoo, their paws silent on concrete, their amber eyes measuring you as if deciding whether you're edible. Every dog alive today — from your neighbor's Pomeranian to the sheepdogs herding flocks in the Caucasus — traces its lineage back to those ice-age hunters.

And the astonishing part? The transformation from wild wolf to loyal dog took a mere twenty or thirty thousand years — a blink of an eye in evolutionary time, the kind of fleeting moment nature normally reserves for adjusting the wingspan of a moth.

It happened almost accidentally. Snow kept falling. Another age of cold crept down from the north, the pale ribbon of the glacier already looming at the horizon like a dull blade. For decades the wolves' hunting grounds shrank: prey migrated south or simply died. Hunger pressed into their ribs. Their world was narrowing.

Meanwhile, at temporary camps of early humans — let's call them *Cro-Magnons* — things were going relatively well. They hunted mammoths and giant deer; near their huts lay generous piles of bones and scraps. Approaching a human settlement was, of course,

terrifying. Humans had spears. They had arrows. They had fire — and no hesitation at all about killing anything that moved.

But in the worst winters, starving she-wolves crept toward the refuse heaps anyway. They gnawed the frozen bones and — against all odds — survived. Naturally, *only the boldest, most curious, and least aggressive* dared to come close. And these, over time, began lingering at the edges of the camps, picking up not only food but also a kind of unintentional partnership: they barked at danger. They kept watch.

Soon enough, the children of the camp were playing with wolf pups.

A hundred generations later, the wolves were chasing prey *for* the humans, not competing *against* them. Hunters kept the pups who were not just affectionate but clever — the ones who understood, somehow, what the humans needed. Gradually, almost imperceptibly, wolves slipped across a boundary and became dogs.

You ask, what happened to the others — the wolves that refused this strange alliance, and the early proto-dogs who didn't quite measure up?

Well... the wolves that stayed away remained wolves, ghosts of the forest, and still are. And the dogs who were too mean, too dim, too unpredictable? Humans simply didn't let them breed. *They faded away.*

Those are the two key words here: selection and offspring

Did hunters keep only the sweetest, smartest pups? Of course not. Their criteria were as varied as their needs: bravery, endurance, sharp senses, the willingness to work with humans, or even a particular talent for hunting a specific kind of prey.

And in the very recent past — practically yesterday on the evolutionary clock — Europeans refined this process with almost obsessive precision. Dogs were selected for distinct physical traits, temperaments, and specialized abilities. That's how we got everything from dachshunds and spaniels to Great Danes and German shepherds. It took not millennia, not even centuries upon centuries — but a couple of hundred years. Two. Hundred. Years. Less time than most cathedrals have been standing.

A *breed*, in this context, simply means a deliberately narrowed population with inherited traits: quick minds, fierce dispositions, short legs, long hair, whatever humans found useful or delightful. We repeated this trick again and again with other animals: from wild buffalo we sculpted dairy and beef cattle; from squat little wild ponies we coaxed magnificent horses built for speed³².

Naturally, not everything worked. Take zebras, for instance — nature's beautifully striped troublemakers. We tried. They declined. Bad temper, unpredictable kicks, zero cooperation. Well, fine. Let them keep their stripes and their opinions.

The words *breed* and *selection* do a remarkable job of describing how fast and how powerfully within-species sorting works among domesticated animals like dogs and horses. They tell us why one animal turns out a spaniel and another a shepherd — both stable, recognizable populations shaped by human choice, each carrying its own distinctive suite of inherited traits.

But, of course, those same words sit awkwardly when we try to apply them to humans. We hesitate — not because the underlying evolutionary process is fundamentally different, but because our language for talking about ourselves has become emotionally loaded, politically radioactive. Still, the mechanism is the same: variation, selection, inheritance, repetition.

So, for people — for our long story of migrations and interwoven destinies — let us choose a gentler, more graceful word: *culture*.

Take the *Gravettian culture*, for instance — the archaeological signature of late Paleolithic Cro-Magnons. We date it from roughly 28,000 to 21,000 BCE, and we name it after the cave of La Gravette in France. We know, with some confidence, that it replaced the earlier *Aurignacian culture* in Western Europe when new groups arrived from the Near East. In plain language: *new people with different skills, different tools, and different minds appeared — and the old culture vanished beneath the new one.*

So, let's agree on a convention. Whenever we speak of cultural layers replacing one another — or merging, or erasing, or transforming — we can simply call the incoming group *new people*: people with better tools, sharper cognitive abilities, or more effective social systems. It is an honest, neutral way to talk about human evolution and within-species selection without stepping on conceptual landmines.

Agreed?

Good. Then now is the moment to pick up the fine, delicate brushes — the sort painters use for the smallest strokes — and begin sketching the world around our “*new people*.” Let's trace the outlines of continents, the curving edges of seas, the great shoulders of mountain ranges. And on this map, we'll place every major civilization we know: the ones that rose, thrived, collided, collapsed, and left us inventions and ideas so powerful that we still carry them today.

Some of their cities are still standing. Some of their temples still cast shadows. You just need to know which country to travel to — and where to look once you're there.

Ready? Let's begin.

Mesopotamia

This was the first truly urban civilization on Earth. By 3000 BCE, the earliest cities had already risen from the dust — names that still hum with antiquity today: Eridu, Uruk, Nippur, Kish, Nineveh, and later the mighty Babylon. Five thousand years ago, here, for the

first time in human history, people learned to smelt bronze, to hitch oxen to solid-wheeled carts, to shape clay on a potter's wheel, and — perhaps the most revolutionary of all — to write.

The land between the Tigris and Euphrates, laced with a breathtakingly intricate web of irrigation canals, was known to its neighbors as Sumer. And what remains of Sumer today is scattered like breadcrumbs across time: the stern statues of priests and priest-kings (and priest-queens), the *lugal's* who governed city-states; thousands of clay tablets; law codes carved into stone stelae; and, most spectacularly, the fragments of the Ishtar Gate from Babylon³⁴.

If you ever find yourself in slightly unkempt, rumbling Istanbul — the city that was once Emperor Constantine's glorious Constantinople — step into the archaeological museum tucked beside Hagia Sophia. There you'll find these miracles. And look closely: somewhere among the lions, bulls, and glazed bricks hides a small statue of a smiling man in a skirt.

That's one of the *lugals* —  in cuneiform. The word itself is just two signs pressed into wet clay: *great* + *man*. Which, frankly, tells you everything you need to know about what his job description must have been.

Egypt

Egyptian civilization inherited the plants and animals first domesticated in Sumer, and by around 4000 BCE small farming settlements began appearing — first in the Nile Delta, then slowly creeping upstream along the long green corridor of Upper Egypt. The Nile Valley is a ribbon of perpetual oasis running through the desert, and its annual summer floods fed a brilliant civilization for thousands of years, right up until the arrival of the Arabs.

Egypt became the beating heart of the world's trade routes. Imagine this: when nomads drifted past farming villages, or when forest hunters encountered pastoralists at the edge of the steppe, they traded whatever extra they had — grain, hides, smoked meat, fish, furs — for whatever might be useful. This was still simple barter.

But trade, real trade, began when nomads started acquiring goods *not* for themselves but to carry to far-off markets and fairs, where craft guilds were busy forging blades, shaping axes, or mining copper and salt. Soon there were merchants — entire families or tribes who made their living roaming between marketplaces. By the early second millennium BCE, caravans, sailing ships, and oared galleys tied together the civilizations of Crete, Egypt, Mesopotamia, and the Indus Valley.

And another five hundred years later, the trade web stretched even farther: bronze-traders from Spain, steppe nomads with their herds and chariots, and Achaean kings from ancient Mycenae selling wine and olive oil — all firmly linked Egypt to Europe³⁵.

The goods themselves were varied — furs and amber, emery stone and gemstones — but the trade system *ran* on weapons-grade metals: copper and, most importantly, tin. Tin came from the northern slopes of the Alps and from the far south of what is now England. It was smelted on site and shipped south in heavy ingots toward Egypt, often with metal tools and expensive weapons in tow. Many such items were sold not by merchants but by the smiths themselves — wandering metalworkers with soot on their hands and whole civilizations depending on their skill.

After the long wars in the Levant, the Mediterranean blossomed with trading cities — Byblos, Tyre, Sidon — and Egypt experienced a spectacular golden age in the 16th–13th centuries BCE, under the descendants of the great Ahmose. You remember their names: Thutmose, Amenhotep, Akhenaten, and his luminous queen Nefertiti. The markets of that era seemed bottomless, an endless appetite for everything crafted, carved, woven, or forged.

You can still see the fruits of that ancient prosperity. If it were up to me, I'd begin a journey along the Nile in Abu Simbel, at the monumental temple of Ramses II, carved around 1300 BCE. Back then the region was called Nubia. From there, drift north to Aswan, to the temples of Kom Ombo, and then onward to Luxor, the Valley of the Kings and Queens, and finally Cairo, with its pyramids crouching at the edge of the desert.

I've always had a fondness for the modest *Bent Pyramid* — the one where ancient architects practiced the art of building a true stone mountain. The burial chamber beneath it was never used, and you can still climb down the long-slanted shaft into its silent core, provided you don't mind a narrow descent into darkness.

In the old Egyptian Museum in Cairo — assembled by Auguste Mariette — you'll find nearly everything you need to understand the civilization of Upper and Lower Egypt. One of its newest treasures is a terracotta head of a queen, unearthed from the ashes of a workshop before it was ever painted. It was found only recently³⁶.

If your mother hasn't misplaced it, ask her for the small bust of Nefertiti. After three days wandering the museum in a kind of trance, I commissioned a jeweler to recreate that face as a pendant. I didn't believe it would work, but he managed it: a smear of enamel over gold, capturing the curve of her profile.

Hold that pendant in your palm and there is nothing left to explain about Egypt. The queen's face is a fusion of will, serenity, and power. She looks at you — and past you, toward the mountain on the west bank of the Nile opposite Luxor, where the dead reside. Her tall crown flows seamlessly into a black-gray pearl from the Southern Ocean. She was only a little over forty in this world.

China

All the early Chinese kingdoms — and with them Burma, the Khmer lands, and Siam — were built by people whose ancestors stepped out of Africa roughly 50,000 years ago.

Remember that small band of hunters who first discovered human speech? They skirted India along the ocean shore and then followed the great rivers inland — the Brahmaputra, the Irrawaddy, the Salween, the Mekong — moving northward into the body of the continent.

Chinese civilization first took shape in the fertile country between the Wei and the Yangtze, and from there it unfurled southward like an ink wash spreading across silk. In the relatively cool valley of the Huang He, people began cultivating millet about 9,500 years ago. Later, in the warmer southern lands, soybeans appeared — and then rice, the grain that would become the empire's heartbeat. By the 5th century BCE, Chinese metalworkers had learned to build true blast furnaces capable of smelting iron on a scale other cultures would not achieve for centuries³⁷.

The fractured kingdoms of early China were first united around 250 BCE by the triumphant Qin dynasty, and their successors — the Han — began assembling the Great Wall. The empire brought peace and prosperity to a territory so vast that it took months to cross. More than 1,500 years ago, Chinese artisans unlocked the secrets of porcelain of impossible delicacy, and of silk spun from the cocoons of mulberry-fed caterpillars — materials so exquisite that they seemed to belong to another world entirely.

By the early 2nd century CE, the Roman Empire and the Han Empire had become unlikely twins separated by deserts and mountain chains. Each held roughly 50 million people. Each controlled about five million square kilometers. Rome built roads and ships; China carved canals through its landscape. The Grand Canal, completed in 609 CE, stitched together the rice fields of the south with the cities and armies of the north — a man-made Mediterranean, a blue artery of civilization.

Both empires fortified their borders against the barbarians who pressed in from all sides. Yet even the strongest dynasties suffered through intervals of famine and violent uprisings. Under the Tang dynasty, China entered a radiant golden age of poetry, painting, and scholarship — a cultural brilliance that dazzles even now. But a century later the empire faltered. Expenses swallowed revenues, rebellions erupted, and the An' Lushan uprising tore through the land. Famine, terror, and war consumed fifty million lives — more than the entire population of Europe at the time³⁸⁻³⁹.

If you ever travel there, begin your journey in Xi'an, the ancient capital and the starting point of the Silk Road. In earlier centuries it was called Chang'an, and by the year 700 it held a million souls — a metropolis glowing with temples, markets, scholars, and soldiers. The Big and Small Wild Goose Pagodas rise there still, serene against the sky, and just outside the city lies the mausoleum of Qin Shi Huangdi, guarded by his silent terracotta army, each soldier modeled with an individuality that feels almost uncanny.

And since you'll be in that part of the world, you might as well see beautiful Burma too. Book a small cabin aboard one of the *Pandaw* riverboats — just a dozen passengers, a polished crew, and a cook who knows exactly what he's doing — and follow the Irrawaddy

upstream from Bagan to Katha, the very town where George Orwell once served. The river winds through landscapes so still and luminous that they feel half-imagined, as though you've slipped briefly into an older world.

Greece

In the development of philosophy, mathematics, and the technologies of the mind, Ancient Greece leapt far ahead of Crete and Egypt. The idea that the universe might hold *countless inhabited worlds* traces all the way back to Anaximander of Miletus, who, in the 6th century BCE, suggested something astonishing: that Earth floats freely in space, supported by nothing, anchored to nothing — simply suspended in the cosmic dark. It is one of those bold thoughts that seems almost modern, a spark that somehow leapt across millennia.

But brilliance alone doesn't guarantee an audience. History is cruel to its pioneers. How many musicians, inventors, and painters have sprinted so far ahead of their age that no one could follow — or even understand — until long after they were gone? A perfect example is Heron of Alexandria, who in the 1st century CE built what was essentially the first steam turbine. His contemporaries dismissed it as a charming toy for the wealthy. Can you imagine? In his workshop lay the seed of the Industrial Revolution — ignored. And there were catapults, ballistae, ingenious automata... a whole mechanical universe centuries too early for its time.

If you ever find yourself in chaotic, sun-bleached Athens, there are two things you must see with your own eyes: the Parthenon and the Herodes Atticus Odeon (you can even buy tickets to a concert there — the acoustics are otherworldly). Both crown the Acropolis, that rocky plateau of gods and democracy. And do not, under any circumstances, skip the National Archaeological Museum. Somewhere inside, lit like a stage set, stands the bronze statue of the Artemision Jockey — a young boy riding a galloping horse, rescued from a shipwreck after two thousand years under the sea. The saltwater preserved every line of muscle, every ounce of motion. It is as if the horse is still trying to leap out of the waves.

But perhaps the most haunting place in Greece is far from the crowds — in the north, near Thessaloniki. There you must make the pilgrimage to the Royal Tombs at Aigai (the *Museum of the Royal Tombs at Aigai*). Excavations were completed only about twenty years ago, and the site feels astonishingly fresh, as if history were still smoldering beneath the soil. This is where Alexander the Great buried his father, Philip II, surrounding him with shimmering silver vessels taken from the villas of the Macedonian countryside and with real battlefield weapons — not ceremonial imitations, but blades and armor that had tasted war.

When Alexander was a child, Philip hired him a tutor — a certain philosopher you might have heard of: Aristotle. And as you can see, that investment paid off rather spectacularly...

Rome and Constantinople

The great empire that carried the flame of Greek culture across continents lived, in a sense, a double life. It had two capitals: Rome, set like a keystone in the center of the Apennine Peninsula, and its brilliant continuation — Constantinople, rising above the Bosphorus, the narrow strait that funnels the waters of the Black Sea into the Sea of Marmara. Rome emerged sometime in the 6th century BCE, and Constantinople finally fell in 1453. Between these dates — two thousand years — the empire held a long, astonishing conversation with the world.

During its reign, Rome gave countless peoples something they had never known at such scale: law and tranquility. Across Europe spread a level of civilization that must have felt like stepping from twilight into sunlight — roads that held firm beneath the wheels of carts for centuries, aqueducts that carried mountain water across valleys, a fully articulated legal system, a new architectural language, and long decades almost untouched by war under the shield of Rome's unshakable legions⁴⁰⁻⁴².

To imagine the Roman Empire, think of a giant living cell: a central nucleus and powerful mitochondria, wrapped in a tough membrane. The legions were that membrane — the armor that protected a vast interior, allowing knowledge, prosperity, and invention to bloom in peace. Even when battles raged at the frontiers, life inside the empire's borders remained mostly steady, predictable, and safe. After the fragmented chaos of the pre-imperial world, this stability must have felt miraculous, like the sudden quiet after a lifetime of storms.

Many of Rome's inventions and technologies still survive today — and you can go see them with your own eyes. If you rent a small apartment on an upper floor overlooking Campo de' Fiori, you can step out early in the morning and wander into the Pantheon before the crowds arrive. There it is: one of the world's first great domes, open to the sky through its perfect circular oculus, balanced on thick brick walls bound together with the world's earliest concrete. Arches, vaults, domes, smooth architectural curves — Rome was the great engineer-artist that first imagined these things at monumental scale.

No wonder Emperor Hadrian, who adored architecture the way some people adore puzzles, played with these ideas like a child with a new toy.

Just two steps away lies the ghost of Domitian's stadium, now transformed into Piazza Navona. From there you slip westward, toward the river, and cross Ponte Vittorio Emanuele. Once across, and with Hadrian's vast round mausoleum — Castel Sant'Angelo — on your right, you follow Via della Conciliazione straight toward St. Peter's Basilica, one of the most overwhelming spaces on Earth. You could spend the rest of the day inside, drifting beneath its colossal vaults, or you might simply sit with a glass of wine until night settles over Piazza di Pietra, just off the Brownian bustle of the Corso, watching passersby and the slow ballet of seagulls gliding over the colonnade of yet another of Hadrian's architectural gifts.

Rome, in the end, is not just a civilization you study — it's a civilization you *walk through*, one stone at a time.

Or perhaps, instead of climbing hills and palaces, wander downward — all the way to the riverbank. Along the edges of the Tiber, you can still spot the dark stone mouths of the Cloaca Maxima, the city's first sewer. You'll probably laugh, but this is the oldest surviving structure in the Eternal City, still functioning much as it did when Rome was barely more than a cluster of huts on seven hills.

The Cloaca wasn't originally built to carry away waste; it was meant to drain the valleys between the hills. The Tiber often spilled its banks, and after rain the small streams that braided the lowlands would wake and flow again, turning everything into swamp — and swamp meant mosquitoes, and mosquitoes meant malaria. And besides, you can't expand a city on marshland, and Rome in those days was already sliding off its hills into the low spaces.

Construction began in the late 6th century BCE, during the reign of the last three Etruscan kings — a period when a wave of new technologies washed into Rome. Later, when the Romans invented aqueducts and suddenly had to dispose of all the excess water, public baths sprang up and the first multi-seat toilets appeared. And, of course, in the uniquely Roman fashion of assigning a deity to every civic function, even the sewer had its own goddess: Venus Cloacina.

On one of your trips to Europe, do yourself a favor: stop in Istanbul for a couple of days — the city once simply called *The City*, the axis around which world politics and long-distance trade revolved for fifteen hundred years.

Here's why. In the 4th century CE, Emperor Constantine embraced Christianity and decided to move the capital of his fading empire to a small Greek town perched on the Bosphorus. You can stay in the old Boukoleon Palace, built right on the Sea of Marmara. Little of the original structure has survived, but if you take a room on the third or fourth floor of the tiny boutique hotel *GLK Mansion Suites*, whose walls rest on the ruins of the palace, the emperor's grand gallery will lie directly beneath your windows.

The impregnable Theodosian Walls, built from Roman brick, still guard Constantinople from the landward side. When a brutal earthquake in 740 toppled several towers — just as the city faced the looming threat of Attila's wild steppe armies — thousands of chariot-race fans from the Hippodrome, the Blue and Green fractions alike, joined the imperial troops to repair the shattered sections. They made it in time. And the city survived³³.

Every time I'm there, I feel an absurd urge to pick up a giant sponge soaked with warm soapy water and scrub away the grime of the gloomy Ottoman centuries that replaced brilliant Byzantium — at least from Sophia's stones and from the main road, the ancient Mese (today's Divan Yolu), which once ran from the Milion — the very heart of the empire — westward to the Hippodrome and beyond, all the way to the walls.

As for **Hagia Sophia**, her story is less a legend than a long, trembling breath of history. Step inside and the air itself seems to quiver with memory — a vast chamber of drifting light where shadows carry the whispers of Justinian and Theodora. They imagined this place not merely as a church, but as a bridge between the earthly and the celestial, a dome broad enough for heaven to rest upon it, a sanctuary where light could become architecture and architecture could become prayer.

Stand beneath that great vault and you may feel a strange sensation, as if the centuries have thinned and something ancient is softly placing its hand upon your shoulder. Gold flickers at the edges of your vision. The silence feels inhabited.

But go soon. Since its reconversion into a mosque, much of its inner radiance has been dimmed. The marble floor — once a silver-gray sea where moonlight wandered like a quiet pilgrim — is now hidden beneath woven carpets. And high in the gallery, the mosaic of *Christ Pantocrator*, that calm, watchful face assembled from pieces of eternity, is veiled whenever possible, allowed to shine only in memory.

Yet if you cannot climb to those upper mosaics, if time or circumstance keeps you from touching that golden hush — do not let sorrow settle on you for long.

For Justinian left another star burning on this earth.

Travel to Ravenna, a city gentle as a half-forgotten dream, and step into his other masterpiece: *San Vitale*. There, one entire nave is clothed in the gold of early Christianity — not a flat, obedient gold, but one that flames softly, like the inside of a chalice when sunlight pours through it. The saints along the walls stand in quiet procession, their faces tender and unguarded, as if they have been waiting for you.

In San Vitale, color becomes breath; gold becomes warmth; the centuries lean close and whisper in your ear. See it tenderly. See it slowly. Some places do not merely endure — they listen, they remember, and for a moment, they let you belong to their remembering...

The Empire of Water

The Age of Discovery began on the ragged western edge of Europe — the Iberian Peninsula, a place forever shaped by conquest and reconquest. In 711, the Umayyad Caliphate surged north across the narrow strait and seized the peninsula, and for the entire Middle Ages the land lay under Moorish rule. It took centuries of the Reconquista for the Christian kingdoms to claw their way back, inch by bloody inch, until at last, in 1492, Castile and Aragon united into what we now call Spain.

And after so many generations of war — real war, with burned fields and broken cities — the Spaniards and Portuguese emerged as utterly transformed people. The Portuguese went so far as to cross the Strait of Gibraltar and seize the port of Ceuta, turning it into a stronghold from which they launched maritime trade in gold and slaves⁴³. After hundreds of years of combat, as you might imagine, they did not look kindly upon Muslims or Africans —

and no one at the time considered the slave trade morally shocking. That judgment would come centuries later.

From Ceuta, you could sail south along the jagged coastline of West Africa, hunting for gold fields and imagining whether Africa might be circumnavigated — whether there might be a sea-route to India and, beyond it, to the mythical islands overflowing with spices (today known as Sulawesi and the Moluccas).

The first real breakthrough came from Bartolomeu Dias, who managed to round the southern tip of Africa — a violent, wind-scoured headland he named the Cape of Storms (although the Portuguese king, dreaming of trade, quietly renamed it the Cape of Good Hope). Dias shattered the ancient illusion of Ptolemy, who believed Africa stretched into an endless southern continent. Suddenly the world opened: if Africa had an end, then a ship from Europe could slip around its coastline and reach the riches of India without passing through the Muslim world. And in 1498 the unbreakable Vasco da Gama proved it.

While the Portuguese crept carefully down Africa's spine, one Genovese sailor — a stubborn man named Cristóbal Colón — begged for funds to sail in the opposite direction: west, straight into the unknown Atlantic. He was convinced that heading west would be the quickest way to reach India. Eventually the Italian adventurer won the backing of Queen Isabella of Castile, and on August 3, 1492, three small ships set sail from Palos de la Frontera, pointed toward the deep blue emptiness.

Two and a half months later, Columbus's caravels struck landfall in the Bahamas, then skirted the shores of Cuba and the island the sailors named La Española — today's Haiti and the Dominican Republic.

Think about it: the Portuguese had spent nearly a century carving a fragile path along Africa's coastline toward India. But once Columbus revealed the existence of a New World, it took only one generation for sailors to map the world's oceans and launch the first circumnavigation of the Earth. A planetary revolution began — the birth of a global economy.

If the story of seafaring captures your imagination (and it should — it's one of humanity's greatest epics), I'd recommend Lewis Dartnell and Lincoln Paine⁴³, who write about it with the clarity and wonder it deserves.

Europe

Europe is the heir to the Roman Empire and to the crystalline thought-world of Greek philosophy⁴⁴⁻⁴⁶. The old idea of infinite worlds — first whispered by Anaximander — resurfaced timidly in the writings of William of Ockham in the 14th century, only to roar back in full force during the Renaissance in the minds of Nicholas of Cusa and Giordano Bruno. For daring to claim that the universe held innumerable suns and countless inhabited

worlds, Bruno was burned alive in Rome — on Campo de' Fiori, a place I hope you already know well.

Everything Europe achieved — every bold stroke of a brush, every philosophical revelation, every attempt to describe the cosmos — now lives in the great museums: the Prado, the Louvre, the Uffizi Gallery on Florence's Piazza della Signoria, and the Musei Vaticani. If you ever find yourself wandering through Rome, add one more sanctuary of beauty to your list: the Villa Borghese, where a single cardinal with impeccable taste gathered some of the most breathtaking marble sculptures ever carved by human hands.

And then there is music — real music, the kind that only came into being after the Renaissance. The magical organ, the piano, the violin — these are the summit of centuries of craftsmanship. As for composers, you already know the essential ones by heart. Take Mozart's *Requiem* in D Minor, performed by Arslys Bourgogne, or Vitali's *Chaconne* soaring under Sarah Chang's bow with a full orchestra behind her. Once you hear them, truly hear them, they never leave you.

Europe was also the stage on which the Industrial Revolution unfolded. Steam engines fed by coal pushed ships across oceans and sent trains thundering over continents. Then came the internal combustion engine — a small, explosive miracle powered by refined petroleum — which suddenly allowed humanity to traverse vast distances at astonishing speed.

The modern era of cheap, mass-produced steel began in Europe in the mid-19th century, when the Bessemer process was invented: air blasted through molten iron to burn away carbon and other impurities, leaving pure, workable iron that could then be precisely alloyed into any grade of steel you wanted. A simple idea — yet it reshaped the world.

And of course, Europe also gave us the machine gun, devised by the British and almost immediately rendering cavalry charges obsolete. It's worth reading Niall Ferguson's excellent book on the British Empire⁴⁷ for this chapter of history — or simply revisiting that unforgettable scene in Spielberg's *War Horse*, where Benedict Cumberbatch leads a magnificent cavalry charge that dissolves, helplessly, against a line of guns.

Japan

I should confess something upfront: the history of civilization is, at its core, the history of technology — and above all, the history of weapons. Nowhere on Earth did the rules of society snap into a new configuration as abruptly as they did in Japan.

Picture the scene: the mid-16th century, a storm-battered Portuguese ship drifting toward the small island of Tanegashima. Its arrival must have looked like an omen — a sliver of Europe washed onto the southern edge of Asia. But the island's young daimyo, Tanegashima Tokitaka, only fifteen years old, was not distracted by omens. He was distracted by *technology*. The Portuguese carried odd-looking metal tubes that spat fire and thunder.

Arquebuses. And the boy understood, instantly and intuitively, what entire generations of samurai would refuse to admit: a new age had arrived.

The moment his smiths learned to cast barrels and mix powder — a process they mastered with almost suspicious speed — the elegant hierarchy of the samurai world collapsed. Spears and katanas, weapons refined over centuries, became quaint ornaments in the face of bullets. Armor that once embodied honor and lineage folded like paper under a single shot. Entire cavalry lines could be shredded from a safe distance by troops who barely knew which end of the gun to hold. The transformation wasn't philosophical or spiritual; it was brutally mechanical. Metal defeated muscle. Chemistry defeated tradition.

If you want to see this turning point crystallized into a single morning, look up a short video about the *Battle of Nagashino* — June 28, 1575. There, the warlords Oda Nobunaga and Tokugawa Ieyasu demonstrated the full, terrifying power of massed gunfire. Using carefully arranged ranks of arquebusiers protected behind wooden palisades, they annihilated the charging cavalry of the Takeda clan. Centuries of samurai warfare ended before lunch. And from the smoking ruins of that battlefield emerged the political unification of Japan under a new military order.

And then, of course, came the *second* time foreign technology rewrote Japan's destiny — Hiroshima and Nagasaki — when, in two blinding flashes, an ancient warrior culture met the final, terrible expression of the modern age.

Two Models of How Societies Develop

You may be surprised, but there are really only two models of societal development. Just two. And you don't need to memorize all the tedious footnotes of the Meiji Restoration in Japan or dissect the gloomy biography of the last, not-very-bright Russian tsar in search of the "true causes" of the bloody revolution that opened the previous century. Humanity already wrote down both models long ago, each crystallized in a single foundational text.

The first is the *Magna Carta Libertatum*, drafted in June 1215 — a list of demands shoved under the nose of King John by a coalition of very irritated English barons. It defended the rights and privileges of the free population of medieval England and, more importantly, planted an idea that would grow for centuries. If you want a beautifully filmed fairy tale about that moment, watch Ridley Scott's *Robin Hood*, with Russell Crowe, Cate Blanchett and the incomparable Max von Sydow. Some articles of the Charter limited royal power by creating new institutions: a *Great Council* of the realm and a committee of twenty-five barons who, remarkably, had the authority to force the king to honor his word.

The second text was born far away, on the windswept steppe: the *Yassa* — *the Law of Great Authority* — compiled by *Temujin*, known to the world as Genghis Khan. After uniting the fractious tribes of Mongolia, he was elected Great Khan in 1206, and the *Yassa* became the spine of a new imperial order.

From the *Magna Carta* grew everything we now associate with freedom: parliaments, elections, due process — eventually not just for hereditary barons, but for craftsmen, merchants, and the “small people” who had once existed only to pay taxes. In this model, the state ceased to be a sacred, fearsome, quasi-divine entity. It became a tool — a practical organization that existed *for its citizens*. The state monopolized violence, protected its borders, enforced the law, and — critically — defended the sanctity of private property. People became valuable in and of themselves, endowed with rights that expanded or contracted depending on the country and the age, but were no longer optional decorations granted by a monarch.

From the *Yassa*, however, emerged something entirely different: a brilliant, unstoppable military machine that allowed nomadic Mongols to conquer half the known world in the span of a single lifetime. China, India, Baghdad, Sogdiana, and the entire caravan route from East to West fell before them. There was only one catch: the *Yassa* made the Khan, his army, and the perpetual hunt for plunder the very purpose of the nation’s existence. A society built for conquest must keep conquering — or fracture.

If you’re curious, watch Sergei Bodrov’s excellent 2007 film *Mongol: The Rise of Genghis Khan* — a beautifully shot portrait of how an empire of horsemen began.

Now, the first path is the one Europe and America walk today. The second path has already been marched — quite enthusiastically — by the Mongols, by the Germans under Hitler, by the Japanese under Emperor Hirohito, and by the Soviet Union, which dreamed of uplifting every worker it could find between Siberia and the Atlantic. I won’t bore you with further commentary on the quirks of either model; the point is simply that there are only two. The details are beautifully laid out in several classic monographs⁴⁹⁻⁵⁰.

As for all the other “paths of development,” you honestly don’t need to waste time on them. We’re not here to catalog remote tribes or the colonial margins of advanced civilizations, right? When someone solemnly assures you — usually with great moral weight and knitted brows — that *all cultures are equally valuable*, you should absolutely nod along. Smile politely. Don’t start a fight. And then, when you’re alone, sit quietly and ask yourself: what exactly did the peoples of the Americas and Africa leave behind before Europeans arrived?

Let’s count on our fingers. For thousands of years, the rather gloomy inhabitants of those continents bashed each other’s heads in with wooden clubs and cut the hearts out of living captives atop pyramids in Tikal and Tenochtitlan. In my opinion, a questionable achievement.

“But what about the Maya calendar, corn, tomatoes, squash, and those small pyramids in the jungle?” you’ll ask. Fair point. Decide for yourself whether that balances the ledger.

On the neighboring continent: barefoot dances in grass skirts, more head-smashing with wooden clubs, and the cheerful sale of one’s own kin to Portuguese slave traders. Not exactly the hallmarks of a high civilization, wouldn’t you say?

And Australia? Also no. We already discussed the origins of Aboriginal Australians and Papuans in the section on “*Real biological differences between branches of Homo sapiens.*” I suspect the gulf between true civilizations and the many tribes that never independently grasped even the simple concept of the wheel is roughly the same as the gulf between the luminous halls of Velázquez and El Greco in the Prado, and the tiny storefront galleries on Geary Street in San Francisco where freshly printed abstract canvases are sold still warm from the printer.

Professors, with the best of intentions, keep inventing ever more elaborate terminology to explain the simplest things — mostly in an attempt to convince you that there is no real gap between the achievements of different civilizational models. One of the freshest examples is that famous notion of “*multiculturalism,*” which preaches the equal worth of Western law and technology and all the miscellaneous rituals practiced by tribes still living in the Stone Age. Another is the so-called “*theory of convergence,*” which insists that the free societies of the West and socialist regimes like the late Soviet Union or the still-breathing Cuba will eventually merge into some unified and just social order — something resembling Campanella’s imaginary “*City of the Sun.*”

The intellectual climate on many Social & Political Science departments reminds me painfully of an old, rather tasteless joke about a modern artist and his son debating the newest terminology of contemporary art:

“Dad, I still can’t figure out the difference between an *installation* and a *performance.*”

“Nothing easier, son. An installation is static. Say you leave a pile of... let’s call it ‘conceptual material’ on the neighbors’ doormat, ring the bell, and run away.”

“Okay, but what about a performance?”

“Oh, that’s even simpler! A performance is dynamic — a process. First you ring the doorbell, and *then*, when they open the door, you deposit your... masterpiece right there on the threshold.”

Chapter 6. The Nineteenth and Twentieth Centuries: From Britain to America

Forgive me, we are riding fast — spurring our horses harder than is proper — but there is simply too much ground behind us, and too little time to cross it.

Over the five centuries that followed the great voyages of the Spanish and Portuguese explorers, the world changed so radically that people living at the start and end of this span would have scarcely recognized each other's civilizations. First came the large, ocean-worthy sailing ships, their canvas bellies swollen with wind as they cut across the Atlantic. Then appeared steel — real steel — followed by cannon, muskets, and rifles. Military rivalry, along with the relentless refinement of weaponry, reshaped every society in Europe.

To build fortifications, maintain standing armies, and keep entire fleets afloat, states were forced to invent increasingly intricate internal systems. Their bureaucracies grew sinewy and disciplined; their treasuries learned to squeeze every available coin; their rulers learned to mobilize hundreds of thousands at a moment's notice. And all the while this race — this feverish, accelerating contest — kept pushing technology and industrial production forward.

Nations that ran along this development track as frantically as Alice sprinting beside the Red Queen gained coherent, functional governments capable of collecting taxes and raising enormous armies whenever danger loomed. Those that failed to run, or tired halfway through, quietly vanished from the map, absorbed into the empires of the more relentless.

And what were these countries fighting for? The same things humans have always fought for: fertile land, new markets for their goods, taxable populations, and sometimes simply to justify the expense of maintaining armies they had already hired and trained. All this continued — grinding, predictable, tragic — until the English looked down and noticed coal lying right beneath their feet⁴⁷.

Just as the agricultural revolution had begun in one particular place ten thousand years earlier, the serious exploitation of fossil fuel also started at a specific point on the map: England. And at a very specific moment in time: barely two centuries ago.

Industrialization offered, almost overnight, incomprehensible power to the first people who figured out how to harness coal's stored sunlight. It was as if the universe had quietly handed them a second source of fire — one that never went out. With this new energy, they could overwhelm continents. They could build machines stronger than storms. They could remake the world to match the geometry of their ambitions.

The Industrial Revolution — whose outlines we began to trace in the chapter on Europe — tore a gash in human history so deep that its consequences rival the ancient biological revolution we discussed earlier: the cosmic partnership between archaea and aerobic bacteria that gave birth to mitochondria, the power plants hidden inside every one of our cells.

In exactly this way, England — just a small, rain-soaked island in the gray waters of the North Sea — expanded into something infinitely larger. It became Great Britain.

A true industrial world had been born — one fueled first by coal and steam, and later by oil. If you ever want to understand how oil seized the helm of history and steered global politics for nearly two centuries, read Daniel Yergin's masterful account; no one has told that story with greater clarity or breadth⁴⁸.

The English were the first to conjure steam engines into existence — hissing, iron-ribbed creatures that multiplied the output of workshops tenfold, accelerated travel, and replaced a fickle and fragile labor force with something that did not fall ill, demand wages, or tire at dusk. These machines were stubborn, dependable, and terrifyingly strong. Bit by bit, the revolution radiated outward across Northwestern Europe and then beyond: as steamships grew larger, maritime trade blossomed; cities and countries drifted closer on newly shrunk maps; exotic goods became ordinary.

Silk and cotton, sugar and spices flowed into Europe from distant colonies, and in return Europe exported what it had learned to produce better than anyone else — textiles, machinery, and weapons. Industrialization did not merely enrich nations; it rewired the entire global order. It gave technologically advanced societies the ability not only to subdue agricultural-age peoples, as Europeans had been doing since the sixteenth century, but to govern them, tax them, and weave them into a functioning imperial economy.

This was the genesis of the British Empire, over whose domains — like those once held by the Spanish crown — the sun truly never set. India, Australia, Malaysia, Hong Kong... a scatter of continents and archipelagos stitched together by ships, telegraph cables, and an unwavering sense of destiny.

It is difficult today, in our fractured, multipolar world, to grasp just how colossal Britain appeared by the mid-nineteenth century. But imagine the Colossus of Rhodes rising over the Mediterranean — now scale it up, stretch it across oceans, and give it steel hulls and global trade routes. That was Great Britain.

After the Industrial Revolution of the nineteenth century, Europe produced several generations of people who suddenly found themselves chafing at the limits of the Old World — its scarce resources, its crushing taxes, its rigid, hierarchical social scaffolding. Life on the continent felt cramped, over-scripted, predetermined. And so, America flung its doors wide open to anyone who longed for a plot of land to till or a business to call their own. For millions of immigrants — clever, stubborn, hungry for reinvention — the New World offered a chance to be born a second time.

America's founders, who studied Roman law with almost monastic devotion, built a new state virtually from scratch. Anchoring themselves in the spirit of the *Magna Charta Libertatum*, they wrote a constitution — lean, ambitious, and nearly sacred — and then forbade anyone from tampering with it. The young country was built on freedom, the rule of law, equal opportunity for all its citizens, and the daring of private enterprise. That was the

alchemy that transformed a former British colony into a global power by the early twentieth century.

And when Europe finally tied itself into knots — descending into two catastrophic world wars that seemed, at first glance, to have erupted out of nothing and for no good reason — America stepped in. She helped Britain and France endure and prevail, and when the fighting ended, she did something far stranger: she stayed.

She rebuilt what had been shattered. Cities rose again from ash. Industries, twisted and broken, were coaxed back to life. And then, in a gesture that would have seemed unthinkable only a few years earlier, she extended that same hand to Germany, to Italy, to Japan — to those who had set the world on fire.

By then, a harsher truth had revealed itself. Peace, trade, and the fragile architecture of international law do not simply exist on their own. They are upheld — quietly, persistently — **by force**. By the military power of those who possess not only weapons, but the means to create them, sustain them, and project them across the world. By alliances, carefully woven and constantly maintained.

And then, almost inevitably, something happened that changed the scale of everything.

It did not arrive by accident. It had been gathering for decades — in laboratories, in universities, in half-formed ideas carried across oceans by restless minds. It was the product of new physics, of American industry learning to move not just goods, but entire worlds of machinery and material, and of a logistical effort so vast it bordered on the improbable. All of it converged in one place, under conditions of secrecy so complete that even those within it could scarcely see the whole.

The Manhattan Project.

There, in the desert — in a landscape that had once been a volcanic caldera — theory became matter. Calculation became force. And a new kind of weapon entered the world, one that did not merely end a war, but altered the very grammar of conflict.

The atomic bomb did more than bring the Second World War to its abrupt and final close. It drew a line under an entire era. After it, war between great powers could no longer unfold as it had before. The cost had become absolute.

It is difficult to overstate the transformation. It was as profound, in its own way, as the first chariots that gave speed to violence, or the first iron blades that made it efficient and scalable. From that moment on, history did not simply continue. It began to move under a new and far more dangerous set of rules.

From 1942 to 1946 the project was commanded by General Leslie Groves of the U.S. Army Corps of Engineers. The gifted physicist Robert Oppenheimer directed the Los Alamos laboratory. The Manhattan Project moved with breathtaking speed: at its peak nearly 130,000 people were involved, and its cost approached two billion U.S. dollars — about

twenty-six billion in today's terms. The greatest technical challenge was producing plutonium. In 1942 the first nuclear reactor in world history, Chicago Pile, was built at the University of Chicago. The second, called X-10 or the Graphite Reactor, followed soon after, and eventually the Hanford site began the first industrial-scale plutonium production.

All this culminated in the assembly of an implosion-type plutonium device codenamed *the Gadget*, tested in the Alamogordo desert on July 16, 1945 — a sunrise blown into existence by human hands.

In parallel with the development of nuclear weapons, the United States built the heavy aircraft capable of carrying them and trained the crews who would fly the missions no one dared imagine only a few years earlier. And then, just weeks after the Trinity test, the plutonium bomb *Fat Man* fell over the city of Nagasaki. That single blast effectively ended the Second World War, sparing — harsh as it sounds — the lives of perhaps a couple million soldiers and preventing the unimaginable civilian slaughter that would have accompanied an inevitable invasion of the Japanese home islands.

I don't want to linger here on the moral dimension; of course, it was horrific. But try, for a moment, to imagine how history would have turned if Hitler — or Emperor Hirohito — had assembled a bomb first. You remember the kamikaze pilots, don't you? If you become curious about the atomic project, you can watch Christopher Nolan's rather middling film *Oppenheimer*. It's imperfect, but serviceable as an introduction.

The Soviet Union always proclaimed that it had developed its own nuclear weapons independently, fully equal to America's. But that was never true. The data on enriched materials and even the bomb designs themselves were stolen almost immediately and passed to Stalin by Soviet agents — David Greenglass, the Rosenbergs, Julius and Ethel, and Klaus Fuchs — who managed to slip into the Manhattan Project despite all its layers of secrecy. And so, just three or four years later, Stalin, too, had the bomb. That was the moment the Third World War began.

Most people now call it the Cold War because direct battles occurred only through proxy conflicts — Vietnam, Korea — where American and Soviet clients fought in place of their patrons. But in reality, for nearly fifty years, it was an undeclared war between the entire free, if somewhat loose and fractious, democratic world and a vast totalitarian empire known as the Soviet Union. The first model of the state was locked in a direct, existential struggle with the second.

Relying on nuclear deterrence and technological superiority, America — joined by half the free nations of Europe, which would eventually form the European Union — contained the USSR and its satellites. Recall that after World War II the countries of Eastern Europe fell under Soviet control and were refashioned into dependent socialist regimes. They became the outer palisade of an empire built on coercion, suspicion, and a very different idea of what human society should be.

After the civilizational and technological triumph of the West, history across nearly all developed nations — China being the lone exception — began to tilt away from dictatorships and autocracies and toward freer societies built on private property and open markets. Europe clawed its way back from the devastation of war with almost superhuman effort. In the United States, the postwar decades became an age of industrial abundance, an era during which education, upward mobility, and small business ownership seemed not just possible, but practically within reach for anyone willing to put in the work.

President Lyndon Johnson even unveiled his sweeping “*Great Society*” programs, as though a prosperous, equitable democracy could now be engineered into permanence. The mood of the time was buoyant; many believed the great crises were behind them, the great wars finished. And towering over this landscape were the giants of the free world: Sir Winston Churchill, the indomitable lady Margaret Thatcher, Presidents John F. Kennedy, Ronald Reagan, and George H. W. Bush. Their vision and resolve helped bring the Soviet Union to its long-delayed collapse in 1991 — though the intellectual debris of socialism, and the more general allure of left-wing utopianism, unfortunately endured.

Over the last thirty years, slowly and almost imperceptibly, the atmosphere in developed societies has shifted. Here is what happened.

Left-wing ideas migrated into academia and, by the early 2000s, had become the mainstream

Somehow, the dying Soviet Union still managed to scratch America with its poisoned rapier. The irony is that a free society — like the one built in the United States — *naturally* produces inequality and stratification. That is normal. People differ in ability: the clever and industrious grow wealthy; the idle and limited do not. Enter, right on cue, left-wing parties, sprouting like mushrooms from rotting wood, proclaiming that such inequality is intolerable and must be corrected. But not by expanding equal rights and preserving open competition — rather, by pursuing “*justice*,” their preferred term. And in this new lexicon, **equity** has only one meaning: **wealth redistribution**. The instruments are always the same — raise taxes on the property and businesses of the competent and the productive, and use the proceeds to support those who decline to participate in the economy. Welfare, basic income, endless subsidies... a familiar litany.

The structure of developed economies changed

All the simple, tedious, industrial work relocated to Asia. As Western societies transitioned into information economies, a person’s true capital became their skills and talent. The number of self-made billionaires reached historic highs, even as redistribution ceased to reduce poverty and instead began to preserve it.

Unemployment in Europe rose from 1–3% in the early 1970s to 8–13% today. In the United States, the number of people receiving food stamps ballooned from nine million to forty-three million — almost a fivefold increase.

And the central crisis: a catastrophic collapse in birth rates across the developed world

By the 1970s and 1980s, European states were forced — demographically — to open their borders to immigrants from former colonies, whose numbers grew rapidly. In time, they became the primary recipients of social benefits. Combine this with the ideological drift of the era, and you get a volatile mixture: left-wing politics gradually transformed from the ideology of the “oppressed worker” into the ideology of the *dependent*, while the parasitic class was no longer “the owners of factories, newspapers, and ships,” as Karl Marx once thundered, but rather those unwilling to join any socially productive activity at all.

The strangest, almost surreal outcome of this cultural shift in the West has been the cultivation of a *sense of guilt* — a deep, almost ritualistic shame — toward society’s underachievers of every stripe. Those who are lazy or weak now tend to see the strong and successful not as examples to aspire to, but as threats, as walking reminders of what they themselves will never become. And so any financially secure person is reimagined as a kind of human dairy cow, expected to provide, indefinitely and unquestioningly, for whatever comforts the perpetually aggrieved might desire.

Soon enough, entire categories of people were recast as oppressors by default. Do you remember the BLM wave, when black gangs looted stores with impunity while police officers were instructed to stand down? Then came *all men* (#MeToo), then all the wealthy, then anyone simply healthy and sane. The moral compass flipped: it became advantageous to be a perpetually whining dependent, endlessly demanding attention and money. And presiding over it all were the progressives, waving their dreary DEI banners.

Today, activists seriously propose that major private corporations restructure their boards to include more women and LGBT members — not based on expertise, but on identity alone. I’ve watched this firsthand. In fact, I personally participated in one such initiative.

About fifteen years ago, when I was an Associate Director of Research & Development at the enormous biotech company BD Biosciences in Silicon Valley, five of us directors were told to brief our new company president. He had just moved into a vast office with glass walls stretching from floor to ceiling. His style was extravagant, deliberately theatrical; he fashioned himself as a flaming gay icon — manicured nails, of course, and a car with a violet sheen. He had been placed with us directly from the company’s East Coast headquarters. His name — please don’t laugh — was Jay Glasscock. Honestly.

We each delivered our short reports and wished him luck. I had been asked to summarize our department’s work on new antibodies conjugated to quantum dots (*QDots*) for leukemia diagnostics. So, I cautiously walked him through the ideas in the book we’d published the year before, doing my best to translate the science into simple language.

The next day Jay appeared in my office doorway, wearing a guilty smile.

“I didn’t understand a damn thing you said yesterday about quantum crystals.”

I jumped out of my chair, flustered.

“Jay — let’s sit down together. I can explain everything, draw diagrams, make it visual —”

“Don’t waste your time,” he interrupted gently. “Just update me once a quarter and handle everything yourself.”

Then he paused at the door and added: “They sent me here for a couple of years to serve as president. To manage you all. As a lesson. You know what times we’re living in...”

And the truth is, he turned out to be a pretty good president: he always secured generous annual bonuses for us and, most importantly, didn’t interfere with any serious decisions.

Please don’t misunderstand me. I *want* women on corporate boards — they are often smarter, calmer, more focused than men. And I have nothing against men loving other men; for many, it’s genetically predestined. Besides, they usually cook well.

But to create a dedicated *Compliance Office* within a major private business — an office specifically designed to ensure that queer employees and women receive director-level positions *regardless* of experience or credentials — is not normal. It’s the equivalent of advising shareholders at a large transportation company to appoint a dump-truck driver as CEO solely because he prefers boys in his private life, while sidelining the existing board members — ordinary white, heterosexual men — who actually understand how to run the business profitably.

Let’s rewind the tape for a moment — back to a time when common sense hadn’t yet been smothered under layers of ideological perfume. Before the semantic shell game played by the architects of the “new ethics,” the West understood its founding principle in one clear word: equality. And not the shapeless, weaponized version we hear today, but the original phrase — **equality of rights**.

It meant something beautifully simple: *the most qualified person gets the job*. If that person happened to be a black gay man, or a straight white one, or a catholic, or a protestant — it didn’t matter. None of those categories were supposed to obstruct his path. That was the whole point: equality meant removing irrelevant barriers, not manufacturing new ones.

What we now call “politically correct hiring practices” — the quota-driven crusades designed to “protect the vulnerable” — are, in fact, the opposite of equality. They *institutionalize* inequality. And a company president who barely understands the business he’s meant to lead? That isn’t compassion. It’s an insult to the shareholders — and a slap in the face to every competent professional in the room.

Why, you might wonder, do left-leaning parties everywhere — whether in America or across Europe — so passionately champion the cause of underachievers, the chronically idle, and the perpetually aggrieved? Why the strange, unshakable enthusiasm for unregulated mass immigration?

The answer is brutally simple: **it’s their electorate**.

The more people depend on welfare instead of work, the larger the natural constituency for socialist and left-wing movements. These parties thrive on clients — people to “save,” to “defend,” and ultimately, to mobilize. Human suffering becomes political capital, and votes become an extractable natural resource. Supporting illegal immigration, for any hypothetical socialist — let’s call him *berniesanders* — is merely a matter of **importing new voters**. Nothing personal, just arithmetic.

Now imagine an alternate world — just for a moment. A world where the billions spent on welfare had instead been funneled into reforming our dismal public schools, or funding grants for new antibiotics, or repairing America’s crumbling infrastructure... or even into those whimsical private space companies whose rockets now outperform the lumbering, bureaucratic dinosaurs of NASA. Within a decade, we’d have witnessed a quantum leap forward in technology and medicine.

Perhaps the time has finally come to ask several uncomfortable but necessary questions. Is it sustainable to maintain an ever-growing class of adults who contribute nothing to society yet live entirely on public assistance? Should individuals who take more from the system than they give to it have the same political power as those who fund it? And isn’t it reasonable — perhaps even fair — to introduce a voting threshold for people who have never worked a day in their lives and have relied solely on welfare?

No taxes, no vote. It sounds stark, but fairness often is.

And consider this: in the ancient *Greek poleis*, the birthplace of democracy, voting rights belonged only to those who defended their city in war. Citizenship demanded participation, not mere existence.

These questions are anything but abstract. Without confronting them honestly, the Western world may one day wake up to find that the very people — and the very genetic and cultural inheritance — that built its great civilizations have quietly dissolved into the surrounding swell of chaos and entropy. The same slow dissolution that, tragically, swallowed Rome and Constantinople.

How Western Civilization Will Evolve Next?

So, what happens now? Let’s sketch, however roughly, the contours of the next act. Even if we can no longer unwind the demographic shifts reshaping the developed world, we can still imagine — perhaps even hope for — the emergence, in Act Three, of a stern but rational “*global policeman*.” Someone who steps onto the stage before the set collapses.

The Next Ten Years

It seems that the blissful Francis Fukuyama celebrated the “end of history” a little too early. Democracy did not triumph everywhere. History, as it turns out, refused to end. And so, we

are likely headed into a small war — small only in the sense that the world will continue turning afterwards, though perhaps not with the same innocence.

Here's why.

A handful of authoritarian or Islamist regimes — Iran, Pakistan, North Korea — have not only stolen nuclear technologies from developed states but have now manufactured a workable supply of warheads. The moment China or Russia passes them the missile systems capable of delivering those warheads a few thousand kilometers, nuclear confrontations with their neighbors (India, Japan, Israel, South Korea) will become nearly unavoidable.

Notice that proxy forces aligned with Iran — Houthi fighters in Yemen and Hezbollah in Lebanon — have already begun harassing commercial shipping in the Red Sea and chipping away at regional stability, especially in Israel. This is only the prologue. The rehearsal before the real performance begins.

What can the West do?

At best, America, Britain, and — if it can muster the courage — the European Union will have to assume the mantle of a “*global police force*” and impose some form of external governance on these destabilized regions. That is, if anything remains to govern after the first generation of tactical nuclear strikes.

And let's be honest with ourselves: the world *needs* such a policeman, because the United Nations — created to manage regional conflicts after World War II — has long since ceased to function. The UN today is little more than a hollow stage set: the lights are on, the microphones work, but nobody is listening.

Meanwhile, at home, the West desperately needs to begin tamping down the ideological fuse of anti-colonialism that now burns at the center of contemporary liberal politics — especially within universities. Thousands of kindhearted young men and women, all “*for everything good and against everything bad*,” have been swept up in this fashionable moral fever. Inhaling the warm, sweet vapors of the “new ethics,” they have quietly mutated to fit the climate — after all, government jobs, grant money, and academic promotions still matter.

It is time — past time — to return to basic common sense.

Modern Western civilization is not responsible for the dysfunction, corruption, stagnation, or predatory politics of authoritarian regimes in Africa, Latin America, or the Middle East. These outcomes are not the legacy of colonialism, capitalism, or Western sin. They simply reflect the realities of very different societies, with very different histories, incentives, and cultural constraints. There is nothing racist in recognizing this — no more than in acknowledging that desert soil differs from mountain soil.

The purest, most unambiguous demonstration of this truth was delivered by a single laboratory experiment: *North Korea and South Korea*. Two populations, one people

genetically, separated only by political systems — and within a few decades one became a technological powerhouse while the other sank into famine, repression, and darkness literally visible from space.

That is the world we must navigate over the next decade. And unless the West wakes up — unless it stops apologizing for its own existence — the navigation may be far rougher than anyone wishes to admit.

The Next 20 Years

Once, when life in Europe was harsher and the winters felt closer, people seemed to possess an instinctive understanding of how a society endures. They knew, almost without articulation, what must be protected, which foundations must be reinforced, and where vigilance cannot be allowed to lapse. Echoes of that older intuition still live in Europe's fairy tales.

Take *The Three Little Pigs*. It is a simple story, nearly childish in its structure, and yet the anonymous storyteller captures something essential about communities. One piglet builds a sturdy house of brick — deliberate, resilient, meant to withstand the unforeseen. The other two piece together light huts of straw and sticks, trusting that luck will be kinder than preparation.

Why do we remember the bricklayer as the wise one? Not because masonry is inherently noble, but because every functioning society exists under some form of external pressure — a force that tests its perimeter, probes its weaknesses, and sometimes consumes what is fragile. In the fairy tale, that force appears as a single wolf, patient and hungry, stalking the piglets one by one. You know the story; it hardly needs retelling. But with every passing year, it resembles the European Union more than a nursery rhyme.

Across the continent, governments have grown comfortable — at times complacent — prioritizing social projects, ambitious energy transitions, and layers of regulation while quietly assuming that security, industrial capability, and geopolitical stability will somehow maintain themselves. Europe's neighbors, distant and near, once merely puzzled by the closure of nuclear plants, the slow erosion of industrial competitiveness, and the growing bureaucratic inertia, now watch with a more focused attention. They can already sense the looseness in those improvised timbers — and they are drawing breath.

Because if two-thirds of the houses are still built of straw, the wolf does not need much imagination. We may see the rest of the story unfold over the next twenty years. And unless Europe — the real Europe, the one inhabited by citizens who vote, build, decide — chooses to rebuild with brick rather than straw, the ending may not resemble the comforting version we narrate to children.

Yet Europe's vulnerability is only half the picture. Across the Atlantic, the United States faces its own moment of decision — and the timeline is not much longer.

If America fails to restore the full production chain of advanced electronics at home — microchips, semiconductor components, high-end chip lithography — or at least relocate it to stable, allied nations through serious nearshoring, the consequences will extend far beyond economic disruption. And if, in parallel, the United States does not secure breakthroughs in energy storage (the next generation of large-capacity batteries) and high-precision defense technologies (autonomous drones with AI systems, next-gen fighters, resilient missile platforms for air and space defense), its global position may erode with surprising speed.

Let us be even more direct: America could cease to be the world's leading technological and military power.

Who steps into that vacuum?

For the next fifteen to twenty years, the primary contender appears to be the authoritarian system governed by the Chinese Communist Party — a state that fuses digital surveillance, ideological discipline, and strategic ambition with an efficiency that is both impressive and unsettling. Whether China ultimately stumbles under the weight of its demographic crisis or its suppressed internal contradictions is another question, but for now it remains a formidable challenger.

And here, inevitably, we must speak of another shadow on the horizon — the grave and gathering threat posed by the nuclear dreams of the hardline Islamist regime that has gripped Iran. You surely remember the brief but thunderous twelve-day war, *Operation Midnight Hummer*, when Israel, together with American strategic bombers, shattered Iran's nuclear complexes with the cold finality of a gavel striking oak.

To me, that operation remains a kind of grim parable: a reminder that when dealing with dictators, persuasion begins only after the hammer has fallen. First you break the fingers reaching for the bomb; then, with immaculate courtesy, you offer a treaty — one that demands every grain of enriched uranium, every bolt and filament of missile technology, be carried out of the country like toxic relics bound for exile.

If the developed world can learn to answer such dangers with clarity and speed — without the usual fog of hesitation — even China's modern-day "emperor Xi", will soften into something far more manageable, as silk softens when touched by warm water. And in that case, no cities will need to burn, no runways will need to glow at midnight.

After all, the old American proverb endures because it is true: *a kind word and a revolver will always travel farther than a kind word alone.*

The coming twenty years will not reward passivity — neither in Europe nor in the United States. The wolf may not be at the door today, but the sound of his footsteps can already be heard on the path.

Inflation, Deglobalization, and Debt

At this point, every halfway sane economic analyst agrees on one thing: the world is drifting toward a new kind of multipolarity — a constellation of competing power centers (the US, China, and the EU, in roughly that order) — and toward a slow but steady unwinding of globalization. According to most forecasts, international flows of capital and goods will begin to contract fifteen to twenty years from now.

Paradoxically, this retreat from global interdependence may help restore employment in developed countries, yet it will erase many of the advantages those countries have grown accustomed to — cheap Chinese labor first among them. Goods and services produced “at home,” rather than in Guangdong or Penang, will become significantly more expensive. Companies will be forced to shoulder enormous costs for skilled labor, and those costs will flow directly into prices. Inflation will rise as predictably as steam from a boiling kettle.

Yes, I’ve read the optimistic essays about AI, robots and automation swooping in to save the day, but robots aren’t free. The capital required to deploy them at scale will itself fuel inflationary pressure. And over all this hangs another heavy cloud: the aging of populations in both the United States and China. That demographic shift becomes a second, powerful inflation driver — swelling the cost of healthcare, retirement benefits, and social support.

In other words, the air around us for the next twenty years will almost certainly be thick with inflation — a slow, persistent erosion of the purchasing power of money.

And that’s before we even talk about America’s internal issues: the ballooning public debt, social programs inflated far beyond their original scope, and the emergence of a nearly critical mass of voting *ochlos* — that polite Greek word for “idle crowds,” now repurposed for people inventing two dozen new genders while never quite getting around to finding a job. Sooner or later, there simply won’t be enough work for them.

You should absolutely look at the charts and long-term cycles in Ray Dalio’s remarkable book — a sober, data-rich mirror held up to America’s future⁵². As I’ve already mentioned, reviving the idea of a voting threshold — a political literacy test, or even a simple rule that **those who have never paid taxes should not have the power to allocate other people’s money** — might be the only way to reverse this trend.

Around that same timeline — fifteen to twenty years — we are likely to face several severe debt crises, which Dalio also describes with unnerving clarity. The relentless expansion of Social Security, Medicare, and Medicaid widens the fiscal deficit and piles up domestic debt that must be continuously issued as US Treasuries — which then require interest payments year after year.

And as always happens in the late stages of long debt cycles, asset prices will crumble at the end of each phase.

This leads to a simple conclusion: buying land and housing during future real-estate downturns, or acquiring shares of reliable US companies when their prices collapse in sharp market corrections, will likely be the **best long-term investments** you can make.

If you don't have time to dive into the fundamentals of America's largest companies, then build your investment portfolio — or your 401(k) — slowly and calmly during serious market pullbacks. Choose stable, broad exchange-traded funds such as *SPY* (which tracks the S&P 500) or *QQQ* (which shadows the NASDAQ-100 tech index). Don't rush. We have no shortage of financial crises and regional wars ahead of us.

Just make sure you choose your stocks and funds thoughtfully, after taking the time to understand the market and consulting investors who've lived through a few cycles. As for land and houses — I'm sure you'll navigate that terrain perfectly well on your own.

The Next 30 Years

The creation of the European Union at the twilight of the twentieth century promised a kind of gilded political sunrise — a unified continent, rational governance, the long-awaited triumph of integration over centuries of feuding. And yet, in the end, its own bureaucracy smothered that promise the way a sow accidentally crushes her piglets in the straw. We'll talk later — in the next chapter — about why large systems behave this way, what structural hazards grow inside the machinery of government.

For now, though, let's admit the obvious: wars, climate shocks, and the increasingly weightless, almost self-parodying quality of European leadership (take a long, honest look at Emmanuel Macron and his grandmotherly consort, or at Boris Johnson's operetta theatrics) will likely open the gates to something Europe is radically unprepared for.

Within a few decades, a vast wave of refugees from Africa and the Middle East — regions governed by kleptocracies at best and by outright cannibalistic regimes at worst — will surge toward the continent. And let's be clear: *this* wave will not resemble today's trickle of migrants slipping quietly across the Mediterranean toward Greek islands or the Italian coast.

Today's migrants mostly hope to land on European welfare rolls, exploiting the naïveté of ordinary voters and the sentimental architecture of social policy. But the next wave will be something else entirely — larger, hungrier, destabilizing by its very nature.

Given that outlook, I would strongly advise against tying too much of your financial future to European corporations or real estate. Demographic collapse is already baked into the soil of many Western European states, and within about twenty-five years it will almost certainly trigger a partial replacement of Europe's aging, well-educated, but low-fertility population with incoming Arab and black Muslim communities. And from that shift will follow all the predictable economic and cultural consequences.

The result?

Hold on — let me try to put this gently, or at least poetically. In sommelier terms, it will be a bit like taking a bottle of Bordeaux and cutting it fifty-fifty with beer.

Chapter 7. An Introduction to Weapons, the State, and Other Serious Matters

All the chapters until now were written for a smart girl in a good family — which meant that, almost unconsciously, I softened my tone. I blurred the sharp edges, skipped the goriest pages of human history, and painted everything with the gentler colors on the palette: whites, pinks, warm daylight blues. But suddenly I felt uneasy. *What if a young man, still at the beginning, reads this one day?* A decade from now he'll be grown, and the world will demand something entirely different from him — something far more rigid than poetry and lace. So today I'm switching modes. Time to put the pretty porcelain back on the shelf and take out the iron toys. We'll start with the simplest and oldest of them all.

Weapons

You don't actually need to read about stars or mitochondria. Soon enough you'll have a woman you trust and children who depend on you — yes, it happens fast. But don't make the mistake of assuming that life in our country will always be peaceful, predictable, or cushioned by systems that never fail. Don't think that the tidal flow of illegal immigration or the swelling ranks of the chronically idle will forever be kept in check.

Type it into Google. Search YouTube. *LA riots, summer 1992. Hamas breach, Israel, October 2023. Now*, watch the faces of the attackers. Watch the terror in their victims — ordinary people ambushed in their kitchens and bedrooms, waiting for help that never arrived. Watch how police vanished from the streets of Los Angeles the moment they were needed most. Watch how even the formidable Israeli army froze, disoriented, as the gates cracked open.

And understand a simple truth: if it happened there, it can happen here. Not in some distant, theoretical future — but in *your* lifetime. You don't need to be afraid. Fear is wasted energy. But you must be **prepared**.

In my gun safe you'll find a heavy, unpretentious Smith & Wesson .38 revolver — the kind of weapon built long before you or I were born by people who understood what it meant to fend for themselves. Take it out. Load all six chambers. Don't cock the hammer yet.

Now, sit down. Close your eyes.

Feel the weight of it — the cold harmony of steel and purpose. This is a piece of precision engineering, a distilled fragment of human ingenuity. A tool forged by people who lived through harder times than ours, who knew exactly what to do when civilized life cracked open and the police — yes, read my lips — ran away. Because they do run. History shows it, again and again.

And when that day comes — if it ever does — and you or your family are in danger, don't hesitate for even a heartbeat. Shoot first. Those you love depend on your willingness to act before you think. This is your calling. Your spine. Your contract with the future.

If stopping an attacker means killing him — then that is what you must do. The Gospel has gentle words for peaceful times, but soldiers and police do not carry a different edition of the book. Don't romanticize these things. Survival is older than scripture.

A good lawyer will untangle the aftermath. The only irreplaceable thing is that you stay alive — and so do they.

Move with intention. Breathe. Aim carefully. Avoid lighted, open areas. Strengthen your doors and windows *before* the trouble arrives. Install lights outdoors, mark the perimeter where you'll open fire if someone breaches it. Yes, there's a term for this: defining your engagement lines. If you want a cinematic primer, watch Ridley Scott's *Kingdom of Heaven*, the director's cut. It teaches more about proper defense than most real-life instructors.

And if someday the revolver stops being enough, the safe also holds a pair of solid assault rifles and NATO-caliber 5.56 magazines. Plenty of them. Don't be afraid of weapons. Get used to them. Let your father show you the difference between calibers, teach you how optics work, let him place your hands correctly on the grip. Learn to shoot well — not for bravado, not for show, but the way you learn to swim or change a tire. One day you'll teach your own children the same skills, and the circle will close, the way it always has for families that plan to survive. Keep calm and carry on. Not as an Instagram slogan, but as a way of life.

The State

We all live inside a ready-made aquarium of society — glass walls, filtered currents, comfortable water temperature. And we forget that someone had to build that aquarium, fill it, defend it, and keep it from cracking.

So, let's ask the simplest questions: *What is a state? Where did it come from? And what is it actually supposed to do?*

At first glance, the answers seem straightforward. In the developed world — the G7 countries we like to cite — the purpose of the state is almost deceptively clean: *peace, prosperity, safety, and a space in which its citizens can build their lives without fear.*

But, of course, it wasn't always like this.

Europe's history — from the Carolingian Empire to the meat-grinder of the 20th century's three "world" wars — is a tapestry woven with invasion, blood, winter campaigns, and the endlessly repeated struggle to keep one's culture alive while someone else tried to burn it out.

And oddly enough, the first chapter of the European state begins not with *Charlemagne*, but with his grandfather — *Charles Martel*, the pragmatic, unromantic master of logistics who lived about two centuries after Rome collapsed into dust.

By then, the Arab armies had already taken Spain. The land we know as the Iberian Peninsula glowed for centuries as *Al-Andalus*, a distant province of the Umayyad Caliphate. Narbonne fell. Avignon fell. Agde and Nîmes in Provence fell. Soon the Moors raided Aquitaine and burned Bordeaux. Imagine the map of Europe shrinking like a frightened animal.

And then — 732 CE. Charles Martel, who understood that *waiting* is just another form of losing, gathered an army: not only Franks but also Alemanni, Bavarians, Saxons, and Frisians. A coalition of tribes that usually quarreled over cattle and borders, suddenly united by a single truth: if they lost, there would be no “later” left to argue about.

His dusk-forged cavalry met the Moorish forces at the Battle of Poitiers and pushed them back across the Pyrenees. And just like that, the map stopped shrinking. That victory bought something priceless: time. It meant that the emerging Frankish realm could offer its scattered tribes a life *outside* the Caliphate, a life where faith was chosen rather than imposed, where culture had room to breathe.

From this fragile beginning eventually came the patchwork *Carolingian Empire*, and many generations later, after the long sunset of Reconquista, the West forged its identity from a strange alchemy:

- ancient democracy,
- Roman law,
- Christian culture of medieval towns,
- the explosive force of the Industrial Revolution.

That alloy produced the humanist values we now call “Western.” It produced the notions — almost taken for granted today — that law matters, that property matters, that the individual is not a subject, but a citizen. This is the soil from which our present prosperity grew. But we’re drifting. Back to the practical.

How does a developed state stay alive?

Let’s set aside oil monarchies and African dictatorships — those follow entirely different rules. A normal, functioning modern state lives **off taxes**. Only that. Margaret Thatcher put it with the precision of a surgeon: “*The government has no money of its own. It only has what it takes from its citizens and its businesses.*”

In a small town, these taxes pay for the courthouse, the roads, a school, a modest hospital, and the sheriff — because even the calmest places have crime, and always will. Higher up the ladder, federal taxes support the army and navy. A country needs them not for vanity,

but because predators exist — and because trade moves along sea lanes that must be guarded like arteries.

Then come the expenses of governance: the federal administration, the monetary system, the courts, Social Security. And if you climb even higher — you reach the top floors of the state: regulation of giant corporations, oversight of natural resources, public funding for universities and science, space agencies like NASA. That's the rough sketch of the machine that shelters you. Not romantic, not glamorous — but the only reason your aquarium water stays warm.

The Engine of a Modern State Is Business

In every mature Western nation, the real engine humming beneath the floorboards isn't a ministry, a committee, or a president's signature on some polished document. **It's business. Private business.**

The kind built on capitalism, sweat, risk, sleepless nights, investment, and — crucially — private property. From the tiny bakery that opens at dawn to pull its first baguettes from the oven, to the grandmother's biotech lab running assays under fluorescent lights, to the multinational leviathans that float across continents — all of them form the bloodstream of a functioning state.

They create jobs. They generate the taxes that feed every public salary.

Even the uniforms — the police officer's badge, the firefighter's turnout gear, the officer's epaulets — are, in a sense, paid for by these businesses. Every paycheck for every government worker is simply a recycled dollar from the private sector, washed through the filters of local and federal budgets.

And business plays by clear, harsh rules. It lives and dies by competition. It must earn profits for its shareholders, or it disappears — quietly, brutally, and without ceremony. Its motto might as well be carved into stone: *"Let the strongest survive."* No speeches, no committees, no excuses.

The Contradiction at the Heart of Modern Society

Here is where the real tension begins — the kind that bends entire civilizations:

Business and the state want different things.

- Business is built on competition, efficiency, and the survival of what works.
- The state, at least in the Western ideal, must care for the weak, the sick, the unlucky. It must redistribute resources, offer safety nets, and maintain programs that would bankrupt any private company in a week.

The state is essentially a monopoly — there is no competing version of it down the street with better hours and a loyalty program. And the bureaucracy inside it changes slowly, if at all. Every four years, only the very top executive layer is reshuffled; everything beneath it persists like sedimentary rock.

We interact with this vast machine every day — DMV counters, school districts, local courts — and the truth is unavoidable: despite good intentions, modern government is spectacularly inefficient at using the money business earns and citizens surrender as taxes.

Why?

- Many government officials lack the skills, speed, or drive that thrive in the private sector.
- The people who *do* have those traits usually go into business, where excellence is rewarded instead of buried under spreadsheets.
- Politicians rise not through merit or managerial genius but through charisma, networking, and mastery of the political “game.”
- And the real killer: positions are filled not by ability but by party loyalty. Competence is optional. Obedience is not.

A Case Study: Recent History

Want examples? Take a breath — the freshest ones are still warm.

Rewind a little. Look at the long arc of the last two Democratic administrations. It began in 2008 with the election of a young DEI candidate, whose résumé before entering politics amounted mostly to *community organizing* in Chicago — a job description so vague it practically evaporates when you try to touch it.

But he fit the new progressive narrative like a custom-tailored suit: diversity, inclusivity, representational symbolism — the whole package.

And to be fair, he started with a genuinely overdue idea: reforming US healthcare. It needed fixing; everyone knew that. But instead of crisp legislation and decisive action, eight long years dissolved into speeches, panels, town halls, more speeches, and the kind of symbolic politics that burns daylight but produces no actual structure.

Meanwhile, deficits ballooned. Hyperregulation seeped into every corner of public life. And the government opened the firehose of spending:

- A disastrous, endless military campaign in Afghanistan.
- Massive subsidies poured into half-baked “green energy” projects.
- Trillions — with a T — of helicopter money during the pandemic, which anyone with a high-school econ class could have predicted would ignite double-digit inflation.

Did I miss anything?

This wasn't just reckless spending of tax revenues. It was something worse: **shoveling debt onto the shoulders of the next generation**, handing our children and grandchildren a bill for decades of political incompetence. It all ended, as you may remember, with an old man in diapers — barely aware of where he was, yet still clinging tightly to power.

Is it any wonder voters snapped? They swung the pendulum hard in the opposite direction, choosing an almost cartoonishly conservative candidate who promised to slash the bureaucracy, stop the financial hemorrhaging, and end the wars. In truth, they weren't voting for a person. **They were voting for chemotherapy** — a last-resort treatment for a sick political system.

You might ask, “Can't we just write good laws and build a fair society?”

It's a beautiful question — the kind that belongs in a civics class, on a protest poster, or whispered late at night in a college dorm when everyone still believes history bends obediently to goodwill. But the answer is older than any constitution and darker than most political theories dare to admit.

Because all roads, in the end, lead straight into human nature — and *that* doesn't update with a software patch. Cain murdered Abel, the very first siblings of our mythic lineage. And in all the millennia since, with empires rising and collapsing like waves on a beach, remarkably little has changed. Yes, you will meet people whose eyes burn with curiosity, loyalty, generosity — the bright edges of our species. But right next to them lurk the old shadows: laziness, envy, cruelty, appetite without restraint.

And here is the uncomfortable truth: *Any attempt to build a “true social state,” full of free housing, free this, free that — any neo-Marxist experiment in any of its modern costumes — leads in one direction only: toward Argentina, North Korea or Cuba.*

Democracy is fragile. Improvised. Forever imperfect. The best we can do is limit what government tries to manage, guarantee equal rights and opportunities, enforce *law and order* for everyone who lives here legally, and resist the urge to play savior to every dysfunctional regime across the ocean.

You can't fix the entire world. You can't even fix countries that refuse to fix themselves (remember Afghanistan, Iraq and Iran?). If they want honest elections — let them fight for them. If they want good leaders — let them choose them. If they don't — well, wipe your tears, close the tab, and get back to your business, your women, your children, your life.

“But can't we at least fix California?”

There comes a moment — usually late at night, when the windows of the world turn dark — when every thoughtful person in California wonders the same forbidden thought: *surely*

there must be a way to set this place right. Not through slogans, not through outrage, but through the slow, stubborn work of rebuilding something vast and weary.

Imagine, then, that the impossible has already happened. The campaign is over; your name has been etched onto posters, ballots, and in the hesitant admiration of strangers. On the morning after your victory, you walk through the long colonnade of the Capitol, the echo of your steps still unfamiliar to the marble. A young staffer opens the door to the governor's office — a room that has outlived earthquakes, administrations, and a century of promises.

The air inside is still. Sunlight lies across the desk like a silent invitation. Papers wait in neat, meaningless stacks; portraits of predecessors look down with a mixture of fatigue and amusement, as if wondering how long your certainty will last. You do not sit right away. No one ever does. Power, when newly handed to you, feels too heavy to touch.

But eventually you lower yourself into the chair, and the leather exhales — a soft, resigned sound, as though acknowledging that it must now shape itself to yet another back, another set of hopes. And it is there, in that first fragile moment before the machinery of government begins to roar around you, that you understand the truth:

If California can be repaired at all, the work will not begin with speeches, committees, or the choreography of cameras. It will begin with a gesture — firm, unmistakable — a gesture that redraws the horizon rather than rearranging the furniture.

And that gesture might look like this:

First, you redirect the torrents of money that currently evaporate into “environmental initiatives,” endlessly multiplying homeless programs (aka easy money for bums and moochers), and sprawling welfare structures — not to abandon compassion, but to reclaim effectiveness. You channel those funds, boldly and unapologetically, into the creation of entirely new schools. Real schools. Serious schools. Schools with a spine.

Every small town gets one. Half of them devoted to mathematics — ten uncompromising hours a week. The other half to biology or physics — also ten hours a week.

No tuition. But merciless yearly exams. Keep only the students who can — and more importantly, who *burn* to learn. Double teachers' salaries. Recruit young scholars who didn't win the academic lottery of tenure but still carry fire in their chests, who want not prestige but purpose.

Build schools so compelling that students line up at dawn for a chance to enter. So demanding that teachers compete for the privilege of teaching there. So rigorous that parents whisper their names the way older generations spoke of Harvard or Caltech.

Spend what you must. Buy the land around the schools now, while it is still dismissed as unremarkable. In ten years, when the neighborhoods bloom around these bright intellectual cores, the state can sell parcels to developers — and the proceeds will carry the schools forward for decades.

And finally — do not neglect the oldest magic in civilization: the story people tell themselves. Launch a devastatingly effective public campaign. Make these schools legendary, aspirational, unmistakably Californian in their ambition. Give the state a new myth to gather around.

Do that, and the long drift ends right there. Do that, and you may well win re-election until the sun burns out — not because you sought power, but because you finally remembered that California was once a place where impossible things began.

And don't forget — clear the streets. Completely.

And yes, please don't forget to sweep the homeless and the mentally ill off the city sidewalks and into new, inexpensive hospitals and psychiatric facilities with firm rules and locked doors. You'll have to build those quickly too — nothing fancy, just strong walls, clean beds, and enough burly orderlies with arms like tree trunks to keep the peace. And you'll need to update the laws so patients can actually stay long enough to get better.

All strictly in accordance with the will of the people of California, of course. You're not a despot... Not outwardly, anyway.

Next: the UC system. Time to operate.

Move quietly. Start by placing your own smart, serious trustees onto the Boards of Regents — people who aren't afraid to clean house. Let them gently, surgically remove the deadwood: professors who've forgotten what teaching means, ideologues who turned classrooms into therapy circles, entire departments built on slogans instead of science.

No more Gender Studies, African American Studies, Sociology, Ecology, Political Science — not on the state's dime, at least. Replace them with heavy-duty doses of mathematics, biology, genetics, chemistry — the intellectual protein young Californians desperately need.

And then go global. Work with the Foreign Office. Open UC admissions commissions in American embassies and consulates across Europe and Asia. Recruit talent the way pro teams recruit star athletes. Bachelor's, Master's, Ph.D's — whatever brings in sharp minds and skilled hands.

If the budget groans or seats run out, tax stupid marijuana, ridiculous tribal casinos, hard liquor, cigarettes — whatever it takes. Use the money to build three or four new UC campuses. If you build them right, they'll function like industrial pumps — sucking brainpower and craftsmanship out of developed nations and anchoring it permanently here, in California.

Then talk to Homeland Security and get a brand-new immigration pipeline approved — something like a "Tech Lottery" with work visas and green cards baked in.

Immigration is wonderful — when it's legal, selective, and designed for those who make your state better (You do remember that not all people are the same, right?).

But as for the *present-day immigration* — the chaotic, murky, cartel-soaked river spilling over the southern border — you'll need to act. Work with the governor of Texas and shut that border down under some respectable public pretext. No, seriously — shut it *down*.

By now I don't need to explain the difference between a pump that brings in talent and a vacuum cleaner that sucks in trash drifting across the Mediterranean into Italy and Greece. Trash that settles onto welfare checks, opens shady mosques with even shadier "community centers," and builds militant networks behind the scenes. You do remember what happened at the Bataclan nightclub in Paris ten years ago, don't you?

Oh — and double the size of every police department across the state. Just in case. That's one line item where you *never* cut corners.

And while you're at it — read a book

Try to get through *All the King's Men* by Robert Penn Warren if you ever find a quiet evening⁵³. Or ask your secretary to read it to you slowly, with expression — the way children's stories used to be read aloud. A great writer lays out the machinery of state politics with clarity and force, not in this breathless, stumbling shorthand of mine.

Hold your people gently, but firmer than they expect — not with force, but with the steady warmth of a hand that refuses to let anyone drift into darkness. Give each person a small plot of responsibility, a corner of earth or duty to tend, so that the old truth becomes real again: that dignity grows where effort is planted, and that a life worked by one's own sweat becomes a life rooted, upright, and whole.

Do this long enough, and slowly — almost shyly — the world will begin to right itself. Order will return like dawn along the ridgeline, quiet at first, then unmistakable.

That is all. Go on now.

And carry yourself the way young *luggals* once carried themselves — not yet kings, but already walking as though the future had placed a hand upon their shoulder and whispered: *your time is coming*.

Chapter 8. The Forerunners of Our Kind

When I drive west from our sleepy university village toward San Francisco, and then slide south toward San Jose, I see more and more of Silicon Valley's modern "*new people*" — the ones who teach, write books, design rockets, build artificial minds, and invent the next generation of medicine. I know quite a few of them. Some run their own ventures — anything from a tiny plant-pathology lab tucked between farm fields in Vacaville to the sprawling campuses of Google or Nvidia. Others move the great rivers of venture capital.

The Valley itself feels like a shimmering Persian carpet, and its people seem woven from ocean haze — arriving here from every corner of the world, with every imaginable shade of skin and color of eyes, speaking a dozen languages without apology. You look at them and think: once, during the age of the British Empire, they must have opened islands and claimed continents; or, much earlier, defended the frontiers of the Tang dynasty from marauding tribes.

What makes them new? What sets them apart from the others? Their minds.

Look at how the world has accelerated in the last hundred years. They've become sharper, tougher, more relentless than the average human being. A couple of centuries ago — when there were far fewer of them — they founded universities, built manufactories, and were always the first to die in endless wars, the first to lead the cavalry charge. It's a metaphor, of course; hardly anyone today knows how to sit a horse or lead men into battle. But they'd figure it out — if you explained it clearly enough and gave them whatever passes for a cavalry squadron in our era.

Everyone else has changed very little. The minds and the willpower of the rest of humanity remain roughly where they were in the time of the pharaohs or, later, during the French Revolution, when the masses nearly wiped out the aristocracy.

At best, they live quietly in their apartments, weary of their jobs, believing whatever the television tells them. Entire genres of video-game shooters with cartoonish monsters were designed for them; whole neon districts of casinos and amusement parks in the Nevada desert — with cardboard pyramids and Eiffel Towers — are built precisely for this population. They avoid risk and responsibility, which is why they rarely own a home, though they'll gladly lecture you about the virtues of the social state and wealth redistribution.

In another version of the same story, "*the other people*" live compactly on welfare and food stamps in the outer districts of big cities. Out there, at the edges, the popular languages are "social justice" and "reparations for slavery".

What unites all of them — every last one — is a single unmistakable mark: *a faint, persistent aversion to work*. It sits on them like the branded fleur-de-lis on Milady's shoulder in *The Three Musketeers* — indelible, quietly defining their place in the world.

And I'm hardly the first to notice. Three hundred years ago, the divide between people was already so sharp, so painfully visible, that polite society preferred not to speak of it — except, perhaps, to their children. Here is a beautiful letter from Lord Chesterfield to his son⁵⁴, sent in 1748:

My dear boy, look closely at how different people become after they have worked on themselves, compared to those who remain rough and unshaped. I am certain that you will never regret the time and effort spent on cultivating your mind. Some common cart driver may possess organs no worse than those of Milton, Locke, or Newton, but in their development these men surpass him far more than he surpasses his own horse.

Genes and Civilizations

We've come a long way together, you and I. Four great canvases now stand completed — colors drying, meanings settling into place — and it is time to step back, move the easels side by side, and let our eyes wander across the sweep of them. From a distance, faint threads begin to reveal themselves: thin, glimmering lines that run between the rise of ancient civilizations and the inner lives of the people who carried them on their shoulders.

So let us linger a moment in that quiet space. Let us look not at the myths we tell ourselves, but at the deeper currents — the pressures, the trials, the quiet chiseling of history that shaped our ancestors from the wandering, half-dreaming primates who once moved through African forests with no notion of pyramids, cities, alphabets, or laws.

We will speak honestly here. Without arrogance, without disguises, without the soft half-truths people use to soothe themselves when the world grows complicated. Because in the end, the questions before us are not many. They can be counted on the fingers of one hand — but each leads downward, into chambers where the roots of human fate intertwine with the long, patient machinery of time.

Are civilizations in any way connected to the forces that shape people across generations?

It seems impossible to deny. At the very least, civilizations intensify the pressures that history exerts on those who live within them. Weak or poorly governed tribes, cities, even entire kingdoms often sagged under the weight of their own missteps — disease, famine, corruption — or were swept into the orbit of more organized, more ambitious neighbors.

In such upheavals, it was usually the bold, the restless, the adaptable who pushed outward into new worlds. You may remember the paleogenetic histories of Europe and India we explored earlier: archives of bones and buried settlements that whisper the same story again and again — that when cultures collided, when distant peoples met after centuries of isolation, the older patterns often dissolved like sand under a rising tide.

Think, for instance, of Francisco Pizarro, who — in the narrow hinge of just a few years — broke the spine of the Inca Empire and folded its vast population into the dominion of a distant crown. The Spaniards who carried this out were scarcely more than a traveling company: in 1532, when they landed on the Pacific coast of what is now Peru, they counted only two hundred men and twenty-seven horses.

Look more closely: throughout Latin America today you can still see the imprint of those Spaniards, hear the echo of their language. Even their family names have radiated across the continent like ripples from a dropped stone. In many regions of Colombia, 94% of men carry a European-type Y chromosome, while their mitochondrial DNA, passed from mother to child, is 90% indigenous. Which means the paternal lineage came overwhelmingly from the Spaniards, while the maternal line remained rooted in the local tribes.

And yes, we must say it plainly, that conquest was brutal, devastating, hardly the sort of tale one would wrap in bright illustrations for children. But history is often made of such collisions, and the record that survives is rarely gentle.

When and Where the Modern “New People” Appeared

It seems their ancestors began drifting into a half-empty Europe from every direction after the Black Death loosened its grip in 1353, leaving more than half the continent dead. Entire towns had fallen silent. Fields lay untouched. The world felt scraped clean.

There’s an excellent chapter on the plague years in Acemoglu and Robinson’s book, but you don’t need historians to tell you what that kind of vacuum does to a civilization⁵¹. It took nearly a century after the end of the so-called Dark Ages for Europe to repopulate itself and to summon the first timid spark of what would become the Renaissance — *Rinascimento* — in Florence. The name comes from the Latin *renascere*, “to be born again,” and for once the word fits perfectly: Europe really was being reborn.

The survivors of Constantinople and the scattered shards of the Eastern Roman Empire played a special role in this revival. Those who had lived through plague and war carried their libraries on their backs, their medicine in leather satchels, their weaponcraft in memory. They arrived in the Italian cities with crates of Greek and Roman manuscripts unknown to medieval Europe — whole intellectual continents no one west of the Aegean had explored for a thousand years.

It was Greeks—*Romaioi*, as they called themselves — who taught in the first Florentine Platonic Academy founded by Cosimo de’ Medici. And when printing appeared in the mid-fifteenth century, it did with ideas what rivers do with silt: it carried them everywhere. Classical knowledge spread like a second sunrise, and European painting, literature, and philosophy reordered themselves around its new light.

For the next several centuries, alongside the growth of science, finance, and manufacturing, entirely new social castes emerged. The flowering of the city-republics elevated a class of urban people untouched by feudal obligations: artisans and craftsmen, merchants and

bankers, minds sharpened not by swords but by numbers, trade winds, and the discipline of ledgers.

Take, for example, the Jews. This small group of Semitic people — scattered by the Romans from their ancestral lands along the warm southern edge of the Mediterranean — found themselves stretched across Europe from Spain to Poland. The locals regarded them with something between suspicion and hostility. They practiced medicine, trade, and moneylending; they stayed close, guarded their language and religion, and often came to one another's aid. Who, in those centuries, would find that endearing?

And so, their neighborhoods were attacked again and again. In every country. On every border of Christendom. Famines, crop failures, plagues, the frustrations of poverty — everything that tormented the masses somehow found its way onto Jewish shoulders.

I won't recite the dates or the body counts. Instead, rent a small apartment someday in Segovia, near the cathedral. Wander into the Jewish quarter, descend into the crypts of the old underground cemetery across the narrow stream. Step inside the silent synagogues. You'll understand without a lecture.

The selective pressure was immense. The families who survived generation after generation were the cleverest, the most persistent, the most adaptable. Evolution doesn't write poetry, but it keeps excellent ledgers. And a glance at the list of Nobel laureates from the past fifty years will tell you more about the results of that long experiment than any anthropology course ever could.

And when Hitler expelled the Jews from Germany on the eve of his great war with Stalin, German science — once dazzling, fearless, world-leading — collapsed for half a century. Meanwhile, in America, the atomic bomb and genetic engineering took shape in the hands of... well, you can guess who.

But the story, of course, was never about one people alone. Across South Asia, the role that Jews once played in Mediterranean trade fell to the Chinese: merchants who stitched distant ports together with silk, porcelain, and shrewd intuition. In the Ottoman world, that mantle passed to the Armenians — a nation of guildsmen, traders, diplomats, and scribes whose caravans moved across mountains like glowing threads in a vast imperial tapestry.

The labels by which we call these peoples — “Chinese,” “Armenian,” “Jewish,” “Greek” — are, in the end, just convenient shells. Names stamped onto history after the fact. What mattered then, as now, were the **new people** constantly being forged: tempered in the long heat of trade routes, shaped in university halls where ideas echoed like hammer blows, refined in the patient disciplines of mathematics, medicine, navigation, craft.

Their birthplaces were accidents. Their destinies were not.

And then, of course, came the warriors — those who reshaped the world not with ledgers or manuscripts, but with the iron grammar of force. Do you remember how the Vikings tore

open the map of Europe at the end of the first millennium? Barely a hundred thousand of them slipped down from the northern seas in their oar-driven longships — *drakkars*, carved like sea-wolves — yet that small number was enough to plow the destinies of England and France into entirely new furrows.

They moved as if history itself had loosened its moorings for them. Trading, raiding, settling, marrying, ruling — until even the great rivers began to speak their tongue.

Along one such river, wide and slow and threaded with northern commerce, they founded a kingdom of their own. A place where Baltic winds met Byzantine gold, where the steppe and the forest shook hands.

Its capital they named *Kyiv*.

Say, why is this? Wherefore? What should we do?

Human evolution is not slowing. If anything, it is beginning to gather speed again — carried now not by climate or predators, but by the restless machinery of civilization itself.

Selection is still at work. It always is. Only now it favors something subtler: the quickness of thought, the flexibility of mind, the quiet courage to adapt when the ground shifts beneath your feet.

Once, it turned heavy, earthbound dinosaurs into albatrosses that could cross entire oceans without touching land. It reshaped a four-legged mammal into a sperm whale, mastering depths no land animal could survive. And somewhere along that same improbable path, it refined a small, warm-blooded creature into Elizabeth Taylor and Marlon Brando — into beauty, into presence, into that strange, unmistakable intensity we recognize as fully human (Watch him as Hamlet, if you have the patience. And her as Cleopatra. You will see what I mean).

And here we are, you and I, living in a narrow, trembling interval between two ice ages — a brief pause in a much longer story. In this moment, selection has turned inward. It is no longer shaping claws or wings, but something far more delicate: memory, willpower, the fragile and astonishing machinery of thought.

Soon, we will run into the limits of that machinery. Our memory — so proud of itself, so central to everything we build — is, in truth, a narrow bottleneck. Imperfect, unreliable, easily overwhelmed. And the old, patient mechanism of biological evolution will not save us this time. It moves too slowly. The next great shift — climatic or otherwise — will not wait.

So, the solution will come from elsewhere. Someone, somewhere, will build it: an external extension of the mind, an interface so natural it will feel less like a tool and more like a continuation of thought itself. A quiet prosthesis for memory, for reasoning, for imagination. And once that door opens, it will not close again.

From there, the path unfolds almost inevitably. Hereditary diseases, carefully removed. Viral remnants, swept from the genome like dust from an abandoned room. And then, perhaps, we will begin to restore what was once lost. Buried within us, under layers of mutation and time, are traces of older capabilities — fragments of a deeper sensory world. The ability to see beyond the narrow band we now call visible light. Into the infrared, where heat writes its silent signatures. Into the ultraviolet, where patterns invisible to us still shimmer just beyond reach. The genes are there. Ancient, damaged, half-forgotten. Not gone — only waiting.

You have already heard, no doubt, of CRISPR, the first careful tool for rewriting what was once untouchable. What follows from it — what shapes it will take, what futures it will carve into the world — is not yet written. That part, whether you realize it or not, is yours.

In the long view, our species — *Homo sapiens* — has only two roads ahead

Over the long arc of time, our species has only two possible destinies: either we dissolve back into the archaic mass... or we fracture into a constellation of daughter species, each one diverging, mutating, rising or disappearing, just as the Neanderthals and Denisovans vanished only yesterday in evolutionary terms.

The same story, the same branching, the same fading in and out of existence — only now some of our future offshoots will evolve naturally, while others will seize the reins and guide their own evolution deliberately, editing their genomes step by step, sculpting themselves as no creature has ever sculpted itself before.

What's in it for me?

A human life — an active, bright, fiercely lived human life — is astonishingly short. And though metaphors come cheap, I will borrow one more time from the *Quattro Stagioni*, because the comparison still rings true.

The first season, your spring, is largely given to you: the mother and father who raise you, the school that shapes your habits, the university that sharpens your mind and teaches you the discipline of clarity. Language, mathematics, chemistry, biology, music — none of these arrive as gifts. They are earned through repetition, concentration, and the willingness to endure frustration. They form the scaffolding of a life that will not crumble under its own weight.

Once you finish school, you will have — at most — five years. You can spend them well or badly, but remember: whatever you become in those five years is what you will be for the rest of your life. And then, around twenty-five, the real story begins — the dangerous, incandescent part — just as Janis Joplin once promised to the little girl sleeping in her cradle:

*Summertime, time, time
Child, the living's easy
Fish are jumping out
And the cotton, Lord
Cotton's high, Lord so high.*

*Your daddy's rich
And your ma is so good-looking, baby
She's a-looking good now
Hush, baby, baby, baby, baby, baby
No, no, no, no, don't you cry, don't you cry.*

*One of these mornings
You're gonna rise, rise up singing
You're gonna spread your wings, child
And take, take to the sky
Lord, the sky...*

*Big Brother & The Holding Company
George Gershwin cover, studio version*

Follow the signs that the world quietly leaves for you: the scent of a good kitchen behind a half-open door; the drift of conversation in rooms where ideas glitter; the precision of a paragraph written by someone whose mind is fully awake; the quick silver of intelligence in another person's eyes. These will lead you to ***your people*** — the ones whose presence sharpens you rather than dulls you.

Arrive in the right lab at just the right moment, the way someone once walked into the orbit of Watson and Crick. Stand next to a young Steve Jobs while he is still fumbling with the design of the first iPhone. Catch the ship sailing out of Spain toward the raw, unmapped colonies. Slip a little money into Dutch tulips, or — centuries later — into Nvidia or Google before the world has any idea what those names will mean.

These people are not hard to recognize. They read. They write. They build. They imagine. Some paint, some compose, some open new businesses bright with possibility. Wealth is not their defining feature, though it may follow them like a loyal dog. What distinguishes them is subtler: their way of looking at the world, their children, their instinct for how to coordinate effort, how to guide others, how to kindle talent until it burns clean and bright.

And then, when you've learned to recognize such people — and to lead them — your own time will come. You'll assemble a crew for your own ship or skiff: open a restaurant, start a small lab, sketch the design of the next Lululemon. Creating your own business with bright, sharp minds will give you a kind of *independence the outside world can neither grant nor revoke* — an independence from the whims of fashion and the passing storms of “social justice” as it appears in headlines. Imagine building something flawless and austere, like a

katana. Or something intimate and radiant, like a Japanese stone garden tucked behind a strong brick wall.

Among the people around you, there may be someone with whom you will one day want to build a life — someone you might marry, someone with whom you might have children. Remember this: your children will inherit half of themselves from that person. Not just temperament and talent, but the deeper, quieter possibilities written into them from the very beginning. So, choose carefully. Choose someone strong, curious, and resilient — someone who sees the world not as something to endure, but as something to shape.

And now — let's speak of the sadder side of the world. What becomes of the others? The ones who spend their evenings watching CNN or the BBC, dragging themselves to their jobs with no spark of ambition, never even entertaining the idea of building something of their own because risk, to them, is a kind of poison. The drifters and daydreamers who have convinced themselves that every conflict can now be solved politely, or who naïvely believe that all people are born identical in talent and temperament.

They trade the precious, luminous hours of their lives for the small change of “fighting climate change,” or “eagerly funding Palestine ‘victims’ who know exactly how to tug the public’s purse strings,” or whatever the slogan of the week happens to be. They pour money — money that could have supported their children, strengthened their families, or built a foundation for the future — into causes as flimsy and rustling as dry leaves.

What awaits them in the end may be darker than anything Bosch ever painted. Leave them be; you’re unlikely to help them. To borrow Dawkins’s metaphor, their boats are crewed by bad rowers. Don’t argue with them; don’t try to explain how the clockwork of our world functions, or where the hands are moving next.

At moments like this I can only recall one of my favorite passages from the Gospel:

The scribes and Pharisees brought to Him a woman caught in adultery. Setting her before Him, they said, “Teacher, this woman was taken in adultery. Moses commanded us to stone such women. What do You say?” But Jesus bent down and wrote on the ground, paying them no heed. When they pressed Him, He stood and said, “Let him who is without sin cast the first stone.” And again, He bent down and wrote on the earth. Hearing this, they felt the sting of their own consciences and walked away one by one, from the eldest to the last. And Jesus, lifting His eyes and seeing no one but the woman, said: “Woman, where are your accusers? Has no one condemned you?” “No one, Lord,” she replied. “Neither do I condemn you,” Jesus said. “Go, and sin no more.” (John 8:3–11)

And you, do not condemn anyone. But do not follow them either. Your path is elsewhere.

Autumn will be busy for you. You will need to gather provisions: reliable investments, land, a home, the means to educate your children and grandchildren. Help them generously — with money, with houses, with whatever you can. The fashionable notion that children must achieve everything on their own is a pleasant myth. Education will only grow more

expensive. Land and property will rise in value with the persistence of gravity. Buy when you can, and keep such things within the family. Forgive me for the plainness of this advice; its simplicity is the mark of its truth.

And then, gently, winter will come for us. Do not be frightened. When you are surrounded by children and grandchildren, there is no such thing as death.

Look at the Romans. What remains of them? Roman law, the unshakable ranks of the legions, guarding the borders of the Empire, the magnificent Pantheon, the aqueducts — and their descendants, who later gathered these fragments and built the Renaissance. What remains of Castile and Aragon? The luminous canvases in the Prado, the navigators who opened the New World, and their descendants, who followed and ruled this world, a realm that stretched from Argentina to Tijuana.

After us, too, something will remain. Perhaps the newborn science of gene editing. Perhaps wonders we cannot yet imagine. Your kids, with the tools they create, may one day speak with God not as petitioners but as collaborators. Some of them, I suspect, will quietly step out of the old category of “servants of God” into the role of his students — as Leonardo hinted in that miraculous fresco, *The Last Supper*, in the refectory of *Santa Maria delle Grazie*. Life is like a genetic tapestry, unfolding and refining itself with every new generation. Your task is simply to carry your thread with care and pass it forward brighter than you received it.

A list of top-notch books, hand-picked so you don't have to suffer:

1. George Orwell, *1984*, 1949
2. William Shakespeare, *Hamlet*, 1599–1601
3. Guido Tonelli, *Genesis: The Story of How Everything Began*, 2019
4. Stephen Hawking, *The Universe in a Nutshell*, 2001
5. Sean Carroll, *From Eternity to Here*, 2010
6. David Bericovici, *The Origins of Everything*, 2016
7. Zack Scott, *Across the Universe*, 2023
8. Robert Hazen, *The Story of Earth*, 2012
9. Erik Asphaug, *When the Earth Had Two Moons*, 2019
10. David Alt and D. Hyndman, *Roadside Geology of Northern and Central California*, 2016
11. Ted Nield, *Supercontinent: Ten Billion Years in the Life of Our Planet*, 2009
12. Robert Hazen, *Symphony in C*, 2019
13. Nick Lane, *The Vital Question*, 2015
14. Richard Dawkins, *The Selfish Gene*, 1976
15. Henry Gee, *A Short History of Life on Earth*, 2021
16. Николай Кукушкин, *Хлопок одной ладонью*, 2020
17. Nick Lane, *Power, Sex, Suicide*, 2005
18. Lewis Dartnell, *Origins: How Earth's History Shaped Human History*, 2019
19. Neil Shubin, *Your Inner Fish*, 2008
20. Dean Lomax, *Locked in Time*, 2021
21. Geoffrey Miller, *The Mating Mind*, 2003
22. Svante Pääbo, *Neanderthal Man*, 2014
23. Alice Roberts, *The Incredible Unlikeliness of Being*, 2014
24. Nicholas Wade, *Before the Dawn*, 2006

25. Nicholas Wade, *A Troublesome Inheritance: Genes, Race and Human History*, 2014
26. David Reich, *Who We Are and How We Got Here. Ancient DNA*, 2018
27. Dick Swaab, *We Are Our Brains: From the Womb to Alzheimer's*, 2015
28. Richard Dawkins. *The Ancestor's Tale: A Pilgrimage to the Dawn of Evolution*, 2004
29. Tom Higham, *The World Before Us*, 2021
30. Ian Morris, *Foragers, Farmers and Fossil Fuels*, 2015
31. John Kelly, *The Great Mortality: An Intimate History of the Black Death*, 2005
32. Beth Shapiro, *Life as We Made It*, 2021
33. John Julius Norwich, *A Short History of Byzantium*, 1997
34. Paul Kriwaczek, *Babylon: Mesopotamia and the Birth of Civilization*, 2010
35. Toby Wilkinson, *The Rise and Fall of Ancient Egypt*, 2010
36. Chris Naunton, *Searching for the Lost Tombs of Egypt*, 2018
37. Mark Edward Lewis, *The Early Chinese Empires*, 2007
38. Roel Sterckx, *Chinese Thought: From Confucius to Cook Ding*, 2019
39. Линь Мэйцунь, *15 Лекций об археологии Великого шелкового пути*, 2021
40. Robert Hughes, *Rome*, 2011
41. Mary Beard, *SPQR: A History of Ancient Rome*, 2015
42. Theodor Mommsen, *The History of Rome*, 1854–1856
43. Lincoln Paine, *The Sea and Civilization: A Maritime History of the World*, 2013
44. Peter Heather, *The Restoration of Rome*, 2013
45. Peter Heather, *Empires and Barbarians*, 2009
46. Niall Ferguson, *The West and the Rest*, 2011
47. Niall Ferguson, *Empire*, 2003
48. Daniel Yergin, *The Prize. The Epic Quest for Oil, Money & Power*, 1991
49. William McNeil, *The Rise of the West*, 1963
50. Ian Morris, *Why the West Rules—for Now*, 2010

51. Daron Acemoglu and James Robinson, *Why Nations Fail*, 2012
52. Ray Dalio, *Principles for Dealing with the Changing World Order*, 2021
53. Robert Penn Warren, *All the King's Men*, 1946
54. Philip Dormer Stanhope Chesterfield, *Letters to His Son*, 1749